



LAN8840

Gigabit Ethernet Transceiver with RGMII and IEEE 1588v2 Support

Features

- Single-Chip 10/100/1000 Mbps Ethernet Transceiver Suitable for IEEE 802.3 Applications
- RGMII with 3.3V/2.5V/1.8V I/Os
 - RGMII Timing Supports On-Chip Delay According to RGMII Version 2.0, with Programming Options for External Delay and Making Adjustments and Corrections to TX and RX Timing Paths
- Auto-Negotiation to Automatically Select the Highest Link-Up Speed (10/100/1000 Mbps) and Duplex (Half/Full)
- On-Chip Termination Resistors for the Differential Pairs
- On-Chip LDO Controller to Support Single 3.3V Supply Operation – Requires Only One External FET to Generate the Core Voltage
- Jumbo Frame Support Up to 16 KB
- 125 MHz Reference Clock Output
- Energy-Detect Power-Down Mode for Reduced Power Consumption When Cable is Not Attached
- EtherSynch® IEEE 1588v2/PTP
 - Layer 2, UDP/IPv4 and UDP/IPv6 formats
 - Tagged and non-tagged frame formats
 - One-step and two-step modes of operation
- Energy Efficient Ethernet (EEE) Support with Low-Power Idle (LPI) Mode and Clock Stoppage for 100BASE-TX/1000BASE-T and Transmit Amplitude Reduction with 10BASE-Te Option
- Wake-On-LAN (WOL) Support with Robust Custom-Packet Detection
- Programmable LED Outputs for Link, Activity, and Speed
- Baseline Wander Correction
- LinkMD® TDR-based Cable Diagnostic to Identify Faulty Copper Cabling
- Signal Quality Indication
- Parametric NAND Tree Support to Detect Faults Between Chip I/Os and Board
- Loopback Modes for Diagnostics
- Automatic MDI/MDI-X Crossover to Detect and Correct Pair Swap at All Speeds of Operation
- Automatic Detection and Correction of Pair Swaps, Pair Skew, and Pair Polarity
- MDC/MDIO Management Interface for PHY Register Configuration
- Interrupt Pin Option
- Power-Down and Power-Saving Modes
- Operating Voltages
 - Core (VDD, VDDAL, VDDAL_PLL)
 - VDD I/O (VDDIO): 3.3V, 2.5V, or 1.8V
 - Transceiver (VDDAH): 3.3V or 2.5V
- Available in commercial (0°C to +70°C) and extended industrial (-40°C to +105°C) temperature ranges
- 48-pin VQFN (7 mm × 7 mm) Package

Target Applications

- Industrial Control
- Laser/Network Printer
- Network Attached Storage (NAS)
- Network Server
- Gigabit LAN on Motherboard (GLOM)
- Broadband Gateway
- Gigabit SOHO/SMB Router
- IPTV
- IP Set-Top Box
- Game Console
- Triple-Play (Data, Voice, Video) Media Center



LAN8840

支持 RGMII 和 IEEE 1588v2 的千兆以太网收发器

特性

- 适用于 IEEE 802.3 应用的单芯片 10/100/1000 Mbps 以太网收发器
 - 采用 3.3V/2.5V/1.8V 输入/输出的 RGMII
 - RGMII 时序支持符合 RGMII 2.0 版的片上延迟, 并提供外部延迟编程选项, 以及对发送和接收时序路径进行调整和校正的功能
- 自动协商, 可自动选择最高链路建立速度 (10/100/1000 Mbps) 和双工模式 (半双工/全双工)
- 差分对采用片上端接电阻
- 片上 LDO 控制器支持单路 3.3V 电源供电——仅需一个外部场效应晶体管即可生成内核电压
- 支持最大 16 KB 的巨型帧
- 125 兆赫参考时钟输出
- 电缆未连接时进入能量检测掉电模式, 以降低功耗
 - EtherSynch® IEEE 1588v2/精确时间协议
 - 第二层、UDP/IPv4 和 UDP/IPv6 格式
 - 带标签和非标签帧格式
 - 单步和双步运行模式
- 支持节能以太网 (EEE), 采用低功耗空闲 (LPI) 模式, 并为 100BASE-TX/1000BASE-T 提供时钟停止功能, 同时支持 10BASE-Te 选项的发送幅度降低
- 支持网络唤醒 (WOL), 并具备强大的自定义数据包检测功能
- 可编程 LED 输出: 链路、活动和速率

- 基线漂移校正
- LinkMD® 基于 TDR 的电缆诊断, 用于识别有故障的铜缆布线
- 信号质量指示
- 参数化 NAND 树支持, 用于检测芯片 I/O 与电路板之间的故障
- 用于诊断的环回模式
- 自动 MDI/MDI-X 交叉, 可在所有工作速率下检测并纠正线对交换
- 自动检测并纠正线对交换、线对偏斜和线对极性

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1.0 PREFACE

1.1 General Terms

TABLE 1-1: GENERAL TERMS

Term	Description
1000BASE-T	1 Gbps Ethernet over twisted pair, IEEE 802.3 compliant
100BASE-TX	100 Mbps Ethernet over twisted pair, IEEE 802.3 compliant
10BASE-T	10 Mbps Ethernet over twisted pair, IEEE 802.3 compliant
ADC	Analog-to-Digital Converter
AFE	Analog Front End
AN, ANEG	Auto-Negotiation
BYTE	8-bits
DA	Destination Address
DCQ	Dynamic Channel Quality
EEE	Energy Efficient Ethernet
FCS	Frame Check Sequence
FSM	Finite State Machine
GPIO	General Purpose I/O
HOST	External system (Includes processor, application software, etc.)
LDO	Linear Drop-Out Regulator
LFSR	Linear Feedback Shift Register
MAC	Media Access Controller
MAGJACK	Configuration strap for power modes.
MDI	Medium Dependent Interface
MDIX	Media Independent Interface with Crossover
MII	Media Independent Interface
MLT-3	Multi-Level Transmission Encoding (3-Levels). A tri-level encoding method where a change in the logic level represents a code bit “1” and the logic output remaining at the same level represents a code bit “0”.
N/A	Not Applicable
OTP	One Time Programmable
PCS	Physical Coding Sublayer
PLL	Phase Locked Loop
POR	Power on Reset.
PTP	Precision Time Protocol
RESERVED	Refers to a reserved bit field or address. Unless otherwise noted, reserved bits must always be zero for write operations. Unless otherwise noted, values are not guaranteed when reading reserved bits. Unless otherwise noted, do not read or write to reserved addresses.

1.0 前言

1.1 一般术语

表 1-1：一般术语

Term	说明
1000BASE-T	基于双绞线的千兆比特每秒以太网，符合 IEEE 802.3 标准
100BASE-TX	基于双绞线的 100 Mbps 以太网，符合 IEEE 802.3 标准
10BASE-T	双绞线上的 10 兆比特每秒以太网，符合 IEEE 802.3 标准
ADC	模数转换器
AFE	模拟前端
自动协商, 自动协商	自动协商
BYTE	8 位
DA	目的地址
DCQ	动态信道质量
EEE	节能以太网
FCS	帧校验序列
FSM	有限状态机
GPIO	通用输入/输出
HOST	外部系统（包括处理器、应用软件等）
LDO	低压差稳压器
LFSR	线性反馈移位寄存器
MAC	媒体访问控制器
磁性网络接口插座	电源模式配置跳线。
MDI	介质相关接口
MDIX	介质无关交叉接口
MII	介质无关接口
三电平传输编码	多电平传输编码（3 个电平）。一种三电平编码方法，其中逻辑电平发生变化表示编码位 “1”，而逻辑输出保持在同一电平表示编码位 “0”。
N/A	不适用
OTP	一次性可编程
PCS	物理编码子层
PLL	锁相环
POR	上电复位。
PTP	精确时间协议
保留	指保留位字段或地址。除非另有说明，写操作时保留位必须始终为零。除非另有说明，读取保留位时不保证其值。除非另有说明，请勿读取或写入保留地址。

TABLE 1-1: GENERAL TERMS (CONTINUED)

Term	Description
RGMII	Reduced Gigabit Media Independent Interface
SA	Source Address
SFD	Start of Frame Delimiter - The 8-bit value indicating the end of the preamble of an Ethernet frame
SQI	Signal Quality Indicator
UDP	User Datagram Protocol - A connectionless protocol run on top of IP networks

1.2 Buffer Types

TABLE 1-2: BUFFER TYPE DESCRIPTIONS

BUFFER	DESCRIPTION
AI	Analog input
AO	Analog output
AIO	Analog bidirectional
ICLK	Crystal oscillator input pin
OCLK	Crystal oscillator output pin
RGMII_O	RGMII compliant output
VIS	Variable voltage Schmitt-triggered input
VO10	Variable voltage output with 10 mA sink and 10 mA source
VOD10	Variable voltage open-drain output with 10 mA sink
VOS10	Variable voltage open-source output with 10 mA source
PU	70 KΩ (typical) internal pull-up. Unless otherwise noted in the pin description, internal pull-ups are always enabled. Note: Internal pull-up resistors prevent unconnected inputs from floating. Do not rely on internal resistors to drive signals external to the device. When connected to a load that must be pulled high, an external resistor must be added.
P	Power pin

Note: Digital signals are not 5V tolerant unless specified.

1.3 Reference Documents

- IEEE 802.3TM-2022 IEEE Standard for Ethernet, <https://standards.ieee.org/ieee/802.3/10422/>
- Reduced Gigabit Media Independent Interface (RGMII) Specification Version 2.0, https://web.archive.org/web/20160303171328/http://www.hp.com/rnd/pdfs/RGMIIv2_0_final_hp.pdf
- OPEN Alliance TC1 - Advanced diagnostics features for 100BASE-T1 automotive Ethernet PHYs Version 1.0 http://www.opensig.org/download/document/218/Advanced_PHY_features_for_automotive_Ethernet_V1.0.pdf

表 1-1：一般术语（续）

Term	说明
RGMII	简化千兆媒体独立接口
SA	源地址
SFD	帧起始定界符 - 指示以太网帧前导码结束的 8 位值
SQI	信号质量指示器
UDP	用户数据报协议 - 一种运行于 IP 网络之上的无连接协议

1.2 缓冲区类型

表 1-2：缓冲区类型说明

缓冲区	说明
AI	模拟输入
AO	模拟输出
AIO	模拟双向
ICLK	晶体振荡器输入引脚
OCLK	晶体振荡器输出引脚
RGMII O ₋	符合 RGMII 规范的输出
VIS	可变电压施密特触发输入
VO10	可变电压输出，具有 10 mA 灌电流和 10 mA 拉电流
VOD10	可变电压开漏输出，具有 10 mA 灌电流
VOS10	可变电压开源输出，具有 10 mA 拉电流
PU	70 KΩ （典型值）内部上拉。除非引脚说明中另有说明，否则内部上拉始终使能。 注： 内部上拉电阻可防止未连接的输入端处于浮空状态。请勿依赖内部电阻驱动器器件外部的信号。连接到必须被上拉至高电平的负载时，必须添加外部电阻。
P	电源引脚

注意：除非另有说明，数字信号不耐受 5V 电压。

1.3 参考文档

- IEEE 802.3TM-2022 IEEE 以太网标准, <https://standards.ieee.org/ieee/802.3/10422/>
- 精简千兆媒体独立接口 (RGMII) 规范 版本 2.0, https://web.archive.org/web/20160303171328/http://www.hp.com/rnd/pdfs/RGMIIv2_0_final_hp.pdf
- OPEN 联盟 TC1 - 100BASE-T1 汽车以太网 PHY 高级诊断功能 1.0版 http://www.opensig.org/download/document/218/面向汽车以太网的高级_PHY_特性____V1.0.pdf

LAN8840

2.0 INTRODUCTION

2.1 General Description

The LAN8840 is a completely integrated triple-speed (10BASE-T/100BASE-TX/1000BASE-T) Ethernet physical-layer transceiver for transmission and reception of data on standard CAT-5 as well as CAT-5e and CAT-6 unshielded twisted pair (UTP) cables.

The LAN8840 provides the Reduced Gigabit Media Independent Interface (RGMII) for direct connection to RGMII MACs in Gigabit Ethernet processors and switches for data transfer at 10/100/1000 Mbps.

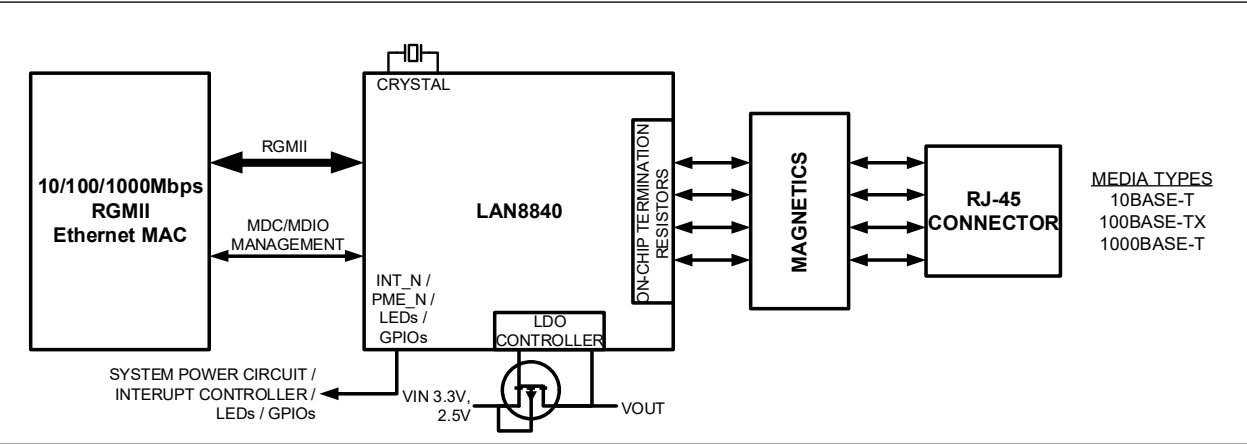
The IEEE1588-2008 PTP functions provide hardware support for the IEEE Std 1588-2008 (v2) Precision Time Protocol (PTP), allowing clock synchronization with remote Ethernet devices, packet time stamping, and time driven event generation. The device supports server or client modes for ordinary and boundary clock operations and a transparent clock per the IEEE Std 1588-2008 specification. End-to-end and peer-to-peer link delay mechanisms are supported as are one-step and two-step operations.

The LAN8840 reduces board cost and simplifies board layout by using on-chip termination resistors for the four differential pairs and by integrating an LDO controller to drive a low-cost MOSFET to supply the core voltage.

The LAN8840 offers diagnostic features to facilitate system bring-up and debugging in production testing and in product deployment. Parametric NAND tree support enables fault detection between LAN8840 I/Os and the board. The LinkMD[®] TDR-based cable diagnostic identifies faulty copper cabling. Remote, external, and local loopback functions verify analog and digital data paths.

The LAN8840 is available in a 48-pin, RoHS Compliant VQFN package.

FIGURE 2-1: SYSTEM BLOCK DIAGRAM



LAN8840

2.0 简介

2.1 概述

LAN8840 是一款完全集成的三速（10BASE-T/100BASE-TX/1000BASE-T）以太网物理层收发器，可通过标准 CAT-5、CAT-5e 和 CAT-6 非屏蔽双绞线 (UTP) 电缆发送和接收数据。

LAN8840 提供简化千兆媒体独立接口 (RGMII)，可直接连接到千兆以太网处理器和交换机中的 RGMII MAC，以 10/100/1000 Mbps 传输数据。

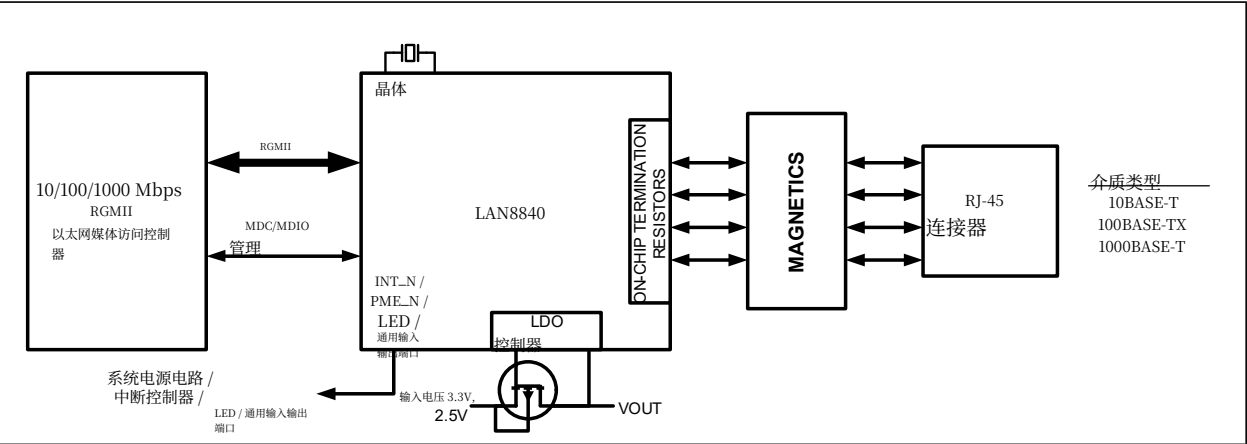
IEEE 1588-2008 PTP 功能提供对 IEEE 1588-2008 标准（v2）精确时间协议（PTP）的硬件支持，可与远程以太网设备进行时钟同步、为数据包添加时间戳，并生成由时间驱动的事件。该器件支持普通时钟和边界时钟的服务器或客户端模式操作，并支持符合 IEEE 1588-2008 标准的透明时钟。它支持端到端和对等链路延迟机制，以及一步和两步操作。

LAN8840 通过为四组差分对采用片上端接电阻，并集成 LDO 控制器来驱动低成本 MOSFET 以提供内核电压，从而降低电路板成本并简化电路板布局。

LAN8840 提供诊断功能，便于在生产测试和产品部署期间进行系统启动调试和调试。参数化 NAND 树支持可检测 LAN8840 输入/输出与电路板之间的故障。LinkMD[®] 基于 TDR 的电缆诊断可识别有故障的铜缆布线。远程、外部和本地环回功能可验证模拟和数字数据路径。

LAN8840 采用 48 引脚、符合 RoHS 标准的 VQFN 封装。

图 2-1：系统框图



5.0 FUNCTIONAL DESCRIPTION

The LAN8840 is a completely integrated triple-speed (10BASE-T/100BASE-TX/1000BASE-T) Ethernet physical layer transceiver solution for transmission and reception of data over a standard CAT-5, as well as CAT-5e and CAT-6, unshielded twisted pair (UTP) cables.

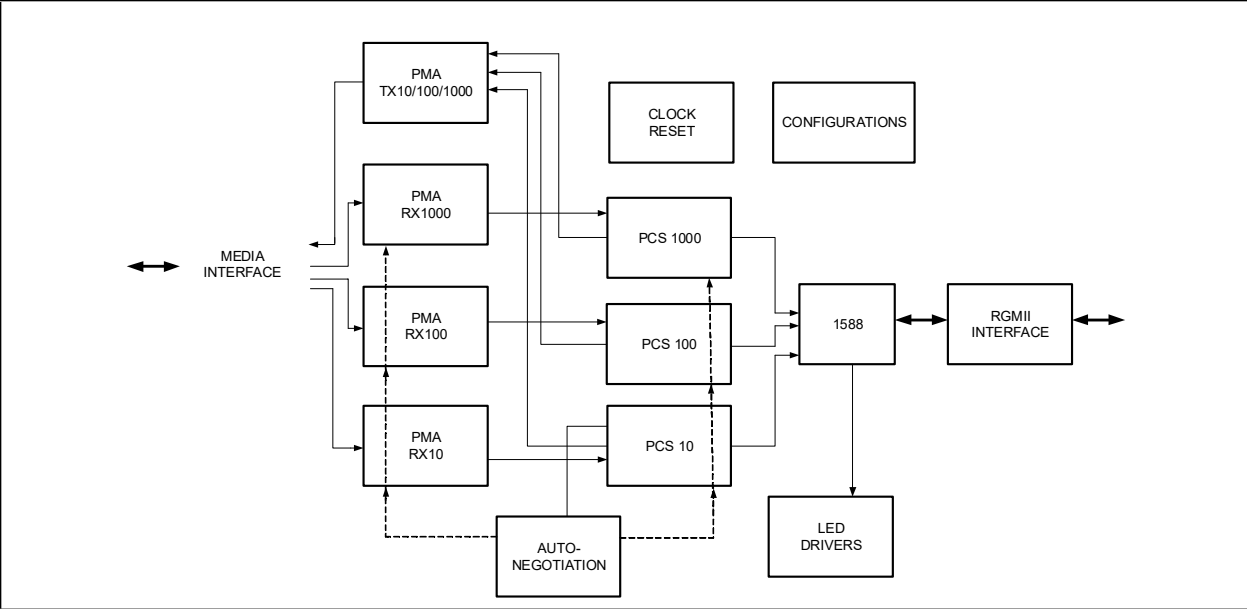
The device reduces board cost and simplifies board layout by using on-chip termination resistors for the four differential pairs and by integrating an LDO controller to drive a low-cost MOSFET to supply the core voltage.

On the copper media interface, the device can automatically detect and correct for differential pair misplacements and polarity reversals, and correct propagation delays and re-sync timing between the four differential pairs, as specified in the IEEE 802.3 standard for 1000BASE-T operation.

The LAN8840 provides the RGMII interface for connection to RGMII MACs in Gigabit Ethernet processors and switches for data transfer at 10/100/1000 Mbps.

Figure 5-1 shows a high-level block diagram of the LAN8840.

FIGURE 5-1: LAN8840 BLOCK DIAGRAM



5.1 10BASE-T/100BASE-TX Transceiver

5.1.1 100BASE-TX TRANSMIT

The 100BASE-TX transmit function performs parallel-to-serial conversion, 4B/5B coding, scrambling, NRZ-to-NRZI conversion, and MLT-3 encoding and transmission.

The circuitry starts with a parallel-to-serial conversion, which converts the MII data from the MAC into a 125 MHz serial bit stream. The data and control stream is then converted into 4B/5B coding, followed by a scrambler. The serialized data is further converted from NRZ-to-NRZI format, and then transmitted in MLT-3 current output. The output current is set by an external 6.04 kΩ 1% resistor for the 1:1 transformer ratio.

The output signal has a typical rise/fall time of 4 ns and complies with the ANSI TP-PMD standard regarding amplitude balance, and overshoot. The wave-shaped 10BASE-T output is also incorporated into the 100BASE-TX transmitter.

5.1.2 100BASE-TX RECEIVE

The 100BASE-TX receiver function performs adaptive equalization, DC restoration, MLT-3-to-NRZI conversion, data and clock recovery, NRZI-to-NRZ conversion, de-scrambling, 4B/5B decoding, and serial-to-parallel conversion.

5.0 功能说明

LAN8840 是一款完全集成的三速（10BASE-T/100BASE-TX/1000BASE-T）以太网物理层收发器解决方案，可通过标准 CAT-5、CAT-5e 和 CAT-6 非屏蔽双绞线（UTP）电缆发送和接收数据。

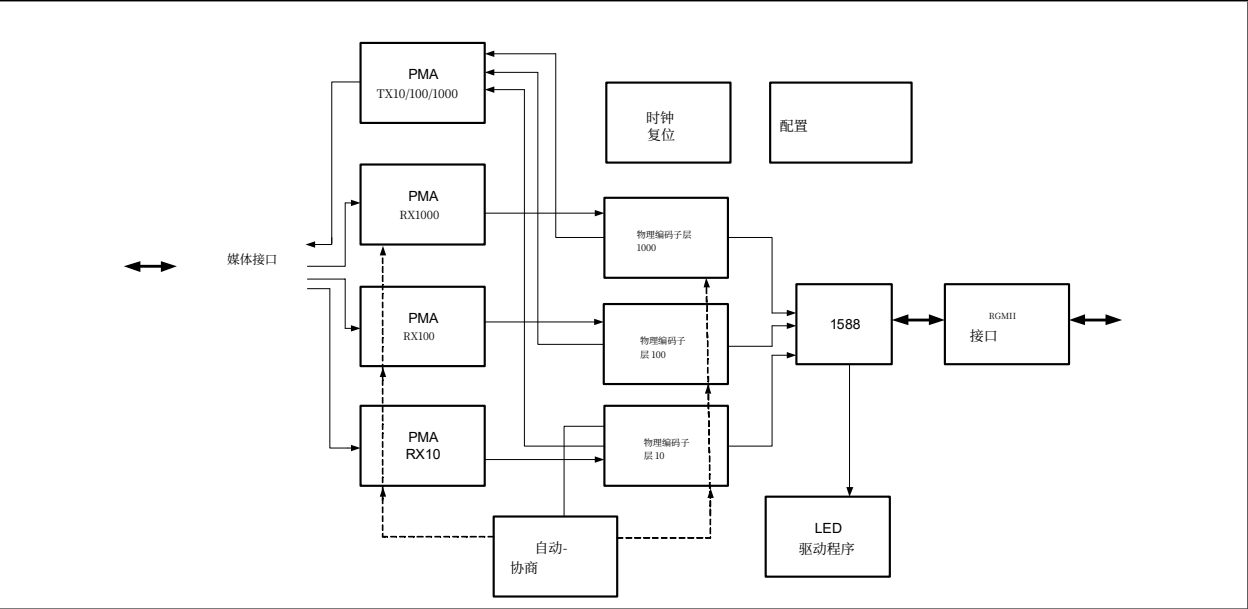
该器件通过为四组差分线对采用片上端接电阻，并集成 LDO 控制器来驱动低成本 MOSFET，以提供内核电压，从而降低电路板成本并简化电路板布局。

在铜缆介质接口上，该器件可自动检测并纠正差分线对错位和极性反转，还可纠正四组差分线对之间的传播延迟并重新同步时序，符合 IEEE 802.3 标准中针对 1000BASE-T 操作的规定。

LAN8840 提供 RGMII 接口，可连接到千兆以太网处理器和交换机中的 RGMII MAC，以 10/100/1000 Mbps 传输数据。

图 5-1 显示了 LAN8840 的高级框图。

图 5-1: LAN8840 框图



5.1 10BASE-T/100BASE-TX 收发器

5.1.1 100BASE-TX 发送

100BASE-TX 发送功能执行并串转换、4B/5B 编码、扰码、NRZ 到 NRZI 转换以及三电平传输编码和发送。

该电路首先进行并串转换，将来自 MAC 的媒体独立接口数据转换为 125 MHz 串行比特流。随后，数据和控制流转换为 4B/5B 编码，并经过扰码器处理。串行化后的数据进一步从 NRZ 格式转换为 NRZI 格式，然后以 MLT-3 电流输出形式发送。对于 1:1 变压器匝数比，输出电流由外部 6.04 kΩ 1% 电阻设定。

输出信号的典型上升/下降时间为 4 ns，且在幅度平衡和过冲方面符合 ANSI TP-PMD 标准。波形整形的 10BASE-T 输出也集成在 100BASE-TX 发射器中。

5.1.2 100BASE-TX 接收

100BASE-TX 接收器功能执行自适应均衡、直流恢复、MLT-3 到 NRZI 转换、数据和时钟恢复、NRZI 到 NRZ 转换、解扰、4B/5B 解码以及串并转换。

LAN8840

The receiving side starts with the equalization filter to compensate for inter-symbol interference (ISI) over the twisted pair cable. Because the amplitude loss and phase distortion are a function of the cable length, the equalizer must adjust its characteristics to optimize performance. In this design, the variable equalizer makes an initial estimation based on comparisons of incoming signal strength against some known cable characteristics, then tunes itself for optimization. This is an ongoing process and self-adjusts against environmental changes such as temperature variations.

Next, the equalized signal goes through a DC-restoration and data-conversion block. The DC-restoration circuit compensates for the effect of baseline wander and improves the dynamic range. The differential data conversion circuit converts the MLT-3 format back to NRZI. The slicing threshold is also adaptive.

The clock-recovery circuit extracts the 125 MHz clock from the edges of the NRZI signal. This recovered clock is then used to convert the NRZI signal into the NRZ format. This signal is sent through the de-scrambler followed by the 4B/5B decoder. Finally, the NRZ serial data is converted to the RGMII format and provided as the input data to the MAC.

5.1.3 SCRAMBLER/DE-SCRAMBLER (100BASE-TX ONLY)

The purpose of the scrambler is to spread the power spectrum of the signal to reduce electromagnetic interference (EMI) and baseline wander. Transmitted data is scrambled using an 11-bit wide linear feedback shift register (LFSR). The scrambler generates a 2047-bit non-repetitive sequence, then the receiver de-scrambles the incoming data stream using the same sequence as at the transmitter.

5.1.4 10BASE-T TRANSMIT

The 10BASE-T output drivers are incorporated into the 100BASE-TX drivers to allow for transmission with the same magnetic. The drivers perform internal wave-shaping and pre-emphasis, and output signals with typical amplitude of 2.5V peak for standard 10BASE-T mode and 1.75V peak for energy-efficient 10BASE-Te mode. The 10BASE-T/10BASE-Te signals have harmonic contents that are at least 31 dB below the fundamental frequency when driven by an all-ones Manchester-encoded signal.

5.1.5 10BASE-T RECEIVE

On the receive side, input buffer and level-detecting squelch circuits are used. A differential input receiver circuit and a phase-locked loop (PLL) perform the decoding function. The Manchester-encoded data stream is separated into clock signal and NRZ data. A squelch circuit rejects signals with levels less than 300 mV or with short pulse widths to prevent noises at the receive inputs from falsely triggering the decoder. When the input exceeds the squelch limit, the PLL locks onto the incoming signal and the device decodes a data frame. The receiver clock is maintained active during idle periods between receiving data frames.

The device removes all 7 bytes of the preamble and presents the received frame starting with the SFD (start of frame delimiter) to the MAC.

Auto-polarity correction is provided for the receiving differential pair to automatically swap and fix the incorrect +/- polarity wiring in the cabling.

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接收端首先通过均衡滤波器补偿双绞线电缆中的码间干扰（ISI）。由于幅度损耗和相位失真取决于电缆长度，均衡器必须调整其特性以优化性能。在此设计中，可变均衡器会将输入信号强度与一些已知的电缆特性进行比较，据此进行初步估算，然后自行调节以达到最佳效果。这是一个持续进行的过程，均衡器会根据温度变化等环境变化进行自我调整。

接下来，均衡后的信号会经过直流恢复和数据转换模块。直流恢复电路可补偿基线漂移的影响并扩大动态范围。差分数据转换电路将三电平传输编码格式转换回 NRZI。判决门限也会自适应调整。

时钟恢复电路从 NRZI 信号的边沿提取 125 兆赫时钟。随后使用恢复的时钟将 NRZI 信号转换为 NRZ 格式。该信号经过解扰器和 4B/5B 解码器处理。最后，NRZ 串行数据被转换为 RGMII 格式，并作为输入数据提供给媒体访问控制器。

5.1.3 扰码器/解扰器（仅适用于 100BASE-TX）

扰码器的作用是扩展信号的功率谱，以减少电磁干扰（EMI）和基线漂移。发送的数据使用 11 位宽的线性反馈移位寄存器（LFSR）进行扰码。扰码器会生成一个 2047 位的非重复序列，然后接收器使用与发射器相同的序列对接收到的数据流进行解扰。

5.1.4 10BASE-T 发送

10BASE-T 输出驱动器集成在 100BASE-TX 驱动器中，因此可以使用相同的磁性元件进行发送。驱动器执行内部波形整形和预加重，标准 10BASE-T 模式下输出信号的典型峰值幅度为 2.5V，节能型 10BASE-Te 模式下则为 1.75V。使用全 1 的曼彻斯特编码信号驱动时，10BASE-T/10BASE-Te 信号的谐波成分至少比基频低 31 dB。

5.1.5 10BASE-T 接收

在接收端，使用输入缓冲器和电平检测静音电路。差分输入接收器电路和锁相环（PLL）执行解码功能。曼彻斯特编码数据流被分离为时钟信号和 NRZ 数据。静音电路会抑制电平低于 300 mV 或脉冲宽度较短的信号，以防止接收输入端的噪声错误地触发解码器。当输入超过静音阈值时，PLL 会锁定输入信号，器件随即解码数据帧。在接收数据帧之间的空闲期间，接收时钟保持活动状态。

器件会移除全部 7 字节的前导码，并将从帧起始定界符（SFD）开始的接收帧传送给媒体访问控制器（MAC）。

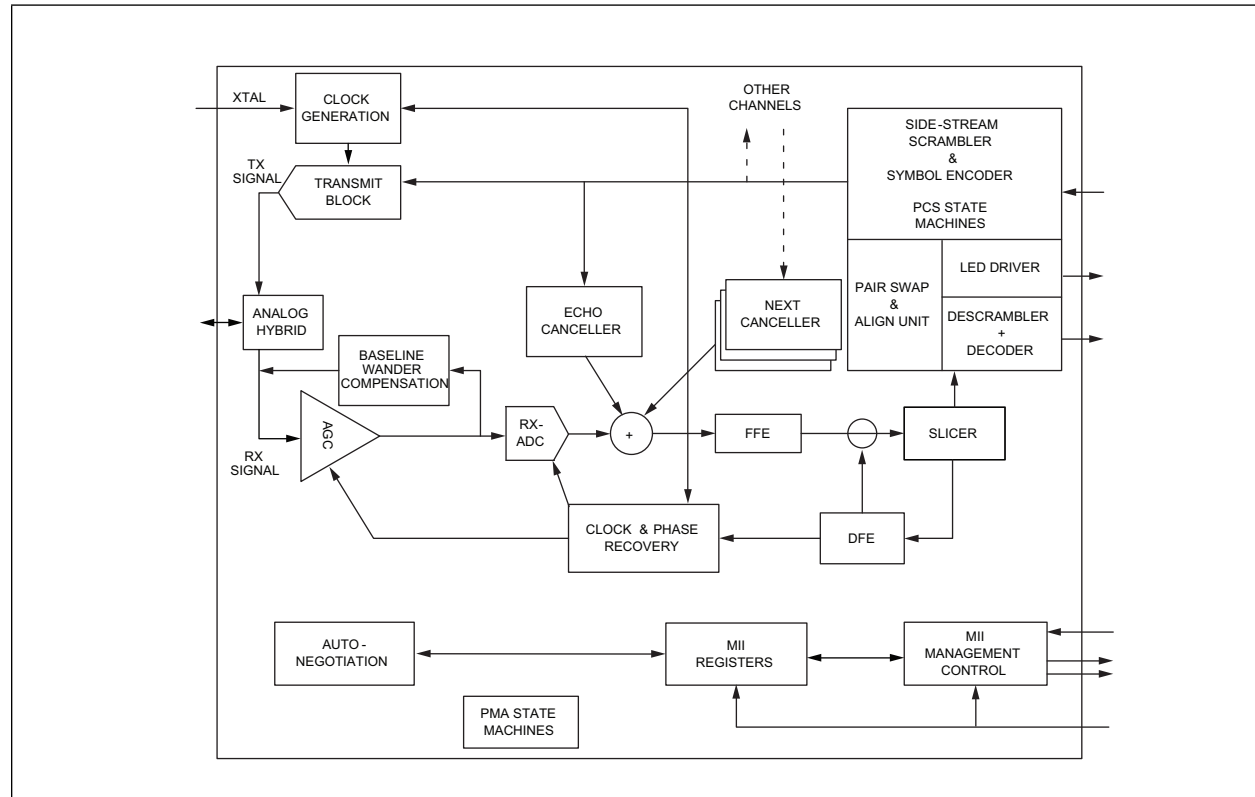
接收差分对具备自动极性校正功能，可自动交换并修正布线中错误的 +/- 极-性接线。

5.2 1000BASE-T Transceiver

The 1000BASE-T transceiver is based-on a mixed-signal/digital-signal processing (DSP) architecture, which includes the analog front-end, digital channel equalizers, trellis encoders/decoders, echo cancelers, cross-talk cancelers, precision clock recovery scheme, and power-efficient line drivers.

Figure 5-2 shows a high-level block diagram of a single channel of the 1000BASE-T transceiver for one of the four differential pairs.

FIGURE 5-2: 1000BASE-T BLOCK DIAGRAM - SINGLE CHANNEL



5.2.1 ANALOG ECHO-CANCELLATION CIRCUIT

In 1000BASE-T mode, the analog echo-cancellation circuit helps to reduce the near-end echo. This analog hybrid circuit relieves the burden of the ADC and the adaptive equalizer.

This circuit is disabled in 10BASE-T/100BASE-TX mode.

5.2.2 AUTOMATIC GAIN CONTROL (AGC)

In 1000BASE-T mode, the automatic gain control (AGC) circuit provides initial gain adjustment to boost up the signal level. This pre-conditioning circuit is used to improve the signal-to-noise ratio of the receive signal.

5.2.3 ANALOG-TO-DIGITAL CONVERTER (ADC)

In 1000BASE-T mode, the analog-to-digital converter (ADC) digitizes the incoming signal. ADC performance is essential to the overall performance of the transceiver.

This circuit is disabled in 10BASE-T/100BASE-TX mode.

5.2.4 TIMING RECOVERY CIRCUIT

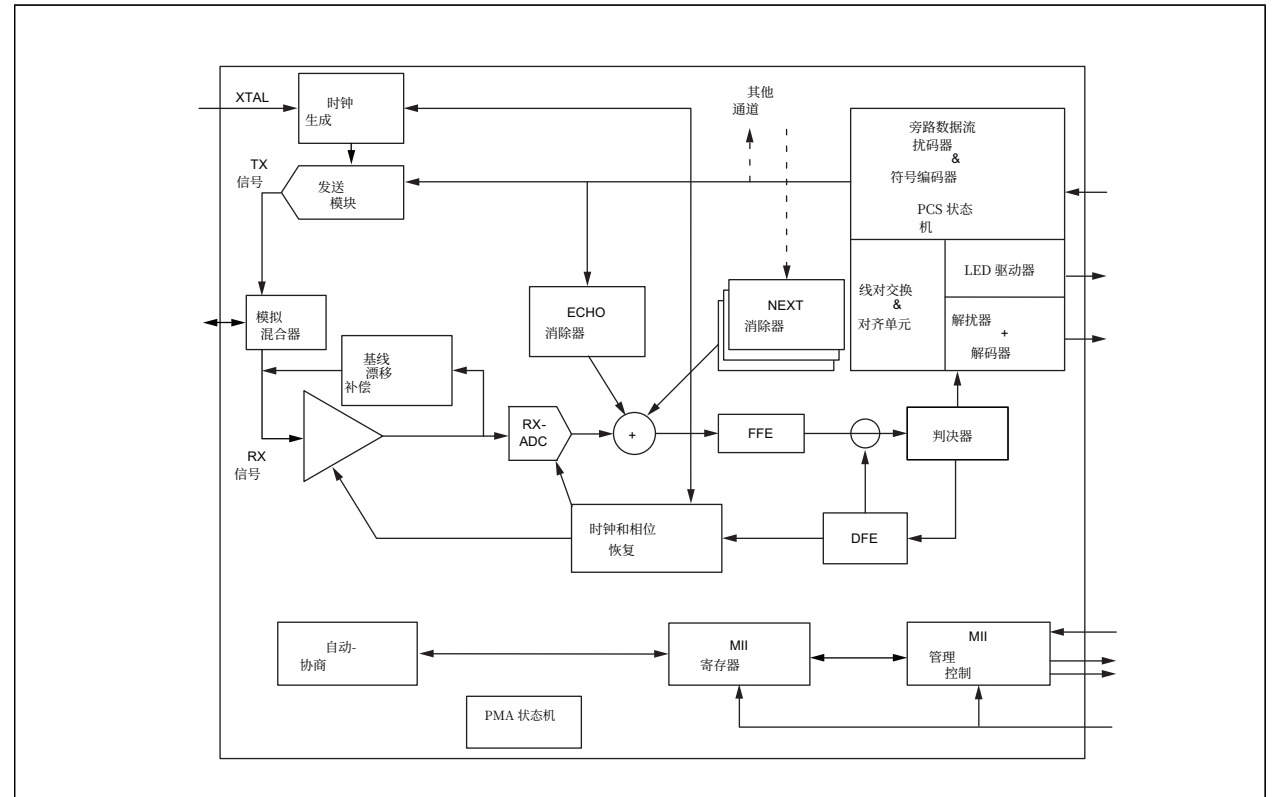
In 1000BASE-T mode, the mixed-signal clock recovery circuit together with the digital phase-locked loop is used to recover and track the incoming timing information from the received data. The digital phase-locked loop has very low long-term jitter to maximize the signal-to-noise ratio of the receive signal.

5.2 1000BASE-T 收发器

1000BASE-T 收发器基于混合信号/数字信号处理 (DSP) 架构, 其中包括模拟前端、数字信道均衡器、网格编码器/解码器、回声消除器、串扰消除器、精密时钟恢复方案和高能效线路驱动器。

图 5-2 显示了 1000BASE-T 收发器中四个差分对之一的单通道高级框图。

图 5-2: 1000BASE-T 单通道框图



5.2.1 模拟回声消除电路

在 1000BASE-T 模式下, 模拟回声消除电路有助于降低近端回声。该模拟混合电路可减轻模数转换器和自适应均衡器的负担。

在 10BASE-T/100BASE-TX 模式下，此电路会被禁用。

5.2.2 自动增益控制 (AGC)

在 1000BASE-T 模式下, 自动增益控制 (AGC) 电路通过提升信号来进行初始增益调整电平。该预处理电路用于提高接收信号的信噪比。

5.2.3 模数转换器 (ADC)

在 1000BASE-T 模式下，模数转换器（ADC）会将输入信号数字化。模数转换器（ADC）的性能对收发器的整体性能至关重要。

在 10BASE-T/100BASE-TX 模式下，此电路处于禁用状态。

5.2.4 定时恢复电路

在 1000BASE-T 模式下, 混合信号时钟恢复电路与数字锁相环配合使用, 从接收数据中恢复并跟踪传入的定时信息。数字锁相环的长期抖动极低, 可最大限度地提高接收信号的信噪比。

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The 1000BASE-T follower PHY must transmit the exact receive clock frequency recovered from the received data back to the 1000BASE-T leader PHY. Otherwise, the leader and follower will not be synchronized after long transmission. This also helps to facilitate echo cancellation and NEXT removal.

5.2.5 ADAPTIVE EQUALIZER

In 1000BASE-T mode, the adaptive equalizer provides the following functions:

- Detection for partial response signaling
- Removal of NEXT and ECHO noise
- Channel equalization

Signal quality is degraded by residual echo that is not removed by the analog hybrid because of impedance mismatch. The device uses a digital echo canceler to further reduce echo components on the receive signal.

In 1000BASE-T mode, data transmission and reception occurs simultaneously on all four pairs of wires (four channels). This results in high-frequency cross-talk coming from adjacent wires. The device uses three NEXT cancelers on each receive channel to minimize the cross-talk induced by the other three channels.

In 10BASE-T/100BASE-TX mode, the adaptive equalizer needs only to remove the inter-symbol interference and recover the channel loss from the incoming data.

5.2.6 TRELLIS ENCODER AND DECODER

In 1000BASE-T mode, the transmitted 8-bit data is scrambled into 9-bit symbols and further encoded into 4D-PAM5 symbols. The initial scrambler seed is determined by the specific PHY address to reduce EMI when more than one device is used on the same board. On the receiving side, the idle stream is examined first. The scrambler seed, pair skew, pair order, and polarity must be resolved through the logic. The incoming 4D-PAM5 data is then converted into 9-bit symbols and de-scrambled into 8-bit data.

5.3 Auto MDI/MDI-X

The Automatic MDI/MDI-X feature eliminates the need to determine whether to use a straight cable or a crossover cable between the device and its link partner. This auto-sense function detects the MDI/MDI-X pair mapping from the link partner, and assigns the MDI/MDI-X pair mapping of the device accordingly.

Table 5-1 shows the device’s 10/100/1000 pin configuration assignments for MDI/MDI-X pin mapping.

TABLE 5-1: MDI/MDI-X PIN MAPPING

Pin (RJ-45 Pair)	MDI			MDI-X		
	1000BASE-T	100BASE-T	10BASE-T	1000BASE-T	100BASE-T	10BASE-T
TXRXP/M_A (1, 2)	A+/-	TX+/-	TX+/-	B+/-	RX+/-	RX+/-
TXRXP/M_B (3, 6)	B+/-	RX+/-	RX+/-	A+/-	TX+/-	TX+/-
TXRXP/M_C (4, 5)	C+/-	Not Used	Not Used	C+/-	Not Used	Not Used
TXRXP/M_D (7, 8)	D+/-	Not Used	Not Used	D+/-	Not Used	Not Used

Auto-MDIX detection is enabled in the device by default.

Auto-MDIX can be disable by setting the swapoff bit in the Digital Debug Control 1 Register. The MDI / MDI-X mode may then be manually selected by the mdi_set bit in the Digital Debug Control 1 Register.

The Auto-MDIX status bits are located in the Digital AX/AN Status Register.

An isolation transformer with symmetrical transmit and receive data paths is recommended to support Auto MDI/MDI-X.

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1000BASE-T 从 PHY 必须将从接收数据中恢复出的精确接收时钟频率发送回 1000BASE-T 主 PHY。否则，经过长时间传输后，主端和从端将无法保持同步。这也有助于实现回声消除和 NEXT 消除。

5.2.5 自适应均衡器

在 1000BASE-T 模式下，自适应均衡器提供以下功能：

- 部分响应信令检测
- 消除 NEXT 和回声噪声
- 信道均衡

由于阻抗失配，模拟混合器未能消除残余回声，从而降低了信号质量。该器件使用数字回声消除器，进一步减少接收信号中的回声分量

在 1000BASE-T 模式下，数据会同时通过全部四对线路（四个通道）进行传输和接收，因此相邻线路会产生高频串扰。该器件在每个接收通道上使用三个 NEXT 消除器，以最大限度地减少其他三个通道引起的串扰。

在 10BASE-T/100BASE-TX 模式下，自适应均衡器只需消除码间干扰，并补偿输入数据中的信道损耗。

5.2.6 网络编码器和解码器

在 1000BASE-T 模式下，发送的 8 位数据会经过扰码，转换为 9 位符号，并进一步编码为 4D-PAM5 符号。为减少同一块电路板上使用多个器件时产生的电磁干扰，初始扰码器种子由特定的 PHY 地址决定。在接收端，首先检查空闲数据流。必须通过逻辑处理确定扰码器种子、线对偏斜、线对顺序和极性。随后，将输入的 4D-PAM5 数据转换为 9 位符号，并解扰为 8 位数据。

5.3 自动 MDI/MDI-X

自动 MDI/MDI-X 功能无需判断器件与其链路伙伴之间应使用直通电缆还是交叉电缆。此自动感应功能可检测链路伙伴的 MDI/MDI-X 线对映射，并相应地设置器件的 MDI/MDI-X 线对映射。

表 5-1显示了器件在 MDI/MDI-X 引脚映射中的 10/100/1000 引脚配置分配。

表 5-1: MDI/MDI-X 引脚映射

Pin (RJ-45 线对)	MDI			介质相关 接口交叉		
	1000BASE-T	100BASE-T	10BASE-T	1000BASE-T	100BASE-T	10BASE-T
发送/接收正负_A (1, 2)	A+/-	发送+/-	发送+/-	B+/-	接收+/-	接收+/-
发送/接收正负_B (3, 6)	B+/-	接收+/-	接收+/-	A+/-	发送+/-	发送+/-
发送/接收正负_C (4, 5)	C+/-	未使用	未使用	C+/-	未使用	未使用
发送/接收正负_D (7, 8)	D+/-	未使用	未使用	D+/-	未使用	未使用

器件默认启用自动 MDI-X 检测。

通过设置数字调试控制 1 寄存器中的 swapoff 位，可以禁用自动 MDIX。然后，可以通过数字调试控制 1 寄存器中的 mdi_set 位手动选择 MDI/MDI-X 模式。

自动 MDIX 状态位位于数字 AX/AN 状态寄存器中。

建议使用发送和接收数据路径对称的隔离变压器，以支持自动 MDI/MDI-X。

5.4 Alignment and Polarity Detection/Correction

In 1000BASE-T mode, the device supports 50 ns ±10 ns difference in propagation delay between pairs of channels in accordance with the IEEE 802.3 standard, and automatically corrects the data skew so the corrected four pairs of data symbols are synchronized.

Additionally, the device detects and corrects polarity errors on all MDI pairs, a useful capability that exceeds the requirements of the standard. Polarity detection and correction applies to 10BASE-T and 1000BASE-T and is not required for 100BASE-TX.

5.5 Wave Shaping, Slew-Rate Control, and Partial Response

In communication systems, signal transmission encoding methods are used to provide the noise-shaping feature and to minimize distortion and error in the transmission channel.

- For 1000BASE-T, a special partial-response signaling method is used to provide the band-limiting feature for the transmission path.
- For 100BASE-TX, a simple slew-rate control method is used to minimize EMI.
- For 10BASE-T, pre-emphasis is used to extend the signal quality through the cable.

5.6 MagJack RJ-45 with Common Center-taps Operation

Normally the center-taps of the device side of the magnetics need to be capacitively coupled to signal ground. To operate with MagJack RJ-45s with common center-taps, the **MAGJACK** strap may be used. This strap sets the default of the MagJack_mode bit in the Operation Mode Strap Override Low Register, which in turn sets the values of the Power Management Mode Registers.

5.7 PLL Clock Synthesizer

The device generates 125 MHz, 25 MHz, and 10 MHz clocks for system timing. Internal clocks are generated from the external 25 MHz crystal or reference clock.

5.8 Auto-Negotiation

The device conforms to the auto-negotiation protocol, defined in Clause 28 of the IEEE 802.3 Specification.

Auto-negotiation allows UTP (unshielded twisted pair) link partners to select the highest common mode of operation.

During auto-negotiation, link partners advertise capabilities across the UTP link to each other, and then compare their own capabilities with those they received from their link partners. The highest speed and duplex setting that is common to the two link partners is selected as the operating mode.

The following list shows the speed and duplex operation mode from highest-to-lowest:

- Priority 1: 1000BASE-T, full-duplex
- Priority 2: 1000BASE-T, half-duplex

Note: The device does not support 1000BASE-T, half-duplex and should not be enabled to advertise such.

- Priority 3: 100BASE-TX, full-duplex
- Priority 4: 100BASE-TX, half-duplex
- Priority 5: 10BASE-T, full-duplex
- Priority 6: 10BASE-T, half-duplex

If auto-negotiation is not supported or the device's link partner is forced to bypass auto-negotiation for 10BASE-T and 100BASE-TX modes, the device sets its operating mode by observing the input signal at its receiver. This is known as parallel detection, and allows the device to establish a link by listening for a fixed signal protocol in the absence of the auto-negotiation advertisement protocol.

5.4 对齐和极性检测/校正

在 1000BASE-T 模式下，器件支持通道对之间传播延迟相差 50 ns ±10 ns，并符合 IEEE 802.3 标准；器件还会自动校正数据偏斜，使校正后的四对数据符号保持同步。

此外，该器件可检测并校正所有 MDI 对上的极性错误，这一实用功能超出了标准的要求。极性检测和校正适用于 10BASE-T 和 1000BASE-T，而 100BASE-TX 不要求具备此功能。

5.5 波形整形、转换速率控制和部分响应

在通信系统中，信号传输编码方法用于实现噪声整形功能，并最大限度地减少传输信道中的失真和错误。

- 对于 1000BASE-T，采用一种特殊的部分响应信令方法，为传输路径提供带宽限制功能。

- 对于 100BASE-TX，采用简单的转换速率控制方法来最大限度地减少电磁干扰。
- 对于 10BASE-T，采用预加重来提升信号通过电缆后的质量。

5.6 采用公共中心抽头的 MagJack RJ-45

通常，磁性元件器件侧的中心抽头需要通过电容耦合至信号地。若要与采用公共中心抽头的 MagJack RJ-45 配合使用，可使用 **磁性网络接口插座** 绑带。此绑带会设置操作模式绑带覆盖低位寄存器中 MagJack_ 模式位的默认值，而该值又会设置电源管理模式寄存器的值。

5.7 PLL 时钟合成器

该器件生成 125 兆赫、25 兆赫和 10 兆赫时钟，用于系统定时。内部时钟由外部 25 兆赫晶体或参考时钟生成。

5.8 自动协商

该器件符合 IEEE 802.3 规范第 28 条中定义的自动协商协议。

自动协商使非屏蔽双绞线（UTP）链路伙伴能够选择双方共同支持的最高工作模式。在自动协商过程中，链路伙伴通过 UTP 链路向彼此通告各自的能力，然后将自身能力与从链路伙伴处收到的能力进行比较。两个链路伙伴共同支持的最高速率和双工设置将被选为工作模式。

以下列表按从高到低的顺序列出了速率和双工工作模式：

- 优先级 1: 1000BASE-T，全双工
- 优先级 2: 1000BASE-T，半双工

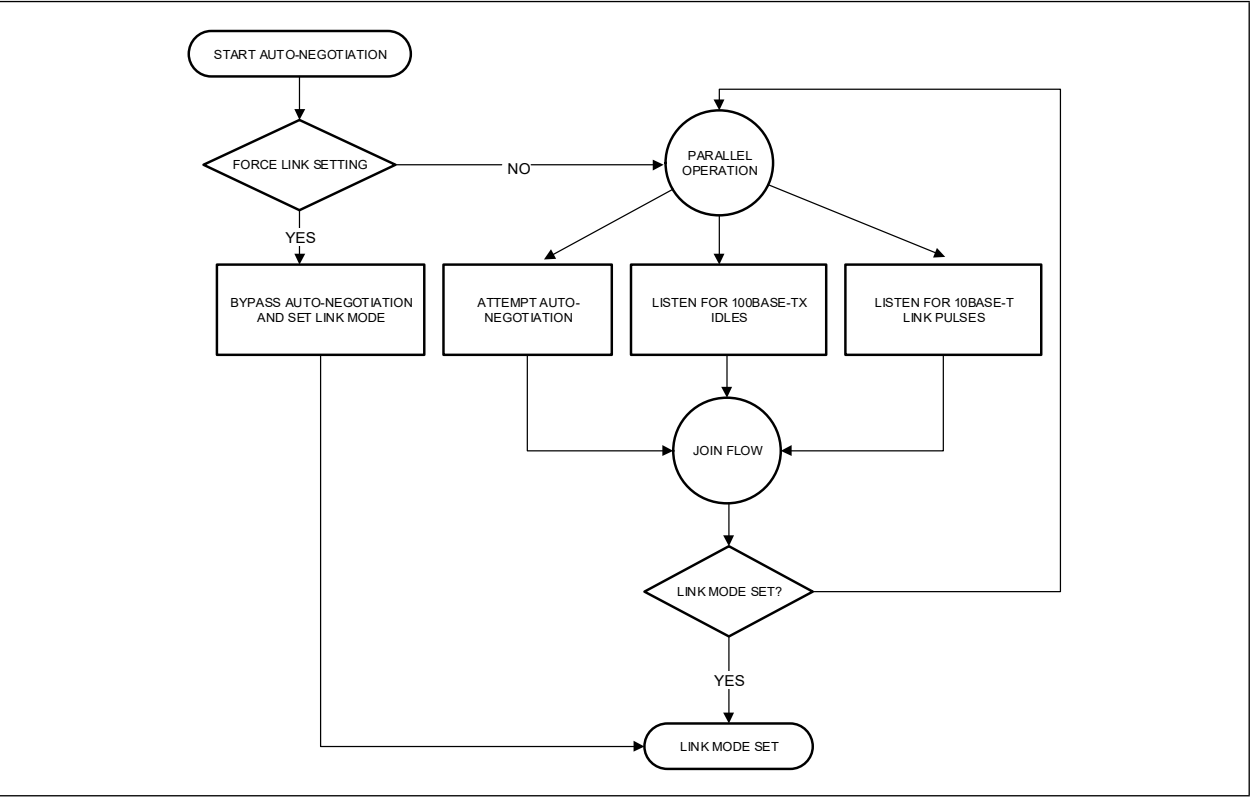
注意：该器件不支持 1000BASE-T 半双工，因此不应使能对其进行通告。

- 优先级 3: 100BASE-TX，全双工
- 优先级 4: 100BASE-TX，半双工
- 优先级 5: 10BASE-T，全双工
- 优先级 6: 10BASE-T，半双工

如果不支持自动协商，或者器件的链路伙伴被强制设置为在 10BASE-T 和 100BASE-TX 模式下绕过自动协商，器件会通过观察接收器上的输入信号来设置其工作模式。这称为并行检测，使器件能够在缺少自动协商通告协议的情况下，通过侦听固定信号协议来建立链路。

The auto-negotiation link-up process is shown in [Figure 5-3](#).

FIGURE 5-3: AUTO-NEGOTIATION FLOW CHART



For 1000BASE-T mode, auto-negotiation is required and always used to establish a link. During 1000BASE-T auto-negotiation, the leader and follower configuration is first resolved between link partners. Then the link is established with the highest common capabilities between link partners.

Auto-negotiation is enabled by default after power-up or hardware reset. After that, auto-negotiation can be enabled or disabled through the Basic Control Register, Bit [12]. If auto-negotiation is disabled, the speed is set by the Basic Control Register, Bits [6, 13] and the duplex is set by the Basic Control Register, Bit [8].

If the speed is changed on the fly, the link goes down and either auto-negotiation and parallel detection initiate until a common speed between the device and its link partner is re-established for a link.

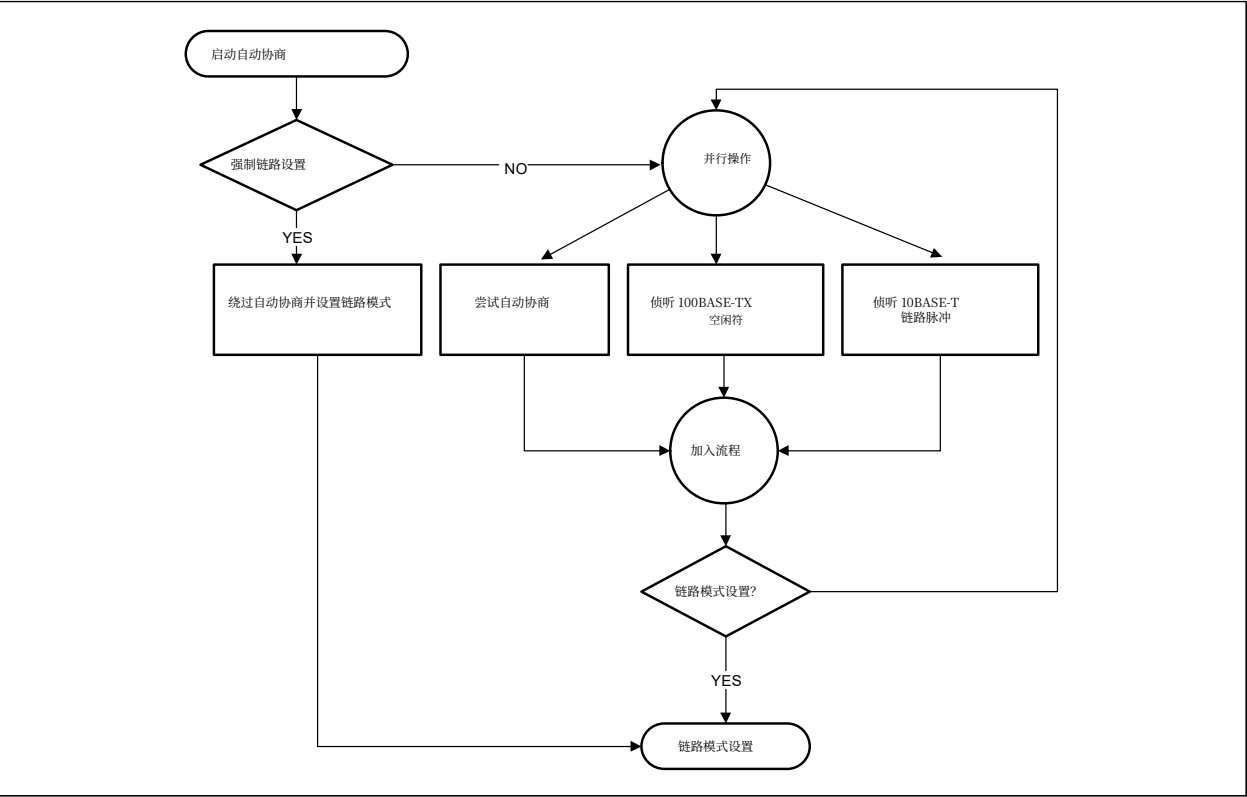
If the link is already established and there is no change of speed on the fly, the changes (for example, duplex and pause capabilities) will not take effect unless either auto-negotiation is restarted through the Basic Control Register, Bit [9], or a link-down to link-up transition occurs (that is, disconnecting and reconnecting the cable).

After auto-negotiation is completed, the link status is updated in the Basic Status Register, Bit [2], and the link partner capabilities are updated in Registers 5h, 6h, 8h, and Ah.

The auto-negotiation finite state machines use interval timers to manage the auto-negotiation process. The duration of these timers under normal operating conditions is summarized in [Table 5-2](#).

自动协商链路建立过程如[图 5-3](#)所示。

图 5-3：自动协商流程图



在 1000BASE-T 模式下，必须使用自动协商来建立链路。进行 1000BASE-T 自动协商时，链路伙伴首先协商确定主从配置，然后以链路伙伴之间最高的共同能力建立链路。

上电或硬件复位后，默认启用自动协商。之后，可以通过基本控制寄存器的位 [12]启用或禁用自动协商。如果禁用自动协商，则通过基本控制寄存器的位 [6, 13] 设置速率，并通过基本控制寄存器的位 [8]设置双工模式。

如果在线更改速率，链路将断开，并启动自动协商和并行检测，直到器件与其链路伙伴重新建立共同速率并恢复链路。

如果链路已建立，且运行过程中速率没有变化，则除非通过基本控制寄存器的位 [9], 重新启动自动协商，或发生链路断开到链路建立的状态转换（即断开并重新连接电缆），否则更改（例如双工和暂停能力）不会生效。

自动协商完成后，基本状态寄存器的位 [2], 会更新链路状态，寄存器 5h、6h、8h 和 Ah 会更新链路伙伴的能力。

自动协商有限状态机使用间隔定时器来管理自动协商过程。在正常运行条件下，这些定时器的持续时间汇总于[表 5-2](#)。

TABLE 5-2: AUTO-NEGOTIATION TIMERS

Auto-Negotiation Interval Timers	Time Duration
Transmit Burst Interval	16 ms
Transmit Pulse Interval	68 μs
FLP Detect Minimum Time	17.2 μs
FLP Detect Maximum Time	185 μs
Receive Minimum Burst Interval	6.8 ms
Receive Maximum Burst Interval	112 ms
Data Detect Minimum Interval	35.4 μs
Data Detect Maximum Interval	95 μs
NLP Test Minimum Interval	4.5 ms
NLP Test Maximum Interval	30 ms
Link Loss Time	52 ms
Break Link Time	1480 ms
Parallel Detection Wait Time	830 ms
Link Enable Wait Time	1000 ms

5.8.1 AUTO-NEGOTIATION NEXT PAGE USAGE

The device supports “Next Page” capability which is used to negotiate Gigabit Ethernet and Energy Efficient Ethernet functionality as well as to support software controlled pages.

As described in IEEE 802.3 Annex 40C “Add-on interface for additional Next Pages”, the device will autonomously send and receive the Gigabit Ethernet and Energy Efficient Ethernet next pages and then optionally send and receive software controlled next pages.

Gigabit Ethernet next pages consist of one message and two unformatted pages. The message page contains an 8 as the message code. The first unformatted page contains the information from the Auto-Negotiation Master Slave Control Register. The second unformatted page contains the Leader-Follower Seed value used to resolve the Leader-Follower selection. The result of the Gigabit Ethernet next pages exchange is stored in Auto-Negotiation Master Slave Status Register.

Gigabit Ethernet next pages are always transmitted, regardless of the advertised settings in the Auto-Negotiation Master Slave Control Register.

Energy Efficient Ethernet (EEE) next pages consist of one message and one unformatted page. The message page contains a 10 as the message code (this value can be overridden in the EEE Message Code Register). The unformatted page contains the information from the EEE Advertisement Register. The result of the Gigabit Ethernet next pages exchange is stored in EEE Link Partner Ability Register.

EEE next pages are transmitted only if the advertised setting in the EEE Advertisement Register is not zero.

APPLICATION NOTE: The Gigabit Ethernet and EEE next pages may be viewed in Auto-Negotiation Next Page RX Register as they are exchanged.

Following the EEE next page exchange, software controlled next pages are exchanged when the Next Page bit in the Auto-Negotiation Advertisement Register is set. Software controlled next page status is monitored via the Auto-Negotiation Expansion Register and Auto-Negotiation Next Page RX Register.

5.8.2 PARALLEL DETECT DUPLEX

Normally, and according to IEEE 802.3, when parallel detection is used to establish the link, the resulting operation is set to half duplex. An option exists to force this result to full duplex. This is enabled by setting the LP Force 100 FD Override and/or LP Force 10 FD Override bits in the Parallel Detect Full Duplex Override Register.

表 5-2：自动协商定时器

自动协商间隔定时器	持续时间
发送突发间隔	16 毫秒
发送脉冲间隔	68 μs
FLP 检测最短时间	17.2 μs
FLP 检测最长时间	185 μs
接收最小突发间隔	6.8 ms
接收最大突发间隔	112 ms
数据检测最小间隔	35.4 μs
数据检测最大间隔	95 μs
NLP 测试最小间隔	4.5 ms
NLP 测试最大间隔	30 ms
链路丢失时间	52 ms
断开链路时间	1480 ms
并行检测等待时间	830 ms
链路启用等待时间	1000 毫秒

5.8.1 自动协商下一页的使用

该器件支持 “下一页” 功能，可用于协商千兆以太网和节能以太网功能，也可支持软件控制页面。

如 IEEE 802.3 附录 40C “附加下一页的扩展接口” 中所述，该器件将自主发送和接收千兆以太网和节能以太网下一页，然后可选择发送和接收由软件控制的下一页。

千兆以太网下一页由一条消息和两个未格式化页面组成。消息页包含一个值为 8 的消息代码。第一个未格式化页面包含自动协商主从控制寄存器中的信息。第二个未格式化页面包含用于确定主从选择的主从种子值。千兆以太网下一页交换的结果存储在自动协商主从状态寄存器中。

无论自动协商主从控制寄存器中通告的设置如何，千兆以太网下一页始终都会发送。

节能以太网（EEE）下一页由一个消息页和一个未格式化页组成。消息页包含值为 10 的消息代码（此值可在 EEE 消息代码寄存器中覆盖）。未格式化页包含 EEE 通告寄存器中的信息。千兆以太网下一页交换的结果存储在 EEE 链路伙伴能力寄存器中。

仅当 EEE 通告寄存器中的通告设置不为零时，才会发送 EEE 下一页。

应用说明： 交换过程中，可在自动协商下一页接收寄存器中查看千兆以太网和 EEE 下一页。

EEE 下一页交换完成后，如果自动协商通告寄存器中的下一页位已置位，则会交换软件控制的下一页。可通过自动协商扩展寄存器和自动协商下一页接收寄存器监控软件控制的下一页状态。

5.8.2 并行检测双工

通常，根据 IEEE 802.3，使用并行检测建立链路时，操作模式会设置为半双工。也可以选择强制将此结果设为全双工。设置并行检测全双工覆盖寄存器中的 LP 强制 100 全双工覆盖位和/或 LP 强制 10 全双工覆盖位即可启用此选项。

5.9 10/100 Mbps Speeds Only

Some applications require link-up to be limited to 10/100 Mbps speeds only.

After power-up/reset, the device can be restricted to auto-negotiate and link-up to 10/100 Mbps speeds only by programming the following register settings:

- 1. Configure the Speed Select[1] bit in the Basic Control Register to '0' to disable the 1000 Mbps speed.
- 2. Configure the 1000BASE-T Full Duplex and 1000BASE-T Half Duplex bits in the Auto-Negotiation Master Slave Control Register to '00' to remove Auto-Negotiation advertisements for 1000 Mbps full/half duplex.
- 3. Write a '1' to the Restart Auto-Negotiation (PHY_RST_AN) bit in the Basic Control Register, a self-clearing bit, to force a restart of Auto-Negotiation.

Auto-Negotiation and 10BASE-T/100BASE-TX speeds use only differential pairs A and B. Differential pairs C and D can be left as no connects.

5.10 Energy Efficient Ethernet

The device implements Energy Efficient Ethernet (EEE) as described in IEEE Standard 802.3az. The specification is defined around an EEE-compliant MAC on the host side and an EEE-compliant link partner on the line side that support the special signaling associated with EEE. EEE saves power by keeping the AC signal on the copper Ethernet cable at approximately 0V peak-to-peak as often as possible during periods of no traffic activity, while maintaining the link-up status. This is referred to as low-power idle (LPI) mode or state.

As set by the **MODE[4:0]** configuration straps, the device has the EEE function enabled or disabled as the power-up default setting. The EEE function can be enabled or disabled by setting or clearing the following EEE advertisement bits in the EEE Advertisement Register (MMD Address 7h, Register 3Ch), followed by restarting auto-negotiation (writing a '1' to the Basic Control Register, Bit [9]):

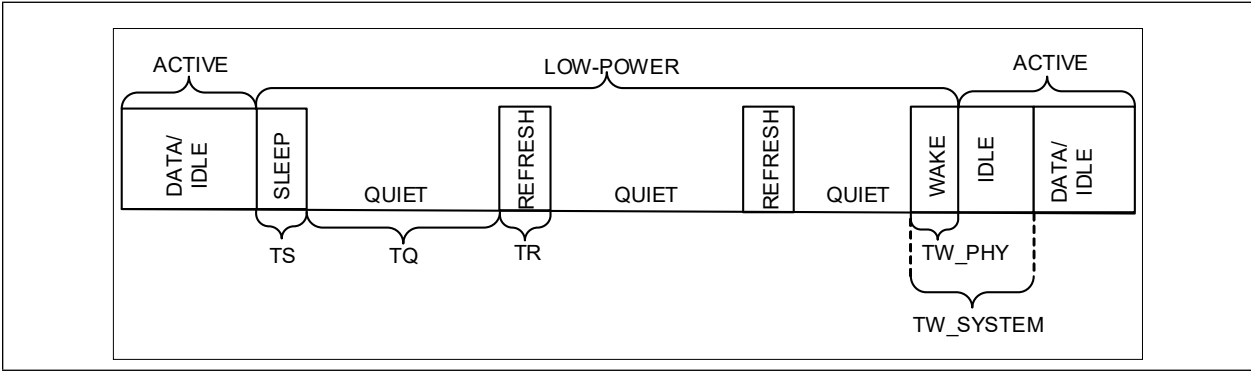
- 1000BASE-T EEE (Bit [2]) = 0/1 // Disable/Enable 1000 Mbps EEE mode
- 100BASE-TX EEE (Bit [1]) = 0/1 // Disable/Enable 100 Mbps EEE mode

During LPI mode, the copper link responds automatically when it receives traffic and resumes normal PHY operation immediately, without blockage of traffic or loss of packet. This involves exiting LPI mode and returning to normal 100/1000 Mbps operating mode. Wake-up times are <16 μs for 1000BASE-T and <30 μs for 100BASE-TX. The LPI state is controlled independently for transmit and receive paths, allowing the LPI state to be active (enabled) for:

- Transmit cable path only
- Receive cable path only
- Both transmit and receive cable paths

During LPI mode, refresh transmissions are used to maintain the link; power savings occur in quiet periods. Approximately every 20 to 22 milliseconds, a refresh transmission of 200 to 220 microseconds is sent to the link partner. The refresh transmissions and quiet periods are shown in [Figure 5-4](#):

FIGURE 5-4: LPI MODE (REFRESH TRANSMISSIONS AND QUIET PERIODS)



Per the IEEE Standard, the device will wait for one second following link up before sending LPI. This is enabled by default via the lpi_1sec_delay bit in the EEE Wait Time Threshold and CRS Block Timer Register.

5.9 仅支持 10/100 Mbps 速率

某些应用要求链路建立仅限于 10/100 Mbps 速率。

上电/复位后，可通过编程设置以下寄存器，使器件仅自动协商并建立 10/100 Mbps 速率的链路：

1. 将基本控制寄存器中的速率选择[1] 位配置为 “0”，以禁用 1000 Mbps 速率。
2. 将自动协商主从控制寄存器中的 1000BASE-T 全双工位和 1000BASE-T 半双工位配置为 ‘00’，以移除针对 1000 Mbps 全双工/半双工的自动协商通告。
3. 向基本控制寄存器中的重启自动协商（PHY_RST_AN）位写入 ‘1’。该位是自清零位，可强制重新启动自动协商。

自动协商和 10BASE-T/100BASE-TX 速率仅使用 A 和 B 线对。C 和 D 差分对可以留空不连接。

5.10 节能以太网

该器件按照 IEEE 802.3az 标准实现节能以太网（EEE）。该规范针对外部系统侧符合 EEE 标准的 MAC，以及线路侧符合 EEE 标准的链路伙伴制定，这两者均支持与 EEE 相关的特殊信令。节能以太网通过在没有流量活动期间尽可能长时间地将铜质以太网电缆上的交流信号保持在约 0V 峰峰值来节省功耗，同时维持链路建立状态。这称为低功耗空闲（LPI）模式或状态。

由**模式[4: 0]** 配置绑带决定，器件的 EEE 功能在上电默认设置中处于启用或禁用状态。通过设置或清除 EEE 通告寄存器（MMD 地址 7h，寄存器 3Ch）中的以下 EEE 通告位，可以启用或禁用 EEE 功能，然后重新启动自动协商（向基本控制寄存器的位 [9] 写入 “1”）：

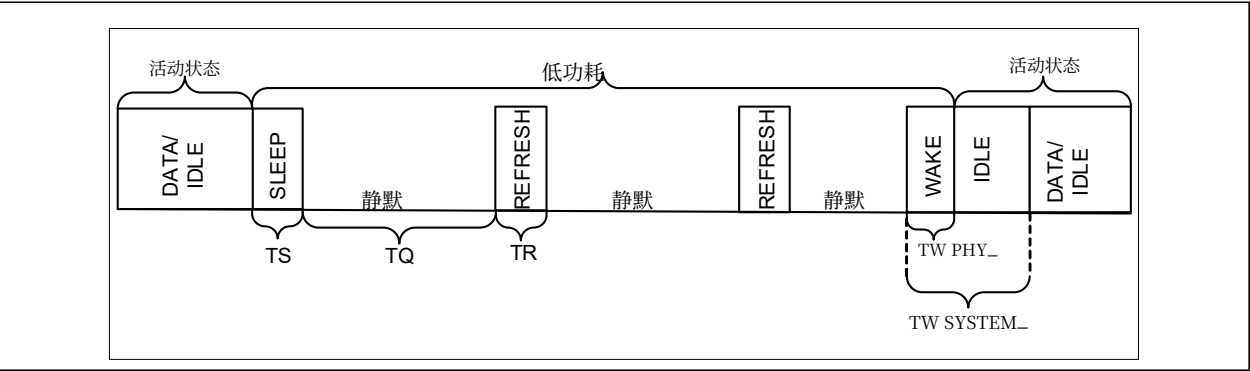
- 1000BASE-T 节能以太网（位 [2]）= 0/1 // 禁用/启用 1000 Mbps 节能以太网模式
- 100BASE-TX 节能以太网（位 [1]）= 0/1 // 禁用/启用 100 Mbps 节能以太网模式

在低功耗空闲模式下，铜缆链路收到流量时会自动响应，并立即恢复正常 PHY 操作，不会阻塞流量或丢失数据包。这包括退出低功耗空闲模式并返回正常的 100/ 1000 兆比特每秒工作模式。1000BASE-T 和 100BASE-TX 的唤醒时间分别为 <16 μs 和 <30 μs。发送和接收路径的低功耗空闲状态可独立控制，因此以下路径的低功耗空闲状态可以处于活动状态（已启用）：

- 仅发送电缆路径
- 仅接收电缆路径
- 发送和接收电缆路径

在低功耗空闲模式下，使用刷新传输来维持链路；静默周期内可节省功耗。大约- 每隔 20 到 22 毫秒，就会向链路伙伴发送一次持续 200 到 220 微秒的刷新传输。刷新传输和静默周期如[图 5-4 所示](#)：

图 5-4：低功耗空闲模式（刷新传输和静默周期）



根据 IEEE 标准，链路建立后，器件会等待一秒再发送低功耗空闲。默认情况下，通过 EEE 等待时间阈值和 CRS 阻塞计时器寄存器中的 lpi_1sec_delay 位启用此功能。

5.10.1 TRANSMIT DIRECTION CONTROL (MAC-TO-PHY)

5.10.1.1 RGMII

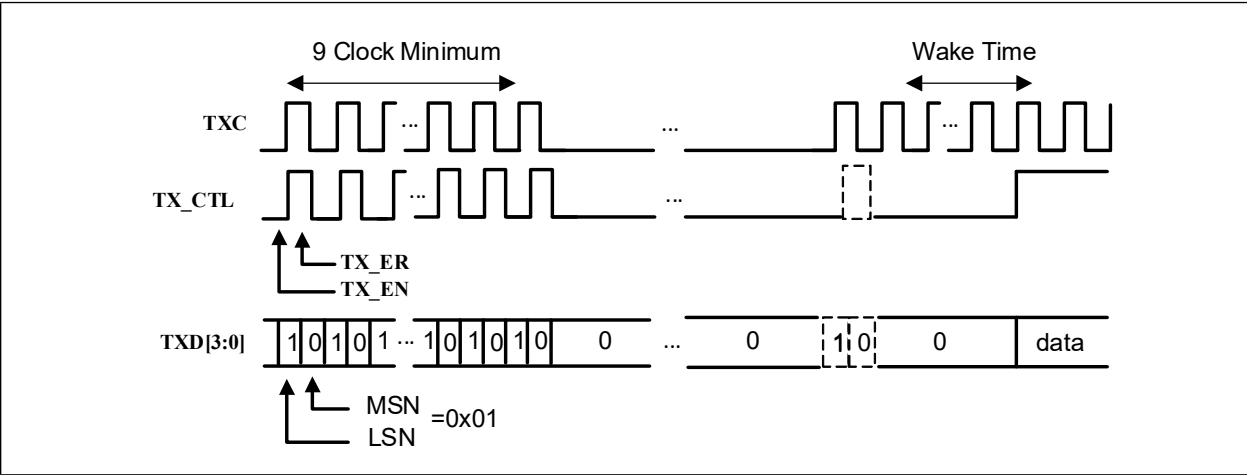
On the RGMII interface at 1000 Mbps, transmission from MAC-to-PHY uses both rising and falling edges of the **TXC** signal. The device uses the **TX_CTL** pin to clock in the logical equivalent of the **TX_EN** signal on the rising edge and the logical equivalent of the **TX_ER** signal on the falling edge. It also uses the **TXD[3:0]** pins to clock in the TX data low nibble bits [3:0] on the rising edge and the TX data high nibble Bits [7:4] on the falling edge.

The device enters LPI mode for the transmit direction when its attached IEEE-compliant MAC de-asserts the logical equivalent of the **TX_EN** signal (the **TX_CTL** pin low on the rising edge), asserts the logical equivalent of the **TX_ER** signal (the **TX_CTL** pin high on the falling edge), and sets TX data Bits [7:0] to 0000_0001 (**TXD[3:0]** pins 0001 on the rising edge and 0000 on the falling edge). The device remains in the 1000Mbps transmit LPI state while the MAC maintains the states of these signals. When the MAC changes any of the logical equivalent **TX_EN**, **TX_ER**, or TX data signals from their LPI state values, the device exits the LPI transmit state.

To save more power, as indicated by the Clock stop capable bit in the PCS Status 1 Register, the MAC can stop the **TXC** signal after the RGMII signals for the LPI state have been asserted for 9 or more **TXC** clock cycles. During the clock stoppage the **TX_CTL** and **TXD[3:0]** pins are most likely held low by the MAC.

Figure 5-5: shows the LPI transition for GMII transmit.

FIGURE 5-5: LPI TRANSITION - RGMII (1000 MBPS) TRANSMIT



RGMII transmission at 100 Mbps from MAC-to-PHY uses both rising and falling edges of the **TXC** signal. The device uses the **TX_CTL** pin to clock in the logical equivalent of the **TX_EN** signal on the rising edge and the logical equivalent of the **TX_ER** signal on the falling edge. It also uses the **TXD[3:0]** pins to clock in the TX data bits [3:0] on the rising edge.

The device enters LPI mode for the transmit direction when its attached IEEE-compliant MAC de-asserts the logical equivalent of the **TX_EN** signal (the **TX_CTL** pin low on the rising edge), asserts the logical equivalent of the **TX_ER** signal (the **TX_CTL** pin high on the falling edge), and sets TX data Bits [3:0] to 0001. The device remains in the 100Mbps transmit LPI state while the MAC maintains the states of these signals. When the MAC changes any of the logical equivalent **TX_EN**, **TX_ER**, or TX data signals from their LPI state values, the device exits the LPI transmit state.

To save more power, as indicated by the Clock stop capable bit in the PCS Status 1 Register, the MAC can stop the **TXC** signal after the RGMII signals for the LPI state have been asserted for 9 or more **TXC** clock cycles. During the clock stoppage the **TX_CTL** and **TXD[3:0]** pins are most likely held low by the MAC.

5.10.1 发送方向控制（MAC 到 PHY）

5.10.1.1 RGMII

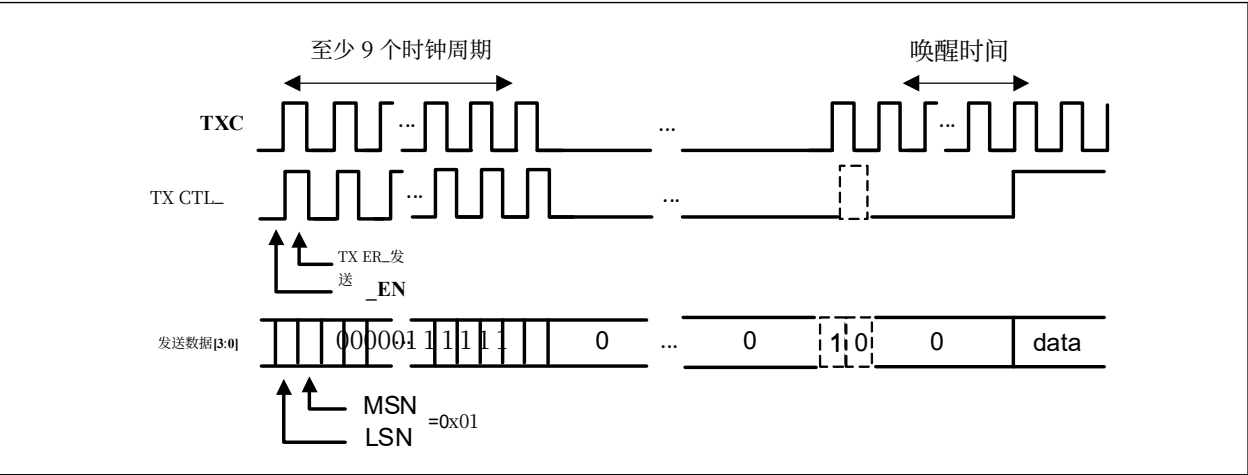
在 1000 兆比特每秒的 RGMII 接口上，MAC 到 PHY 的发送使用**TXC**信号的上升沿和下降沿。器件使用**发送_控制**引脚，在上升沿输入**发送_使能**信号的逻辑等效信号，在下降沿输入**发送_错误**信号的逻辑等效信号。此外，器件还使用**发送数据[3:0]**引脚，在上升沿输入 TX 数据低半字节位 [3:0]，在下降沿输入 TX 数据高半字节位 [7:4]。

当连接的符合 IEEE 标准的 MAC 撤销逻辑等效的 **发送_使能** 信号（在上升沿将 **发送_控制** 引脚拉低，在下降沿将逻辑等效的 **发送_错误** 信号（在下降沿将 **发送_控制** 引脚拉高），并将 TX 数据位 [7:0] 设置为 0000_0001 （在上升沿将 **发送数据[3:0]** 引脚设为 0001，在下降沿设为 0000）时，器件会进入发送方向的低功耗空闲模式。在 MAC 维持这些信号的状态期间，器件会保持在 1000 Mbps 发送低功耗空闲状态。当 MAC 将逻辑等效的 **发送_使能**、**发送_错误**或 T X 数据信号中的任意一项从其低功耗空闲状态值更改时，器件会退出低功耗空闲发送状态。

为进一步节省功耗，如 PCS 状态 1 寄存器中的时钟停止能力位所示，在 RGMII 信号已保持低功耗空闲状态 9 个或更多 TXC 时钟周期后，媒体访问控制器可以停止 **TXC** 信号。在时钟停止期间，媒体访问控制器很可能会将 **发送_控制** 和 **发送数据[3:0]** 引脚保持为低电平。

图 5-5：显示了 GMII 发送的低功耗空闲转换。

图 5-5：低功耗空闲转换 - RGMII（1000 兆比特/秒）发送



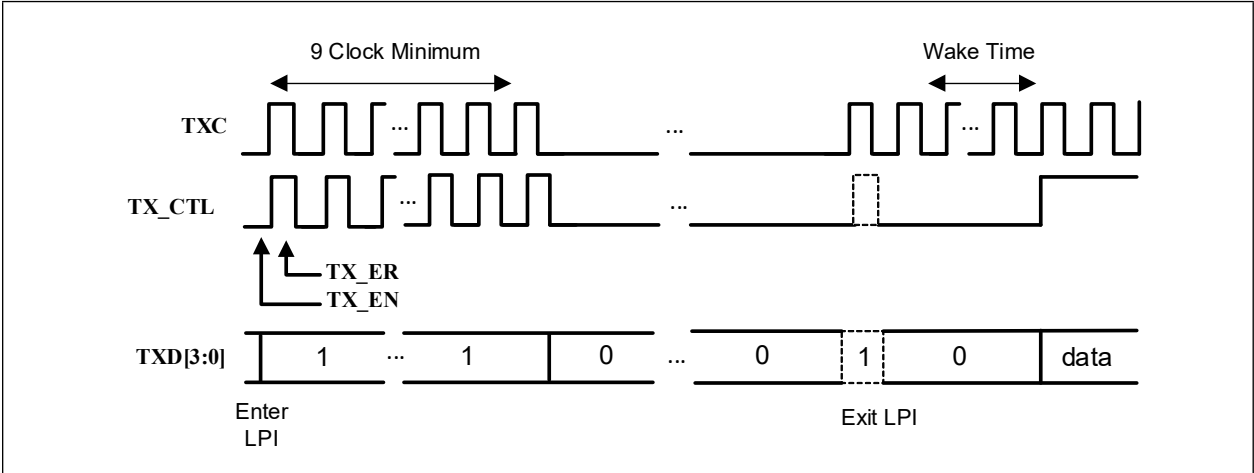
MAC 到 PHY 的 100 Mbps RGMII 发送使用 **TXC** 信号的上升沿和下降沿。器件使用 **发送_控制**引脚，在上升沿采样逻辑等效的 **发送_使能**信号，并在下降沿采样逻辑等效的 **发送_错误**信号。器件还使用 **发送数据[3:0]** 引脚在上升沿采样 TX 数据位 [3:0]。

当与器件连接的符合 IEEE 标准的 MAC 撤销逻辑等效的 **发送_使能**信号（即上升沿时 **发送_控制**引脚为低电平）、置位逻辑等效的 **发送_错误** 信号（即下降沿时 **发送_控制**引脚为高电平），并将发送数据位 [3:0] 设为 0001 时，器件将进入发送方向的低功耗空闲模式。只要 MAC 保持这些信号的状态，器件就会保持在 100Mbps 发送低功耗空闲状态。当 MAC 将任何逻辑等效的 **发送_使能**、**发送_错误**或发送数据信号从其低功耗空闲状态值更改时，器件就会退出发送低功耗空闲状态。

为进一步节省功耗，如 PCS 状态 1 寄存器中的时钟停止能力位所示，在低功耗空闲状态的 RGMII 信号置位 9 个或更多 TXC 时钟周期后，媒体访问控制器可以停止 TXC 信号。在时钟停止期间，媒体访问控制器很可能会将 TX_CTL 和发送数据[3:0] 引脚保持为低电平。

Figure 5-6: shows the LPI transition for RGMII transmit in 100 Mbps mode.

FIGURE 5-6: LPI TRANSITION - RGMII (100 MBPS) TRANSMIT



5.10.2 RECEIVE DIRECTION CONTROL (PHY-TO-MAC)

5.10.2.1 RGMII

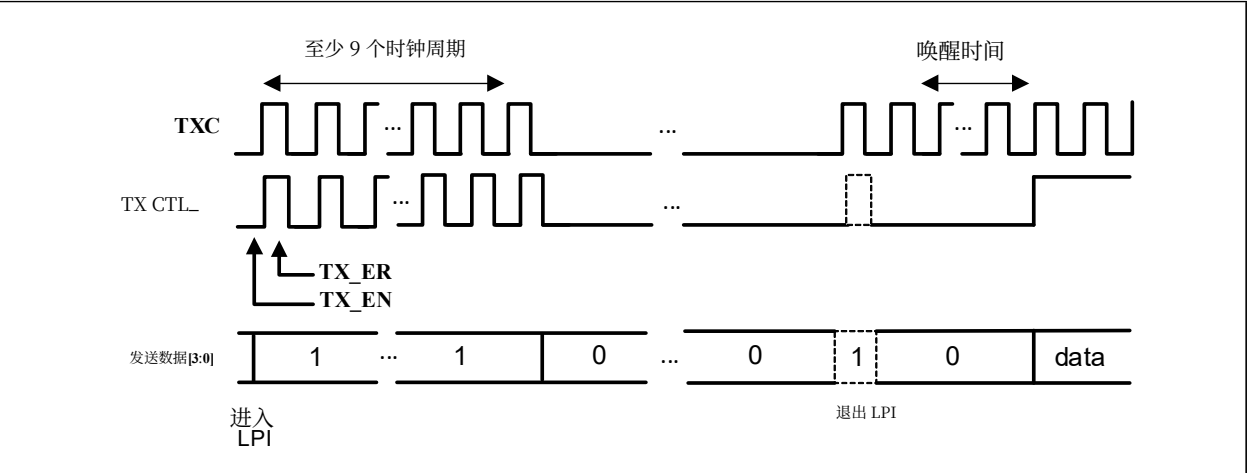
On the RGMII interface at 1000 Mbps, reception from PHY-to-MAC uses both rising and falling edges of the **RXC** signal. The device uses the **RX_CTL** pin to clock out the logical equivalent of the **RX_DV** signal on the rising edge and the logical equivalent of the **RX_ER** signal on the falling edge. It also uses the **RXD[3:0]** pins to clock out the RX data low nibble bits [3:0] on the rising edge and the RX data high nibble bits [7:4] on the falling edge.

The device enters LPI mode for the receive direction when it receives the /P/ code bit pattern (sleep/refresh) from its IEEE-compliant link partner. It then drives the logical equivalent of the **RX_DV** signal low (on the rising clock edge) and the logical equivalent of the **RX_ER** signal high (on the falling edge), to the MAC. Also, the **RXD[3:0]** pins are driven to 0001 on the rising clock edge and 0000 on the falling clock edge to set the RX data bits [7:0] to 0000_0001. The device remains in the 1000 Mbps receive LPI state while it continues to receive the refresh from its link partner, so it will continue to maintain and drive the LPI output states for the RGMII receive output pins to inform the attached IEEE-compliant MAC that it is in the receive LPI state. When the device receives a non /P/ code bit pattern (non-refresh), it exits the receive LPI state and sets the **RX_DV** and **RXD[3:0]** output pins accordingly for a normal frame or normal idle.

To save more power, the device stops the **RXC** clock output to the MAC after 9 or more **RXC** clock cycles have occurred in the receive LPI state. The Clock-stop enable bit in the PCS Control 1 Register controls if the device stops the **RXC** clock output. During the clock stoppage the **RX_CTL** and **RXD[3:0]** pins are held low by the device. The device may restart **RXC** at any time while it is asserting LPI, but shall restart **RXC** so that at least one positive transition occurs before it deasserts LPI.

图 5-6: 展示了 100 Mbps 模式下 RGMII 发送的低功耗空闲转换。

图 5-6: 低功耗空闲转换 - RGMII (100 Mbps) 发送



5.10.2 接收方向控制（PHY 到 MAC）

5.10.2.1 RGMII

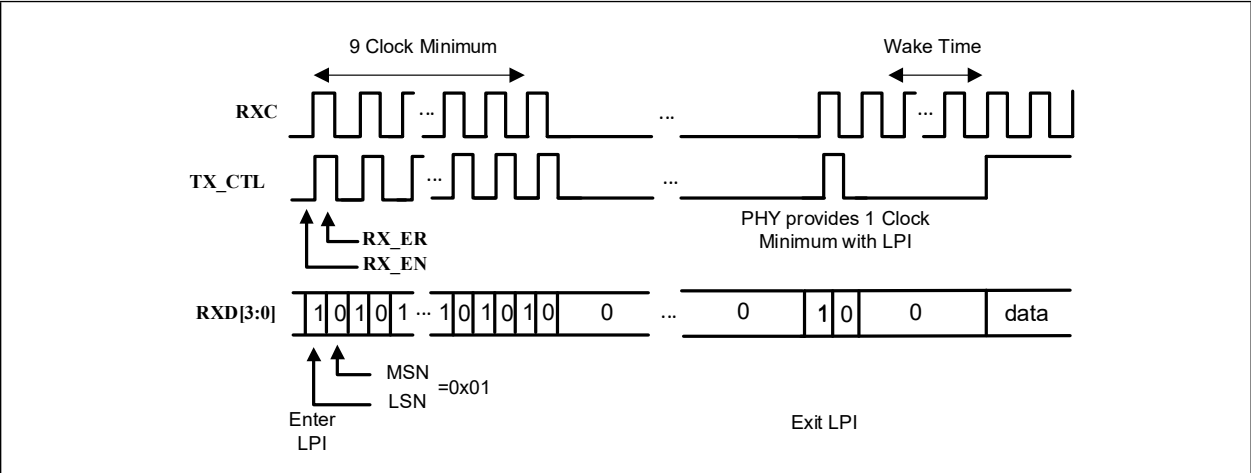
在 1000 兆比特每秒的 RGMII 接口上，PHY 到 MAC 的接收使用**接收时钟**信号的上升沿和下降沿。器件使用**接收_控制**引脚，在上升沿输出**接收_DV**信号的逻辑等效信号，并在下降沿输出**接收_错误**信号的逻辑等效信号。器件还使用**接收数据[3:0]**引脚，在上升沿输出 RX 数据低半字节位 [3:0]，并在下降沿输出 RX 数据高半字节位 [7:4]。

当器件从符合 IEEE 标准的链路伙伴接收到 /P/ 编码位模式（休眠/刷新信号）时，接收方向将进入低功耗空闲模式。随后，在时钟上升沿，器件向媒体访问控制器输出逻辑等效的低电平 **接收_DV** 信号；在时钟下降沿，输出逻辑等效的高电平 **接收_错误** 信号。此外，在时钟上升沿，**接收数据[3:0]** 引脚被驱动为 0001，在时钟下降沿则被驱动为 0000，以便将 RX 数据位 [7:0] 设为 0000_0001。只要器件持续从链路伙伴接收刷新信号，就会保持在 1000 兆比特/秒接收低功耗空闲状态，并持续维持和驱动 RGMII 接收输出引脚的低功耗空闲输出状态，通知连接的符合 IEEE 标准的媒体访问控制器其处于接收低功耗空闲状态。当器件接收到非 /P/ 编码位模式（非刷新信号）时，就会退出接收低功耗空闲状态，并相应地设置 **接收_DV** 和 **接收数据[3:0]** 输出引脚，以输出正常帧或正常空闲信号。

为进一步降低功耗，在接收低功耗空闲状态下经过 9 个或更多接收时钟周期后，器件会停止向媒体访问控制器输出接收时钟。PCS 控制 1 寄存器中的时钟停止使能位用于控制器件是否停止输出接收时钟。在时钟停止期间，器件会将 **RX_控制** 和 **RXD[3:0]** 引脚保持为低电平。器件可在声明低功耗空闲的任何时刻重新启动接收时钟，但必须确保在撤销低功耗空闲之前，接收时钟至少出现一次正跳变。

Figure 5-7: shows the LPI transition for RGMII receive in 1000 Mbps mode.

FIGURE 5-7: LPI TRANSITION - RGMII (1000 MBPS) RECEIVE



RGMII reception at 100 Mbps from PHY-to-MAC uses both rising and falling edges of the **RXC** signal. The device uses the **RX_CTL** pin to clock out the logical equivalent of the **RX_DV** signal on the rising edge and the logical equivalent of the **RX_ER** signal on the falling edge. It also uses the **RXD[3:0]** pins to clock out the RX data bits [3:0] on the rising edge.

The device enters LPI mode for the receive direction when it receives the /P/ code bit pattern (sleep/refresh) from its IEEE-compliant link partner. It then drives the **RX_DV** pin low on the rising clock edge and high on the falling clock edge to de-assert the **RX_DV** signal and assert the **RX_ER** signal, respectively, to the MAC. Also, the **RXD[3:0]** pins are driven to 0001. The device remains in the 100 Mbps receive LPI state while it continues to receive the refresh from its link partner, so it will continue to maintain and drive the LPI output states for the RGMII receive output pins to inform the attached IEEE-compliant MAC that it is in the receive LPI state. When the device receives a non /P/ code bit pattern (non-refresh), it exits the receive LPI state and sets the **RX_DV** and **RXD[3:0]** output pins accordingly for a normal frame or normal idle.

To save more power, the device stops the **RXC** clock output to the MAC after 9 or more **RXC** clock cycles have occurred in the receive LPI state. The Clock-stop enable bit in the PCS Control 1 Register controls if the device stops the **RXC** clock output. During the clock stoppage the **RX_CTL** and **RXD[3:0]** pins are held low by the device. The device may restart **RXC** at any time while it is asserting LPI, but shall restart **RXC** so that at least one positive transition occurs before it deasserts LPI.

Figure 5-8: shows the LPI transition for RGMII receive in 100 Mbps mode.

FIGURE 5-8: LPI TRANSITION - RGMII (100 MBPS) RECEIVE

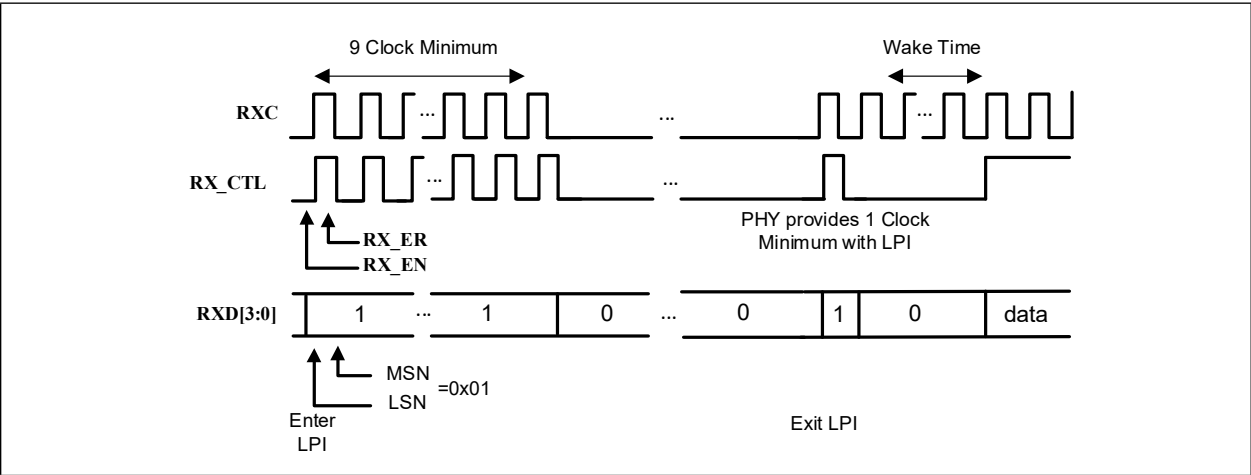
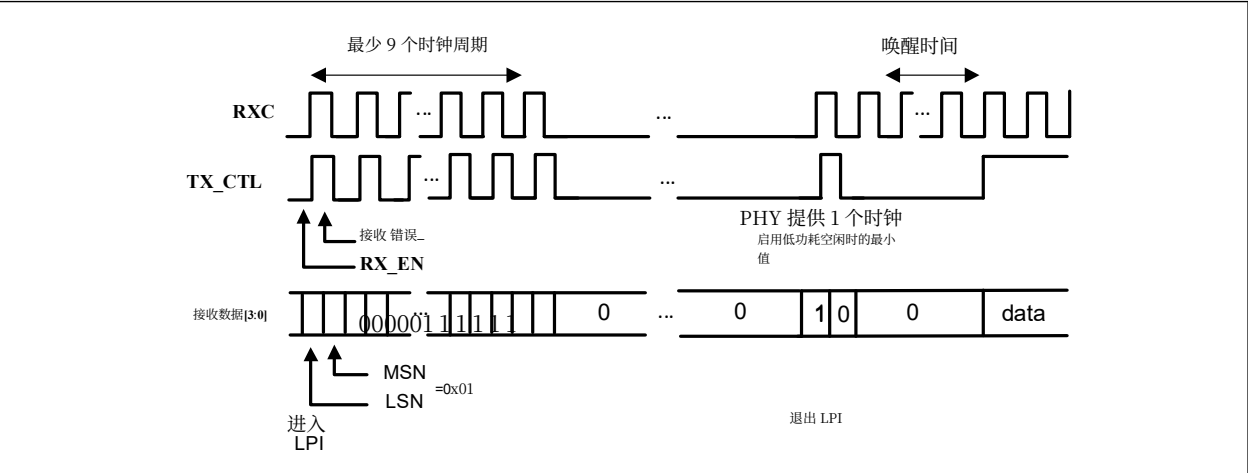


图 5-7: 展示了 1000 Mbps 模式下 RGMII 接收的低功耗空闲转换。

图 5-7: 低功耗空闲转换 - RGMII (1000 兆比特/秒) 接收



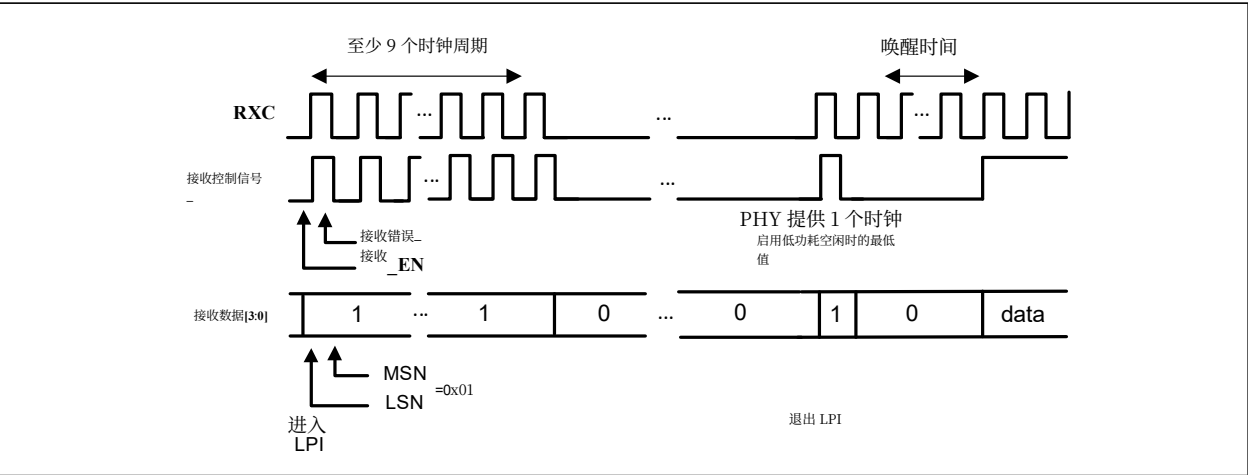
PHY 到 MAC 的 100 Mbps RGMII 接收使用**接收时钟**信号的上升沿和下降沿。器件使用**接收_控制**引脚，在上升沿输出**接收_DV**信号的逻辑等效值，并在下降沿输出**接收_错误**信号的逻辑等效值。器件还使用**接收数据[3:0]**引脚在上升沿输出 RX 数据位 [3:0]。

当器件从符合 IEEE 标准的链路伙伴接收到 /P/ 编码位模式（休眠/刷新）时，会进入接收方向的低功耗空闲模式。随后，器件在时钟上升沿将 **接收_DV** 引脚驱动为低电平，并在时钟下降沿驱动为高电平，分别向媒体访问控制器撤销 **接收_DV** 信号并置位 **接收_错误** 信号。此外，**接收数据[3:0]** 引脚被驱动为 0001。在持续接收链路伙伴发送的刷新信号期间，器件会保持在 100 Mbps 接收低功耗空闲状态，因此会持续维持并驱动 RGMII 接收输出引脚的低功耗空闲输出状态，以通知连接的符合 IEEE 标准的媒体访问控制器其处于接收低功耗空闲状态。当器件接收到非 /P/ 编码位模式（非刷新信号）时，会退出接收低功耗空闲状态，并相应地设置 **接收_DV** 和 **接收数据[3:0]** 输出引脚，以输出正常帧或正常空闲信号。

为进一步节省功耗，当接收低功耗空闲状态下经过 9 个或更多接收时钟周期后，器件会停止向媒体访问控制器输出 **RXC**时钟。PCS 控制 1 寄存器中的时钟停止使能位用于控制器件是否停止输出**RXC**时钟。在时钟停止期间，器件会将**接收_控制**和**接收数据[3: 0]** 引脚保持为低电平。在断言低功耗空闲期间，器件可以随时重新启动**RXC**，但必须确保在撤销低功耗空闲之前，**RXC**至少发生一次正跳变。

图 5-8: 展示了 100 Mbps 模式下 RGMII 接收端的低功耗空闲转换。

图 5-8: 低功耗空闲转换 - RGMII (100 Mbps) 接收



5.10.3 10BASE-Te MODE

For standard (non-EEE) 10BASE-T mode, normal link pulses (NLPs) with long periods of no AC signal transmission are used to maintain the link during the idle period when there is no traffic activity. To save more power, the device provides the option to enable 10BASE-Te mode, which saves additional power by reducing the transmitted signal amplitude from 2.5V to 1.75V. As set by the **MODE[4:0]** configuration straps, 10BASE-Te mode is enabled or disabled by default and can be enabled or disabled by clearing or setting the p_cat3 bit in the AFED Control Register in MMD space.

5.10.4 REGISTERS ASSOCIATED WITH EEE

The following MMD registers are provided for EEE configuration and management:

- MMD Address 3h, Register 0h — PCS Control 1 Register
- MMD Address 3h, Register 1h — PCS Status 1 Register
- MMD Address 7h, Register 3Ch — EEE Advertisement Register
- MMD Address 7h, Register 3Dh — EEE Link Partner Ability Register
- MMD Address 1Ch, Register 9h — AFED Control Register

5.11 802.3br Frame Preemption

IEEE Standard 802.3br-2016 specifies a method for interspersing express traffic by preempting the transmission of a normal packet, transmitting the express packet and then resuming the normal packet. The receiver likewise reassembles the fragmented normal packet.

The preemption and reassembly is performed by a new MAC Merge sublayer, which resides between the MAC layer and the Reconciliation Sublayer and makes use of a newly defined mPacket format. This mPacket format starts with the normal preamble but supplements the normal Start of Frame Delimiter (SFD = 0xD5) with newly defined Start mPacket Delimiters (SMDs) of various values. Also added for certain mPackets is a fragment count octet.

Since these SMDs are effectively part of the data stream and not part of the framing (i.e. they are D codes), the PHY functions will pass these to the GMII/MII/RGMII as normal data.

Section 5.16 defines SOF pulse generation support for the SMDs.

Note: In 10BASE-T, PCS LOOPBACK SWAP/POLARITY CONTROL REGISTER Bit 2 needs to be set so the preamble is preserved.

5.12 Dynamic Channel Quality (DCQ) (TC1 and TC12)

The device provides dynamic channel quality features that include Mean Square Error (MSE), Signal Quality Indicator (SQI), and peak Mean Square Error (pMSE) values. These features are designed to be compliant with Sections 6.1.1, 6.1.2, and 6.1.3 of the *OPEN Alliance TC1 - Advanced diagnostics features for 100BASE-T1 automotive Ethernet PHYs Version 1.0* specification and *TC-12 - Advanced diagnostics features for 1000BASE-T1 automotive Ethernet PHYs Version 1.7*, respectively. These DCQ features are detailed in the following sections:

- [Mean Square Error \(MSE\)](#)
- [Signal Quality Indicator \(SQI\)](#)
- [Peak Mean Square Error \(pMSE\)](#)

MLT-3 modulation is used for data transmission in 100BASE-TX and PAM5 modulation is used for data transmission in 1000BASE-T.

Logically, 100BASE-TX (MLT-3) and 1000BASE-T (PAM5) have signal values of {-1, 0, +1} and {-2, -1, 0, +1, +2}, respectively. These logic levels are mapped to slicer reference levels of {-128, 0, 128} for 100BASE-TX and {-128, -64, 0, 64, 128} for 1000BASE-T. The middle points (the compare thresholds) are {-64, 64} for 100BASE-TX and are {-96, -32, 32, 96} for 1000BASE-T.

Ideally, each receive data sample would be the maximum distance from the compare thresholds, with error values of 0. But because of noise and imperfection in real applications, the sampled data may be off from its ideal. The closer to the compare threshold, the worse the signal quality.

The slicer error is a measurement of how far the processed data off from its ideal location. The largest instantaneous slicer error for 1000BASE-T is +/-32. The largest instantaneous slicer error for 100BASE-TX is +/-64. A higher absolute slicer error means a degraded signal receiving condition.

5.10.3 10BASE-Te 模式

对于标准（非节能以太网）10BASE-T 模式，在没有流量活动的空闲期，会使用长时间不进行交流信号传输的普通链路脉冲（NLP），以维持链路。为了进一步节省功耗，该器件提供了启用 10BASE-Te 模式的选项；该模式通过将发送信号幅度从 2.5V 降至 1.75V 来进一步降低功耗。根据**模式[4: 0]** 配置绑带的设置，10BASE-Te 模式默认启用或禁用；也可以通过清除或置位 MMD 空间中模拟前端控制寄存器内的 p_cat3 位来启用或禁用该模式。

5.10.4 与节能以太网相关的寄存器

以下 MMD 寄存器用于节能以太网的配置和管理：

- MMD 地址 3h，寄存器 0h — PCS 控制 1 寄存器
- MMD 地址 3h，寄存器 1h — PCS 状态 1 寄存器
- MMD 地址 7h，寄存器 3Ch — EEE 通告寄存器
- MMD 地址 7h，寄存器 3Dh — EEE 链路伙伴能力寄存器
- MMD 地址 1Ch，寄存器 9h — AFED 控制寄存器

5.11 802.3br 帧抢占

IEEE 标准 802.3br-2016 规定了一种插入快速流量的方法：通过抢占普通数据包的发送，发送快速数据包，然后继续发送普通数据包。接收器也会重新组装被分片的普通数据包。

抢占和重新组装由新的 MAC 合并子层执行，该子层位于 MAC 层与协调子层之间，并使用新定义的 mPacket 格式。此 mPacket 格式以普通前导码开头，但会使用新定义的、具有不同取值的 mPacket 起始定界符（SMD），补充普通帧起始定界符（SFD = 0xD5）。某些 mPacket 还会添加分片计数八位组。

由于这些 SMD 实际上属于数据流的一部分，而不是成帧的一部分（即它们是 D 码），PHY 功能会将它们作为普通数据传递到 GMII/MII/RGMII。

第 5.16 节 定义了对 SMD 的帧起始脉冲生成支持。

注意： 在 10BASE-T 中，需要设置 PCS 环回交换/极性控制寄存器的位 2，以保留前导码。

5.12 动态信道质量 (DCQ) (TC1 和 TC12)

该器件提供动态信道质量功能，包括均方误差 (MSE)、信号质量指示器 (SQI) 和峰值均方误差 (pMSE) 值。这些功能旨在符合《*OPEN 联盟 TC - 汽车以太网 PHY 高级诊断功能 1.0版*》中的第 6 1、6 1 1 2、6 1 3 *1 1001* and . . 节，以及《*TC-12 - 1000BASE-T1 汽车以太网 PHY 高级诊断功能 版本-1.7*》中的相关要求。这些 DCQ 功能详见以下各节：

- [均方误差 \(MSE\)](#)
- [信号质量指示器 \(SQI\)](#)
- [峰值均方误差 \(pMSE\)](#)

100BASE-TX 使用 MLT-3 调制进行数据传输，1000BASE-T 使用 PAM5 调制进行数据传输。

从逻辑上讲，100BASE-TX（三电平传输编码）和 1000BASE-T（PAM5）的信号值分别为 {-1, 0, +1} 和 {-2, -1, 0, +1, +2}。对于 100BASE-TX，这些逻辑电平映射到判决器参考电平 {-128, 0, 128}；对于 1000BASE-T，则映射到 {-128, -64, 0, 64, 128}。中间点（比较阈值）对于 100BASE-TX 为 {-64, 64}，对于 1000BASE-T 为 {-96, -32, 32, 96}。

理想情况下，每个接收数据样本与比较阈值的距离都应达到最大，错误值为 0。但由于实际应用中存在噪声和不完美因素，采样数据可能偏离理想值。越接近比较阈值，信号质量就越差。

切片器误差用于衡量处理后数据偏离理想位置的程度。1000BASE-T 的最大瞬时切片器误差为 +/-32。100BASE-TX 的最大瞬时切片器误差为 +/-64。切片器误差的绝对值越大，信号接收条件就越差。

5.12.1 MEAN SQUARE ERROR (MSE)

This section defines the implementation of section 6.1.1 of the TC1 and TC12 specifications. The device can provide detailed information of the dynamic signal quality by means of a MSE value. This mode is enabled by setting the sqi_enable bit in the DCQ Configuration Register. This bit must be set for all DCQ measurements.

With this method, the slicer error is converted into a squared value and then filtered by a programmable low pass filter. This is similar to taking the average of absolute slicer error over a long moving time window. For each data sample, the difference between the absolute slicer error (scaled by x2 (before squaring) for 1000BASE-T) and the current filtered value is added back into the current filtered value.

Note: The sqi_squ_mode_en bit in the DCQ Configuration Register must be set to choose square mode.

The sqi_kp field in the DCQ Configuration Register sets the weighting of the add back as a divide by $2^{\text{sqi_kp}}$, effectively setting the filter bandwidth. As the sqi_kp value is increased, the weighing is decreased, and the mean slicer error value takes a longer time to settle to a stable value. Also as the sqi_kp value is increased, there will be less variation in the mean slicer error value reported.

The scale611 field in the DCQ Configuration Register is used to set a divide by factor (divide by 2^{scale611}) such that the MSE value is linearly scaled to the range of 0 to 511. If the divide by factor is too small, the MSE value is capped at a maximum of 511.

The filtered error value is saved every 65,536 symbols (524,288 us).

In order to capture the MSE Value, the DCQ Read Capture bit in the DCQ Configuration Register needs to be written as a high with the desired cable pair specified in the DCQ Channel Number field of the same register. The DCQ Read Capture bit will immediately self-clear and the result will be available in the DCQ Mean Square Error Register.

In addition to the current MSE Value, the worst case MSE value since the last read of DCQ Mean Square Error Register is stored in DCQ Mean Square Error Worst Case Register.

5.12.2 SIGNAL QUALITY INDICATOR (SQI)

The device provides two SQI methods:

- [SQI Method A](#): TC1 Section 6.1.2 compliant
- [SQI Method B](#): Proprietary method

5.12.2.1 SQI Method A

This section defines the implementation of section 6.1.2 of the TC1 and TC12 specifications. This mode builds upon the Mean Square Error (MSE) method by mapping the MSE value onto a simple quality index. This mode is enabled by setting the sqi_enable bit, in the DCQ Configuration Register.

As in the Mean Square Error (MSE) method, the sqi_squ_mode_en bit in the DCQ Configuration Register must be set to choose square mode and the scale611 field in the DCQ Configuration Register is used to set the divide by factor (divide by 2^{scale611}) such that the MSE value is linearly scaled to the range of 0 to 511.The MSE value is compared to the thresholds set in the DCQ SQI Table Registers to provide a SQI value between 0 (worst value) and 7 (best value) as follows:

TABLE 5-3: MSE TO SQI MAPPING

MSE Value		SQI Value
Greater Than	Less Than or Equal To	
	SQI_TBL7.SQI_VALUE	7
SQI_TBL7.SQI_VALUE	SQI_TBL6.SQI_VALUE	6
SQI_TBL6.SQI_VALUE	SQI_TBL5.SQI_VALUE	5
SQI_TBL5.SQI_VALUE	SQI_TBL4.SQI_VALUE	4
SQI_TBL4.SQI_VALUE	SQI_TBL3.SQI_VALUE	3
SQI_TBL3.SQI_VALUE	SQI_TBL2.SQI_VALUE	2
SQI_TBL2.SQI_VALUE	SQI_TBL1.SQI_VALUE	1
SQI_TBL1.SQI_VALUE		0

5.12.1 均方误差 (MSE)

本节定义了 TC1 和 TC12 规范第 6.1.1 节的实现。器件可通过 MSE 值提供动态信号质量的详细信息。设置 DCQ 配置寄存器中的 sqi_en- able 位即可启用此模式。所有 DCQ 测量都必须设置此位。

使用此方法，判决器误差会转换为平方值，然后由可编程低通滤波器进行滤波。这类似于在较长的滑动时间窗口内计算判决器绝对误差的平均值。对于每个数据样本，绝对判决器误差（对于 1000BASE-T，在平方前乘以 x2）与当前滤波值之间的差值会加回到当前滤波值中。

注意： 必须设置 DCQ 配置寄存器中的 sqi_squ_mode_en 位，才能选择平方模式。

DCQ 配置寄存器中的 sqi_kp 字段将加回权重设为除以 $2^{\text{sqi_kp}}$ ，从而有效地设置滤波器带宽。随着 sqi_kp 值增大，权重会降低，平均判决器误差值需要更长时间才能稳定下来。此外，随着 sqi_kp 值增大，报告的平均判决器误差值变化会更小。

DCQ 配置寄存器中的 scale611 字段用于设置除数（除以 2^{scale611} ），从而将 MSE 值线性缩放到 0 至 511 的范围内。如果除数过小，MSE 值将被限制为最大值 511。

每 65,536 个符号（524,288 微秒）保存一次滤波后的错误值。

要捕获 MSE 值，需要将 DCQ 配置寄存器中的 DCQ 读取捕获位置为高电平，并在同一寄存器的 DCQ 通道编号字段中指定所需的电缆线对。DCQ 读取捕获位会立即自动清零，结果可在 DCQ 均方误差寄存器中读取。

除当前 MSE 值外，自上次读取 DCQ 均方误差寄存器以来的最坏情况 MSE 值也存储在 DCQ 均方误差最坏情况寄存器中。

5.12.2 信号质量指示器 (SQI)

该器件提供两种 SQI 方法：

- [SQI 方法 A](#)：符合 TC1 第 6.1.2 节
- [SQI 方法 B](#)：专有方法

5.12.2.1 SQI 方法 A

本节定义了 TC1 和 TC12 规范第 6.1.2 节的实现。此模式基于均方误差 (MSE) 方法，将 MSE 值映射为一个简单的质量指数。通过设置 DCQ 配置寄存器中的 sqi_enable 位，可使能此模式。

与均方误差 (MSE) 方法相同，必须设置 DCQ 配置寄存器中的 sqi_squ_mode_en 位以选择平方模式，并使用 DCQ 配置寄存器中的 scale611 字段设置除数（除以 2^{scale611} ），从而将 MSE 值线性缩放至 0 到 511 的范围。将 MSE 值与 DCQ SQI 表寄存器中设置的阈值进行比较，以得出介于 0（最差值）和 7（最佳值）之间的 SQI 值，具体如下：

表 5-3: MSE 到 SQI 的映射

MSE 值		SQI 值
大于	小于或等于	
	SQI_TBL7.SQI_值	7
SQI_TBL7.SQI_值	SQI TBL6.SQI 值_ –	6
SQI TBL6.SQI 值_ –	SQI TBL5.SQI 值_ –	5
SQI_TBL5.SQI_值	SQI TBL4.SQI 值_ –	4
SQI TBL4.SQI 值_ –	SQI TBL3.SQI 值_ –	3
SQI TBL3.SQI 值_ –	SQI TBL2.SQI 值_ –	2
SQI TBL2.SQI 值_ –	SQI TBL1.SQI 值_ –	1
SQI TBL1.SQI 值_ –		0

In order to capture the SQI value, the DCQ Read Capture bit in the DCQ Configuration Register needs to be written as a high with the desired cable pair specified in the DCQ Channel Number field of the same register. The DCQ Read Capture bit will immediately self-clear and the result will be available in the DCQ SQI Register.

In addition to the current SQI the worst case (lowest) SQI since the last read is available in the SQI Worst Case field. The correlation between the SQI values stored in the DCQ SQI Register and an according signal to noise ratio (SNR) based on AWG noise (bandwidth of 80MHz) is shown in [Table 5-4](#). The bit error rates to be expected in the case of white noise as interference signal is shown in the table as well for information purposes.

A link loss only occurs if the SQI value is 0.

TABLE 5-4: SQI VALUE CORRELATION

SQI Value	Recommended BER for AWG Noise Model
0	BER>10^-10
1	
2	
3	
4	
5	BER<10^-10
6	
7	

5.12.2.2 SQI Method B

With the SQI Method B, the slicer error is converted into an absolute value and then filtered by a programmable low pass filter. This is similar to taking the average of absolute slicer error over a long moving time window. This mode is enabled by setting the sqi_enable bit, in the DCQ Configuration Register.

For each data sample, the difference between the absolute slicer error (scaled by x2 (before squaring) for 1000BASE-T) and the current filtered value is added back into the current filtered value. The sqi_squ_mode_en bit in the DCQ Configuration Register is used to square the (scaled) slicer error.

The sqi_kp field in the DCQ Configuration Register sets the weighting of the add back as a divide by 2^sqi_kp, effectively setting the filter bandwidth. As the sqi_kp value is increased, the weighing is decreased, and the mean slicer error value takes a longer time to settle to a stable value. Also as the sqi_kp value is increased, there will be less variation in the mean slicer error value reported.

The filtered error value is saved every 65,536 symbols (524,288 us).

In order to capture the current error value, the DCQ Read Capture bit in the DCQ Configuration Register needs to be written as a high with the desired cable pair specified in the DCQ Channel Number field of the same register. The DCQ Read Capture bit immediately self-clears and the result is available in the Mean Slicer Error Register.

A software based lookup table (derived empirically in lab conditions) may be used to report a SQI number.

5.12.3 PEAK MEAN SQUARE ERROR (PMSE)

This section defines the implementation of section 6.1.3 of the TC1 and TC12 specifications. The peak MSE value is intended to identify transient disturbances which are typically in the microsecond range. This mode is enabled by setting the sqi_enable bit, in the DCQ Configuration Register.

With this method, the slicer error is converted into a squared value and then filtered by a programmable low pass filter. This is similar to taking the average of absolute slicer error over a moving time window.

For each data sample, the difference between the absolute slicer error (scaled by x2 (before squaring) for 1000BASE-T) and the current filtered value is added back into the current filtered value.

Note: The sqi_squ_mode_en bit in the DCQ Configuration Register must be set to choose square mode.

The sqi_kp3 field in the DCQ Configuration Register sets the weighting of the add back as a divide by 2^(sqi_kp3), effectively setting the filter bandwidth. As the sqi_kp3 value is increased, the weighing is decreased, and the mean slicer error value takes a longer time to settle to a stable value.

要捕获 SQI 值，需将 DCQ 配置寄存器中的 DCQ 读取捕获位写为高电平，并在同一寄存器的 DCQ 通道编号字段中指定所需的电缆线对。DCQ 读取捕获位会立即自行清零，结果可在 DCQ SQI 寄存器中读取。

除当前 SQI 外，自上次读取以来的最差（最低）SQI 值也可在 SQI 最差情况字段中查看。存储在 DCQ SQI 寄存器中的 SQI 值与基于 AWG 噪声（带宽为 80 兆赫）的相应信噪比 (SNR) 之间的相关性如[表 5-4](#)所示。该表还列出了白噪声作为干扰信号时预期的误码率，供参考。

只有当 SQI 值为 0 时，才会发生链路丢失。

表 5-4: SQI 值相关性

SQI 值	AWG 噪声模型的建议误码率
0	误码率>10^-10
1	
2	
3	
4	
5	误码率<10^-10
6	
7	

5.12.2.2 SQI 方法 B

使用 SQI 方法 B 时，切片器误差会转换为绝对值，然后由可编程低通滤波器进行滤波。这类似于在较长的滑动时间窗口内对绝对切片器误差取平均值。通过在 DCQ 配置寄存器中设置 sqi_enable 位可启用此模式。

对于每个数据样本，绝对切片器误差（在平方前乘以 2，用于 1000BASE-T）与当前滤波值之间的差值会加回到当前滤波值中。DCQ 配置寄存器中的 sqi_squ_mode_en 位用于对（缩放后的）切片器误差进行平方。

DCQ 配置寄存器中的 sqi_kp 字段将加回权重设置为除以 2^sqi_kp，从而有效地设置滤波器带宽。随着 sqi_kp 值增大，权重会降低，平均判决器误差值需要更长时间才能稳定。此外，随着 sqi_kp 值增大，所报告的平均判决器误差值的波动会减小。

每 65,536 个符号（524,288 微秒）保存一次滤波后的误差值。

要捕获当前误差值，需将 DCQ 配置寄存器中的 DCQ 读取捕获位写为高电平，并在同一寄存器的 DCQ 通道编号字段中指定所需的电缆线对。DCQ 读取捕获位会立即自动清零，结果可在平均判决器误差寄存器中读取。

可以使用基于软件的查找表（通过实验室条件下的实测结果得出）来报告 SQI 数值。

5.12.3 峰值均方误差（PMSE）

本节定义了 TC1 和 TC12 规范第 6.1.3 节的实现。峰值 MSE 值用于识别通常处于微秒范围内的瞬态干扰。通过设置 DCQ 配置寄存器中的 sqi_enable 位，可启用此模式。

使用此方法时，切片器误差会转换为平方值，然后通过可编程低通滤波器进行滤波。这类似于在滑动时间窗口内对切片器误差的绝对值取平均。对于每个数据样本，绝对切片器误差（对于 1000BASE-T，在平方前乘以 2）与当前滤波值之差会加回到当前滤波值中。

注意：必须设置 DCQ 配置寄存器中的 sqi_squ_mode_en 位，才能选择平方模式。

DCQ 配置寄存器中的 sqi_kp3 字段通过将加回值除以 2^(sqi_kp3) 来设置其权重，从而有效地设置滤波器带宽。随着 sqi_kp3 值增大，权重会降低，平均判决器误差值需要更长时间才能稳定下来。

Every 65,536 symbols (524,288 us), the highest filtered value over that previous 524,288 us period is saved.

The scale613 field in the DCQ Configuration Register is used to set a divide by factor (divide by $2^{\wedge \text{scale613}+3}$) such that the peak MSE value is linearly scaled to the range of 0 to 63. If the divided by factor is too small, the peak MSE value is capped at a maximum of 63.

In order to capture the Peak MSE Value, the DCQ Read Capture bit in the DCQ Configuration Register needs to be written as a high with the desired cable pair specified in the DCQ Channel Number field of the same register. The DCQ Read Capture bit will immediately self-clear and the result will be available in the DCQ Peak MSE Register.

In addition to the current Peak MSE Value the worst case Peak MSE value since the last read of DCQ Peak MSE Register is stored in the same register.

5.13 Loopback Modes

The device supports the following loopback operations to verify analog and/or digital data paths.

- Digital (near-end) loopback.
- Remote (Far-End) loopback
 - Remote (Far-End) loopback with switch mode
- External connector loopback

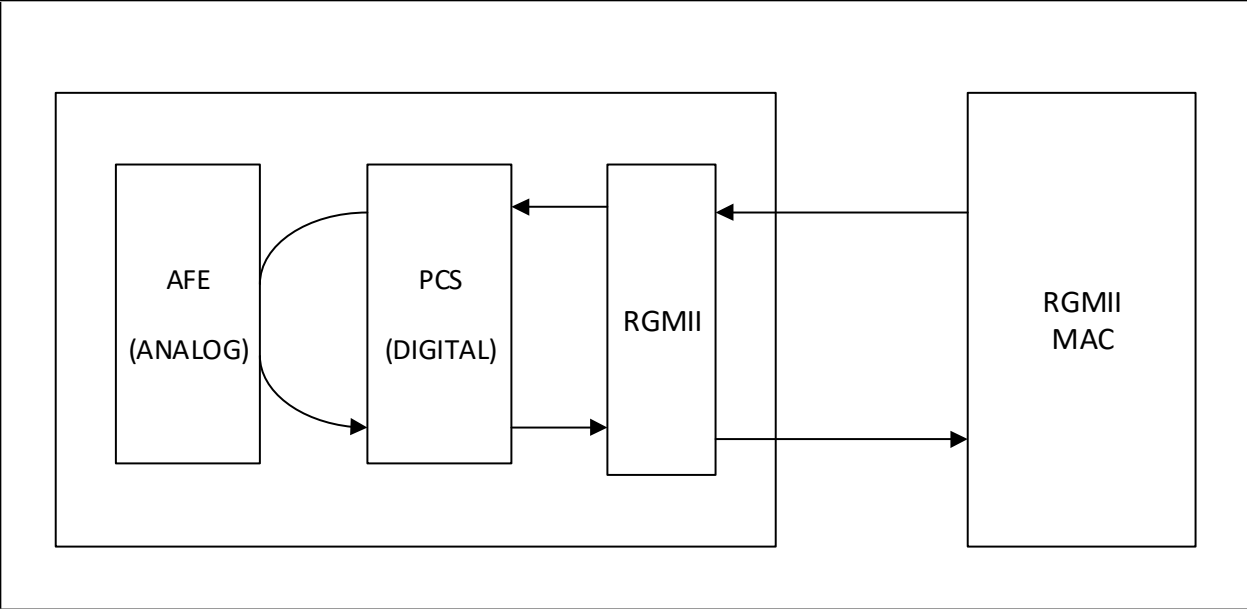
5.13.1 DIGITAL (NEAR-END) LOOPBACK

This loopback mode checks the RGMII transmit and receive data paths between the device and the external MAC, and is supported for all three speeds (10/100/1000 Mbps) at full-duplex.

The loopback data path is shown in [Figure 5-9](#):

1. RGMII MAC transmits frames to the device.
2. Frames are wrapped around inside the device.
3. The device transmits frames back to RGMII MAC

FIGURE 5-9: DIGITAL (NEAR-END) LOOPBACK



每经过 65,536 个符号（524,288 us），都会保存此前 524,288 us 周期内经过滤波的最大值。

DCQ 配置寄存器中的 scale613 字段用于设置除数（除以 $2^{\text{scale613}+3}$ ），以便将峰值 MSE 值线性缩放到 0 到 63 的范围。如果除数过小，峰值 MSE 值将被限制为最大值 63。

要捕获峰值 MSE 值，必须将 DCQ 配置寄存器中的 DCQ 读取捕获位置为高电平，并在同一寄存器的 DCQ 通道编号字段中指定所需的电缆线对。DCQ 读取捕获位将立即自动清零，结果可在 DCQ 峰值 MSE 寄存器中读取。

除当前峰值 MSE 值外，自上次读取 DCQ 峰值 MSE 寄存器以来的最差峰值 MSE 值也存储在同一个寄存器中。

5.13 环回模式

该器件支持以下环回操作，用于验证模拟和/或数字数据路径。

- 数字（近端）环回。
- 远端（远端侧）环回
 - 交换机模式下的远端（远端侧）环回
- 外部连接器环回

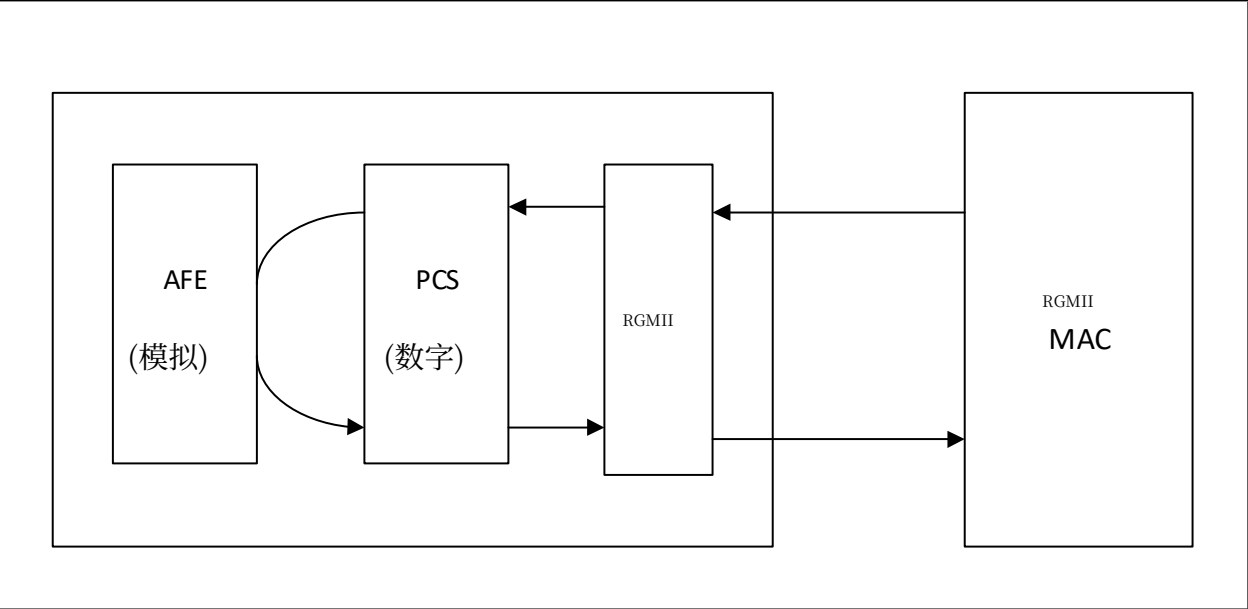
5.13.1 数字（近端）环回

此环回模式用于检查器件与外部媒体访问控制器之间的 RGMII 发送和接收数据路径，并支持全双工下的三种速率（10/100/1000 Mbps）。

环回数据路径如图 5-9 所示。

1. RGMII 媒体访问控制器向器件发送帧。
2. 帧在器件内部回环。
3. 器件将帧发送回 RGMII 媒体访问控制器

图 5-9：数字（近端）环回



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The following programming steps and register settings are used for local loopback mode.

For 1000 Mbps loopback,

- Set Basic Control Register,
 - Bit [14] = 1 // Enable local loopback mode
 - Bits [6, 13] = 10 // Select 1000 Mbps speed
 - Bit [12] = 0 // Disable auto-negotiation
 - Bit [8] = 1 // Select full-duplex mode
- Set Auto-Negotiation Master Slave Control Register,
 - Bit [12] = 1 // Enable leader-follower manual configuration
 - Bit [11] = 0 // Select follower configuration (required for loopback mode)

For 10/100 Mbps loopback,

- Set Basic Control Register,
 - Bit [14] = 1 // Enable local loopback mode
 - Bits [6, 13] = 00 / 01 // Select 10 Mbps/100 Mbps speed
 - Bit [12] = 0 // Disable auto-negotiation
 - Bit [8] = 1 // Select full-duplex mode

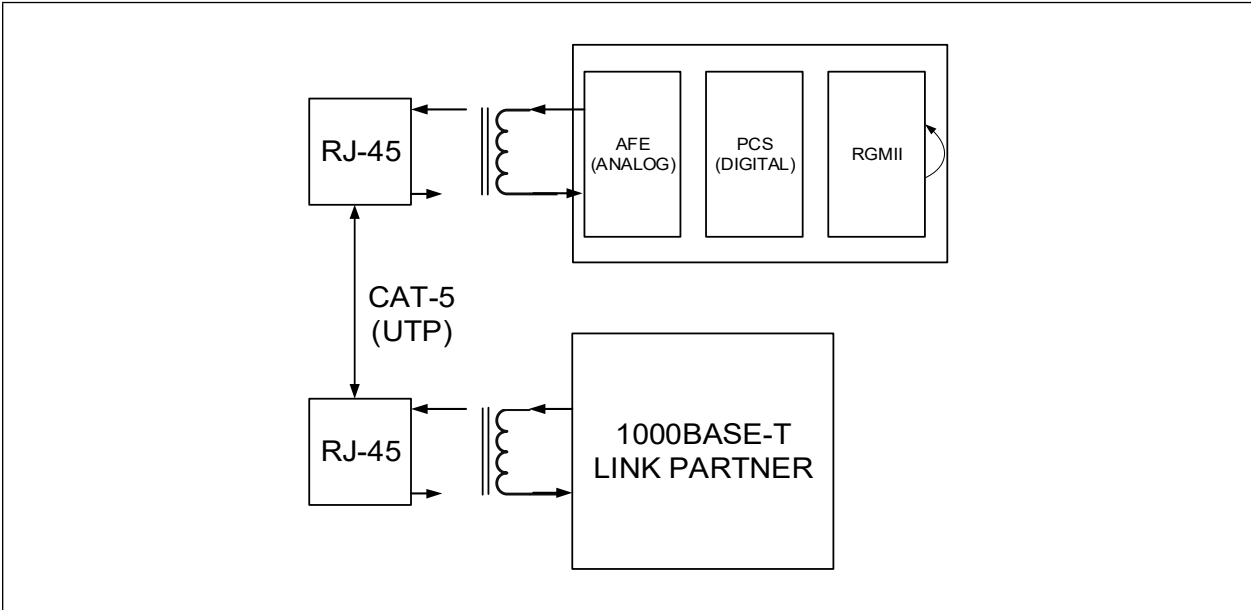
5.13.2 REMOTE (FAR-END) LOOPBACK

This loopback mode checks the line (differential pairs, transformer, RJ-45 connector, Ethernet cable) transmit and receive data paths between the device and its link partner, and is supported for 1000BASE-T full-duplex mode only.

The loopback data path is shown in [Figure 5-10](#):

- The Gigabit PHY link partner transmits frames to the device.
- Frames are wrapped around inside the device.
- The device transmits frames back to the Gigabit PHY link partner.

FIGURE 5-10: REMOTE (FAR-END) LOOPBACK



LAN8840

本地环回模式使用以下编程步骤和寄存器设置。

对于 1000 兆比特每秒环回,

- 设置基本控制寄存器,
 - 位 [14] = 1 // 使能本地环回模式
 - 位 [6, 13] = 10 // 选择 1000 Mbps 速率
 - 位 [12] = 0 // 禁用自动协商
 - 位 [8] = 1 // 选择全双工模式
- 设置自动协商主从控制寄存器,
 - 位 [12] = 1 // 启用领导者-跟随者手动配置
 - 位 [11] = 0 // 选择从端配置 (环回模式所需)

对于 10/100 Mbps 环回,

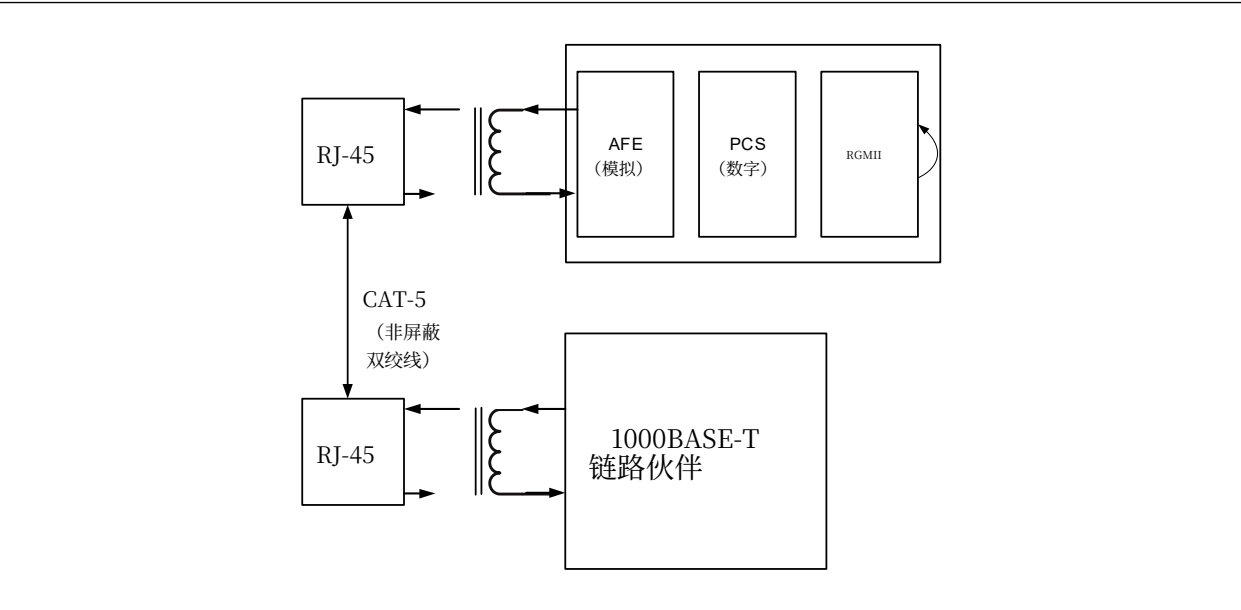
- 设置基本控制寄存器,
 - 位 [14] = 1 // 使能本地环回模式
 - 位 [6, 13] = 00 / 01 // 选择 10 Mbps/100 Mbps 速率
 - 位 [12] = 0 // 禁用自动协商
 - 位 [8] = 1 // 选择全双工模式

5.13.2 远端 (远端侧) 环回

此环回模式用于检查器件与其链路伙伴之间线路 (差分对、变压器、RJ-45 连接器、以太网电缆) 的发送和接收数据路径, 且仅支持 1000BASE-T 全双工模式。环回数据路径如 [图 5-10](#) 所示:

- 千兆物理层链路伙伴向器件发送帧。
- 帧在器件内部被环回。
- 器件将帧发送回千兆物理层链路伙伴。

图 5-10: 远端 (远端侧) 环回



The following programming steps and register settings are used for remote loopback mode.

1. Set Basic Control Register,
 - Bits [6, 13] = 10 // Select 1000 Mbps speed
 - Bit [12] = 0 // Disable auto-negotiation
 - Bit [8] = 1 // Select full-duplex modeOr just auto-negotiate and link up at 1000BASE-T full-duplex mode with the link partner.
2. Set PCS Loop-back Swap/Polarity Control Register,
3. Bit [8] = 1 // Enable remote loopback mode

5.13.2.1 Switch Mode Loopback

During Remote (Far-End) Loopback, an option may be enabled where the MAC DA and SA are swapped and the FCS is recalculated. This mode is useful for testing where a network switch is the link partner. Upon returning the modified frame to the switch, the switch may direct the frame to the original ingress port. This mode is enabled with the Switch Port Loopback bit in the Common Control Register.

5.13.3 EXTERNAL CONNECTOR LOOPBACK

The connector loopback testing feature allows the twisted pair interface to be looped back externally. When using this feature, the PHY must be connected to a loopback connector or a loopback cable. Pair A should be connected to pair B, and pair C to pair D, as shown in Figure 5-11. The connector loopback feature functions at all available interface speeds.

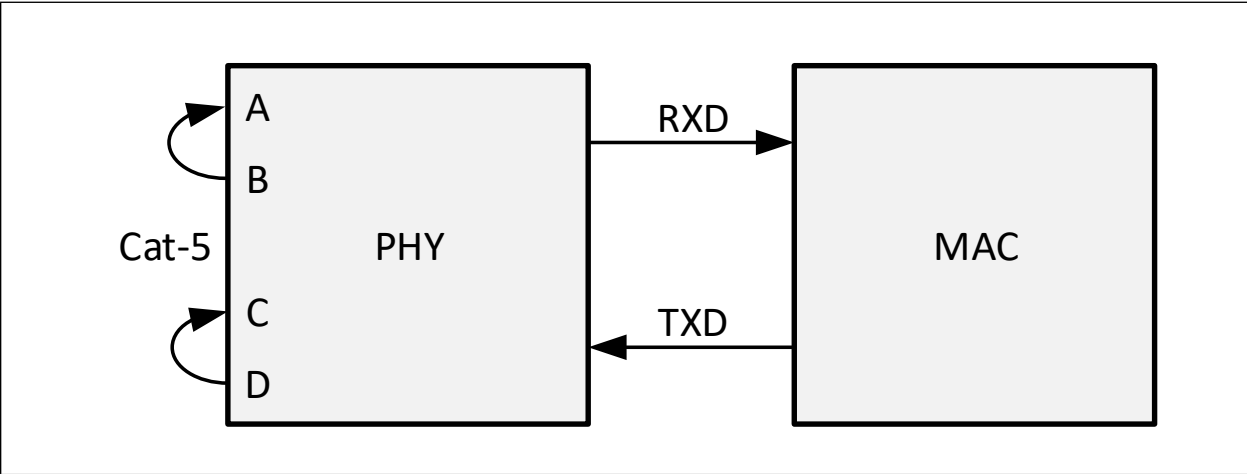
This loopback tests the PHY digital and MAC connectivity.

When using the connector loopback testing feature, the device Auto-Negotiation, speed, and duplex configuration is set using the Basic Control Register, Auto-Negotiation Advertisement Register, and Auto-Negotiation Master Slave Control Register.

For 1000BASE-T connector loopback, the following additional writes are required to be executed in the following order:

- Disable Auto-Negotiation and set the speed to 1000Mbps and the duplex to full by setting the Basic Control Register to a value of 0140h.
- Set the Leader-Follower configuration to leader by setting the Master/Slave Manual Configuration Enable and Master/Slave Manual Configuration Value bits in the Auto-Negotiation Master Slave Control Register.
- Enable the 1000BASE-T connector loopback by setting the Ext_lpbk bit in the Reserved Register.

FIGURE 5-11: EXTERNAL CONNECTOR LOOPBACK



远程环回模式使用以下编程步骤和寄存器设置。

1. 设置基本控制寄存器,
 - 位 [6, 13] = 10 // 选择 1000 Mbps 速率
 - 位 [12] = 0 // 禁用自动协商
 - 位 [8] = 1 // 选择全双工模式或者直接进行自动协商，并与链路伙伴以 1000BASE-T 全双工模式建立链路。
2. 设置 PCS 回环交换/极性控制寄存器,
3. 位 [8] = 1 // 启用远程环回模式

5.13.2.1 交换机模式回环

在远端（远端侧）环回期间，可以启用一个选项来交换 MAC 目的地址和源地址，并重新计算帧校验序列。此模式适用于链路伙伴为网络交换机时的测试。将修改后的帧返回交换机后，交换机可能会将该帧转发到原始入端口。此模式通过通用控制寄存器中的交换机端口回环位启用。

5.13.3 外部连接器环回

连接器回环测试功能允许双绞线接口进行外部环回。使用此功能时，必须将 PHY 连接到回环连接器或回环电缆。应将 A 对线连接到 B 对线，并将 C 对线连接到 D 对线，如图 5-11 所示。连接器环回功能支持所有可用的接口速率。

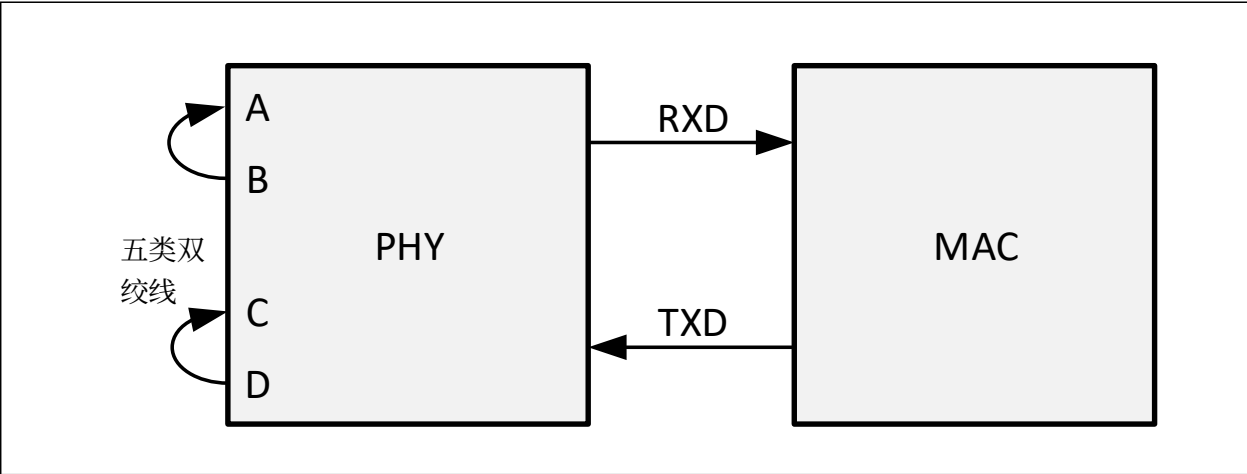
此环回测试 PHY 数字电路和 MAC 连通性。

使用连接器回环测试功能时，通过基本控制寄存器、自动协商通告寄存器和自动协商主从控制寄存器设置设备自动协商、速率和双工配置。

对于 1000BASE-T 连接器环回，必须按以下顺序执行以下附加写入操作：

- 将基本控制寄存器设置为 0140h，以禁用自动协商并将速率设为 1000 Mbps、双工模式设为全双工。
- 通过设置自动协商主从控制寄存器中的主从手动配置使能位和主/从手动配置值位，将主从配置设为主端。
- 通过设置保留寄存器中的 Ext_lpbk 位，启用 1000BASE-T 连接器环回。

图 5-11: 外部连接器环回



5.14 LinkMD® Cable Diagnostic

The LinkMD function uses Time Domain Reflectometry (TDR) to analyze the cabling plant for common cabling problems, such as open circuits, short circuits, and impedance mismatches as well as the distance to the fault. Each of the four twisted pairs are tested separately.

LinkMD operates by sending a pulse of known amplitude and duration down the selected differential pair, then analyzing the polarity and shape of the reflected signal to determine the type of fault: open circuit for a positive/non-inverted amplitude reflection and short circuit for a negative/inverted amplitude reflection. The time duration for the reflected signal to return provides the approximate distance to the cabling fault. The LinkMD function processes this TDR information and presents it as a numerical value that can be translated to a cable distance.

LinkMD is initiated by accessing the Cable Diagnostic Register.

To test each individual cable pair, set the cable pair in the Cable Diagnostics Test Pair (VCT_PAIR[1:0]) field of the Cable Diagnostic Register, along with setting the Cable Diagnostics Test Enable (VCT_EN) bit. The Cable Diagnostics Test Enable (VCT_EN) bit will self-clear when the test is concluded.

The test results (for the pair just tested) are available in the Cable Diagnostic Register. With the Bit[9:0] Definition (VCT_SEL[1:0]) field set to 0, the Cable Diagnostics Status (VCT_ST[1:0]) field will indicate a Normal (properly terminated), Open or Short condition, or a Failed Test.

If the test result was Open or Short, the Cable Diagnostics Data or Threshold (VCT_DATA[7:0]) field indicates the distance to the fault in meters as approximately:

- distance to fault = (VCT_DATA - 22) * 4 / cable propagation velocity

With an accuracy of +/- 2% to 3% for short and medium cables and +/- 5% to 6% for long cables. Inaccuracy is due to cable pitch differences between cable manufacturers, where cable pitch is the number of wire twists per unit cable length.

LinkMD supports diagnostic cable testing in the following three scenarios:

- No Link Partner
- Fully Passive Link Partner
- Partially Active Link Partner

5.14.1 LinkMD WITH NO LINK PARTNER

In this scenario the remote end of the cable is unplugged.

Prior to running the cable diagnostics, perform a software reset (Basic Control Register, Bit [15] = 1). After reset, the following must be configured:

- Auto Negotiation disabled [Basic Control Register, Bit [12] = 0]
- Full Duplex set [Basic Control Register, Bit [8] = 1]
- Link Speed set to 1000Mbps [Basic Control Register, Bit [13] = 0, Bit [6] = 1]
- Auto MDI/MDI-X disabled [Digital Debug Control 1 Register, Bit [6] = 1]
- Leader-Follower configuration set to Follower [Auto-Negotiation Master Slave Control Register = 0x1200]

Wait 10ms prior to testing the pairs.

In this scenario, for each of the four twisted pairs LinkMD should only return Open or Short along with a distance.

- Open with “distance” providing the correct cable length indicates a good cable
- Short or Open with any other distance indicates a bad cable

After running the cable diagnostics, perform another software reset [Basic Control Register, Bit [15] = 1].

5.14.2 LinkMD WITH FULLY PASSIVE LINK PARTNER

In this scenario the remote end of the cable is connected to a Link Partner. The Link Partner must be powered down or configured to be in the following passive state:

- Auto Negotiation disabled
- Auto MDI/MDI-X disabled

Prior to running the cable diagnostics, perform a software reset (Basic Control Register, Bit [15] = 1). After reset, the following must be configured:

- Auto Negotiation disabled [Basic Control Register, Bit [12] = 0]

5.14 LinkMD® 电缆诊断

LinkMD 功能使用时域反射测量（TDR）分析布线设施中的常见布线问题，例如开路、短路和阻抗不匹配，并测量故障点的距离。四对双绞线分别进行测试。

LinkMD 会沿选定的差分线对发送一个幅度和持续时间已知的脉冲，然后分析反射信号的极性和形状，以确定故障类型：正向/非反相幅度反射表示开路，负向/反相幅度反射表示短路。反射信号返回所需的持续时间可提供布线故障的大致距离。LinkMD 功能会处理这些时域反射测量信息，并以数值形式呈现，可将该数值换算为电缆距离。

通过访问电缆诊断寄存器可启动 LinkMD。

要测试每条独立的电缆线对，请在电缆诊断寄存器的电缆诊断测试线对 (VCT_PAIR[1:0]) 字段中设置线对，同时设置电缆诊断测试使能 (VCT_EN) 位。测试结束后，电缆诊断测试使能 (VCT_EN) 位将自动清零。

测试结果（针对刚刚测试的线对）可在电缆诊断寄存器中查看。当位[9:0] 定义 (VCT_SEL[1:0]) 字段设置为 0 时，电缆诊断状态 (VCT_ST[1:0]) 字段将指示正常（正确端接）、开路、短路或测试失败。

如果测试结果为开路或短路，电缆诊断数据或阈值 (VCT_DATA[7:0]) 字段将以米为单位指示故障的大致距离：

- 故障距离 = (VCT_DATA - 22) * 4 / 电缆传播速度

短距离和中距离电缆的精度为 +/- 2% 至 3%，长距离电缆的精度为 +/- 5% 至 6%。误差是由不同电缆制造商生产的电缆绞距差异造成的，其中电缆绞距是指单位电缆长度内线路的扭绞次数。

LinkMD 支持以下三种情况下的诊断电缆测试：

- 无链路伙伴
- 完全被动链路伙伴
- 部分主动链路伙伴

5.14.1 无链路伙伴时的 LinkMD

在此场景中，电缆远端未插接。

运行电缆诊断前，先执行软件复位（基本控制寄存器，位 [15] = 1）。复位后，必须进行以下配置：

- 禁用自动协商 [基本控制寄存器，位 [12] = 0]
- 设为全双工 [基本控制寄存器，位 [8] = 1]
- 链路速度设为 1000 Mbps [基本控制寄存器，位 [13] = 0，位 [6] = 1]
- 禁用自动 MDI/MDI-X [数字调试控制 1 寄存器，位 [6] = 1]
- 主从配置设为从端 [自动协商主从控制寄存器 = 0x1200]

测试双绞线对前，请等待 10 毫秒。

在此情况下，对于四对双绞线中的每一对，LinkMD 应仅返回开路或短路以及距离。

- 若返回开路，且“距离”给出的电缆长度正确，则表示电缆状况良好
- 若返回短路，或开路但距离不符，则表示电缆状况不良

运行电缆诊断后，再执行一次软件复位 [基本控制寄存器，位 [15] = 1]。

5.14.2 LinkMD 与完全被动链路伙伴

在此场景中，电缆远端连接到一个链路伙伴。链路伙伴必须断电，或配置为以下被动状态：

- 自动协商已禁用
- 自动 MDI/MDI-X 已禁用

运行电缆诊断前，执行软件复位（基本控制寄存器，位 [15] = 1）。复位后，必须配置以下内容：

- 自动协商已禁用 [基本控制寄存器，位 [12] = 0]

- Full Duplex set [Basic Control Register, Bit [8] = 1]
- Link Speed set to 1000Mbps [Basic Control Register, Bit [13] = 0, Bit [6] = 1]
- Auto MDI/MDI-X disabled [Digital Debug Control 1 Register, Bit [6] = 1]
- Leader-Follower configuration set to Follower [Auto-Negotiation Master Slave Control Register = 0x1200]

Wait 10ms prior to testing the pairs.

In this scenario, for each of the four twisted pairs LinkMD will return Open or Short along with a distance or Normal.

- Normal indicates a good cable which is properly terminated at the Link Partner. Due to no TDR reflections, no cable distance is available with this result (VCT_DATA[7:0] is invalid).
- Open or Short indicates a bad cable or improper Link Partner termination. VCT_DATA[7:0] indicates the distance to the cable fault.

After running the cable diagnostics, perform another software reset [Basic Control Register, Bit [15] = 1].

5.14.3 LinkMD WITH PARTIALLY ACTIVE LINK PARTNER

In this scenario the remote end of the cable is connected to a Link Partner. The Link Partner is powered up and in the following state:

- Auto Negotiation may be enabled, disabled, or unknown
- Auto MDI/MDI-X is disabled, but we are able to selectively configure MDI or MDI-X, as required for the LinkMD tests

Prior to running the cable diagnostics, perform a software reset (Basic Control Register, Bit [15] = 1). After reset, the following must be configured:

- Auto Negotiation disabled [Basic Control Register, Bit [12] = 0]
- Full Duplex set [Basic Control Register, Bit [8] = 1]
- Link Speed set to 1000Mbps [Basic Control Register, Bit [13] = 0, Bit [6] = 1]
- Auto MDI/MDI-X disabled [Digital Debug Control 1 Register, Bit [6] = 1]
- Leader-Follower configuration set to Follower [Auto-Negotiation Master Slave Control Register = 0x1200]

Wait 10ms prior to testing the pairs.

The LinkMD test is first run with the Link Partner in MDI mode on twisted pairs B, C, and D.

The LinkMD test is then run with the Link Partner in MDI-X mode on twisted pair A.

- Normal indicates a good cable which is properly terminated at the Link Partner. Due to no TDR reflections, no cable distance is available with this result (VCT_DATA[7:0] is invalid).
- Open or Short indicates a bad cable or improper Link Partner termination. VCT_DATA[7:0] indicates the distance to the cable fault.

After running the cable diagnostics, perform another software reset [Basic Control Register, Bit [15] = 1].

APPLICATION NOTE: If the Cable Diagnostics Status (VCT_ST[1:0]) field indicates Failed it is generally due to multiple pulses being received after sending out a single pulse. In this Partially Active test the Link Partner is allowed to be in the Auto-Negotiation state but if the Link Partner is mistakenly in forced 1000BASE-T or 100BASE-TX mode the test will fail.

APPLICATION NOTE: Any signal received from a link partner will interfere with the TDR test. Energy detection should first be checked on each wire pair by using the procedure in Section 7.29.2, "Energy-Detect Power-Down Mode".

5.15 RGMII Interface

The Reduced Gigabit Media Independent Interface (RGMII) supports on-chip data-to-clock delay timing according to the RGMII Version 2.0 Specification, with programming options for external delay timing and to adjust and correct TX and RX timing paths. The RGMII also supports timing modes for the OPEN Alliance SIG TC6 - RGMII EPL (Electrical-Physical Layer) Recommendation for Automotive Application. The on-chip timing delay on the TX and RX clock paths is enabled or disabled as required by the operating mode.

RGMII provides a common interface between RGMII PHYs and MACs, and has the following key characteristics:

- Pin count is reduced from 24 pins for the IEEE Gigabit Media Independent Interface (GMII) to 12 pins for RGMII.

- 已设置为全双工 [基本控制寄存器, 位 [8] = 1]
- 链路速度设置为 1000 Mbps [基本控制寄存器, 位 [13] = 0, 位 [6] = 1]
- 已禁用自动 MDI/MDI-X [数字调试控制 1 寄存器, 位 [6] = 1]
- 主从配置设为从端 [自动协商主从控制寄存器 = 0x1200]

测试双绞线对之前, 请等待 10 毫秒。

在这种情况下, 对于四组双绞线对中的每一组, LinkMD 都会返回开路或短路及其距离, 或者返回正常。

- 正常表示电缆状况良好, 且在链路伙伴端正确端接。由于没有 TDR 反射, 此结果不提供电缆距离 (VCT_数据[7:0] 无效)。
- 开路或短路表示电缆故障或链路伙伴端端接不当。VCT_数据[7:0] 表示到电缆故障点的距离。

运行电缆诊断后, 再次执行软件复位 [基本控制寄存器, 位 [15] = 1]。

5.14.3 与部分活动链路伙伴配合使用的 LinkMD

在此场景中, 电缆远端连接到一个链路伙伴。链路伙伴已通电, 且处于以下状态:

- 自动协商可能已使能、已禁用或状态未知
- 自动 MDI/MDI-X 已禁用, 但我们可以根据 LinkMD 测试的需要, 有选择地配置为介质相关接口或介质相关接口交叉

运行电缆诊断前, 请执行软件复位 (基本控制寄存器, 位 [15] = 1) 。复位后, 必须进行以下配置:

- 禁用自动协商 [基本控制寄存器, 位 [12] = 0]
- 设置为全双工 [基本控制寄存器, 位 [8] = 1]
- 链路速度设置为 1000 Mbps [基本控制寄存器, 位 [13] = 0, 位 [6] = 1]
- 已禁用自动 MDI/MDI-X [数字调试控制 1 寄存器, 位 [6] = 1]
- 主从配置设为从端 [自动协商主从控制寄存器 = 0x1200]

测试线对前等待 10ms。

首先在链路伙伴处于 MDI 模式时, 对双绞线对 B、C 和 D 运行 LinkMD 测试。

随后在链路伙伴处于 MDI-X 模式时, 对双绞线对 A 运行 LinkMD 测试。

- “正常” 表示电缆状况良好, 且在链路伙伴端正确端接。由于没有 TDR 反射, 此结果不提供电缆距离 (VCT_数据[7:0] 无效)。
- “开路” 或 “短路” 表示电缆故障或链路伙伴端端接不正确。VCT_数据[7:0] 表示到电缆故障点的距离。

运行电缆诊断后, 再对基本控制寄存器执行一次软件复位, 位 [15] = 1]。

应用说明: 如果电缆诊断状态 (VCT_ST[1:0]) 字段指示失败, 通常是因为发送单个脉冲后接收到多个脉冲。在此部分激活测试中, 允许链路伙伴处于自动协商状态; 但如果链路伙伴意外处于强制 1000BASE-T 或 100BASE-TX 模式, 测试将失败。

应用说明: 从链路伙伴接收到的任何信号都会干扰时域反射测试。应先按照第 7.29.2 节 “能量检测掉电模式” 中的步骤检查每对线路上的能量检测。

5.15 RGMII 接口

简化千兆媒体独立接口 (RGMII) 支持片上数据到时钟延迟时序, 符合 RGMII 2.0 版规范, 并提供外部延迟时序的编程选项, 以及调整和校正发送和接收时序路径的选项。RGMII 还支持 OPEN 联盟 SIG TC6 - RGMII 电气物理层 (EPL) 汽车应用建议规范中的时序模式。根据工作模式的要求, 可启用或禁用发送和接收时钟路径上的片上时序延迟。

RGMII 为 RGMII PHY 和 MAC 提供通用接口, 并具有以下主要特性:

- 引脚数从 IEEE 千兆介质无关接口 (GMII) 的 24 个减少到 RGMII 的 12 个。

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- All speeds (10 Mbps, 100 Mbps, and 1000 Mbps) are supported at both half- and full-duplex.
- Data transmission and reception are independent and belong to separate signal groups.
- Transmit data and receive data are each four bits wide, a nibble.

In RGMII operation, the RGMII pins function as follows:

- The MAC sources the transmit reference clock, **TXC**, at 125 MHz for 1000 Mbps, 25 MHz for 100 Mbps, and 2.5 MHz for 10 Mbps.
- The PHY recovers and sources the receive reference clock, **RXC**, at 125 MHz for 1000 Mbps, 25 MHz for 100 Mbps, and 2.5 MHz for 10 Mbps.
- For 1000BASE-T, the transmit data, **TXD[3:0]**, is presented on both edges of **TXC**, and the received data, **RXD[3:0]**, is clocked out on both edges of the recovered 125 MHz clock, **RXC**.
- For 10BASE-T/100BASE-TX, the MAC holds **TX_CTL** low until both PHY and MAC operate at the same speed. During the speed transition, the receive clock is stretched on either a positive or negative pulse to ensure that no clock glitch is presented to the MAC.
- **TX_ER** and **RX_ER** are combined with **TX_EN** and **RX_DV**, respectively, to form **TX_CTL** and **RX_CTL**. These two RGMII control signals are valid at the falling clock edge.

On the device, after power-up or reset, the device is configured to RGMII mode if the **RGMII_EN** configuration strap is set to ‘1’. See the [Configuration Straps](#) section.

The device has the option to output a 125 MHz reference clock on the **CLK125_NDO** pin. This clock provides a lower-cost reference clock alternative for RGMII MACs that require a 125 MHz crystal or oscillator. The 125 MHz clock output is enabled after power-up or reset if the **CLK125_EN** strap-in pin is pulled high (via the clk125 Enable bit in Common Control Register).

5.15.1 ISOLATE MODE

When the Isolate (PHY_ISO) bit in the Basic Control Register is set, the device will three-state the RGMII output pins and ignore the RGMII input pins. The optional **CLK125_NDO** output pin is an exception.

5.15.2 RGMII SIGNAL DEFINITION

[Table 5-5](#) describes the RGMII signals. Refer to the RGMII Version 2.0 Specification for more detailed information.

TABLE 5-5: RGMII SIGNAL DEFINITION

RGMII Signal Name	Pin Type (with respect to PHY)	Pin Type (with respect to MAC)	Description
TXC	Input	Output	Transmit Reference Clock (125MHz for 1000Mbps, 25MHz for 100Mbps, 2.5MHz for 10Mbps)
TX_CTL	Input	Output	Transmit Control
TXD[3:0]	Input	Output	Transmit Data[3:0]
RXC	Output	Input	Receive Reference Clock (125 MHz for 1000 Mbps, 25 MHz for 100 Mbps, 2.5 MHz for 10 Mbps)
RX_CTL	Output	Input	Receive Control
RXD[3:0]	Output	Input	Receive Data[3:0]

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- 半双工和全双工均支持所有速率（10 Mbps、100 Mbps 和 1000 Mbps）。
- 数据传输和接收彼此独立，属于不同的信号组。
- 发送数据和接收数据的宽度均为 4 位，即一个半字节。

在 RGMII 工作模式下，RGMII 引脚的功能如下：

- MAC 在 1000 Mbps 下以 125 MHz、100 Mbps 下以 25 MHz、10 Mbps 下以 2.5 MHz 的频率提供发送参考时钟 **TXC**。
- PHY 恢复并提供接收参考时钟 **RXC**，在 1000 Mbps 下为 125 MHz，在 100 Mbps 下为 25 MHz，在 10 Mbps 下为 2.5 MHz。
- 对于 1000BASE-T，发送数据 **TXD[3: 0]**，在 **TXC** 的两个边沿上呈现；接收数据 **RXD[3:0]**，则在恢复的 125 MHz 时钟 **RXC** 的两个边沿上输出。
- 对于 10BASE-T/100BASE-TX，在 PHY 和 MAC 以相同速率运行之前，MAC 会将 **TX_控制** 保持为低电平。在速率转换期间，接收时钟会在正脉冲或负脉冲处延长，以确保不会向 MAC 提供时钟毛刺。

- **发送_错误**和**接收_错误**分别与**发送_使能**和**接收_DV**组合，形成**发送_控制**和**接收_控制**。这两个 RGMII 控制信号在时钟下降沿有效。
- 器件上电或复位后，如果将**RGMIL_使能**配置跳线设为“1”，器件将配置为 RGMII 模式。请参阅[配置绑带](#)一节。

器件可选择在**时钟125_NDO**引脚上输出 125 MHz 参考时钟。对于需要 125 MHz 晶体或振荡器的 RGMII MAC，此时钟可作为成本更低的参考时钟替代方案。如果**时钟125_使能**绑带引脚被拉高（通过通用控制寄存器中的 clk125 使能位），则 125 MHz 时钟输出会在上电或复位后启用。

5.15.1 隔离模式

设置基本控制寄存器中的隔离（PHY_ISO）位后，器件会将 RGMII 输出引脚置为三态，并忽略 RGMII 输入引脚。可选的**时钟125_NDO**输出引脚是个例外。

5.15.2 RGMII 信号定义

[表 5-5](#)介绍了 RGMII 信号。有关详细信息，请参阅 RGMII 2.0 版规范。

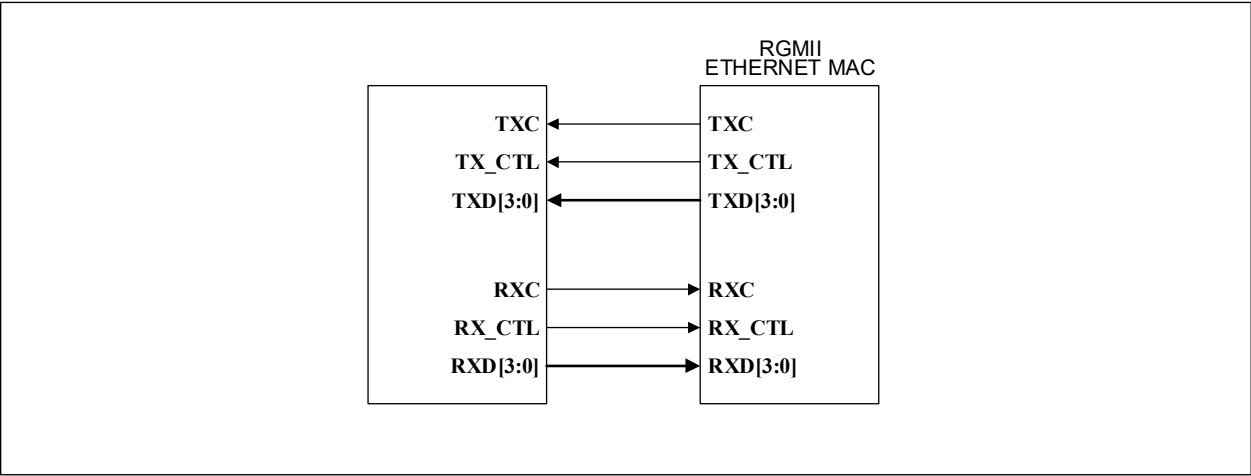
表 5-5: RGMII 信号定义

RGMII 信号名称	引脚类型（相对于 PHY）	引脚类型（相对于媒体访问控制器）	说明
TXC	输入	输出	发送参考时钟（1000 Mbps 时为 125 兆赫，100 Mbps 时为 25 兆赫，10 Mbps 时为 2.5 兆赫）
TX_CTL	输入	输出	发送控制
发送数据[3:0]	输入	输出	发送数据[3:0]
RXC	输出	输入	接收参考时钟（1000 兆比特/秒时为 125 兆赫，100 兆比特每秒时为 25 兆赫，10 兆比特每秒时为 2.5 兆赫）
RX_CTL	输出	输入	接收控制
接收数据[3:0]	输出	输入	接收数据[3:0]

5.15.3 RGMII SIGNAL DIAGRAM

The RGMII pin connections to the MAC are shown in [Figure 5-12](#):

FIGURE 5-12: RGMII INTERFACE



5.15.4 RGMII TIMING

5.15.4.1 RGMII ID Timing / TC6 Delay on Source (DoS) mode

As the default, after power-up or reset, the RGMII timing conforms to the timing requirements in the RGMII Version 2.0 Specification for Internal Delay mode as well as the timing requirements for the TC6 Delay on Source (DoS) mode.

For the transmit path (MAC to PHY), the device does not add any delay locally at its **TXC**, **TX_CTL** and **TXD[3:0]** input pins, and expects the **TXC** delay to be provided on-chip by the MAC.

APPLICATION NOTE: If MAC does not provide any delay or insufficient delay for the **TXC**, the device has two options. The device may add a fixed 2ns, DLL based delay to the **TXC** input. In addition to this fixed delay, pad skew registers can provide up to a 5.8 ns on-chip delay (~4.5 ns additional can be added to **TXC**, ~1.3 ns can be subtracted from **TX_CTL** and **TXD[3:0]**).

For the receive path (PHY to MAC), the device adds a fixed 2ns, DLL based delay to the **RXC** output pin with respect to **RX_CTL** and **RXD[3:0]** output pins.

APPLICATION NOTE: If necessary, the device has pad skew registers that can adjust the **RXC** on-chip delay by up to an additional 5.8 ns (~4.5 ns additional can be added to **RXC**, ~1.3 ns can be subtracted from **RX_CTL** and **TXD[3:0]**).

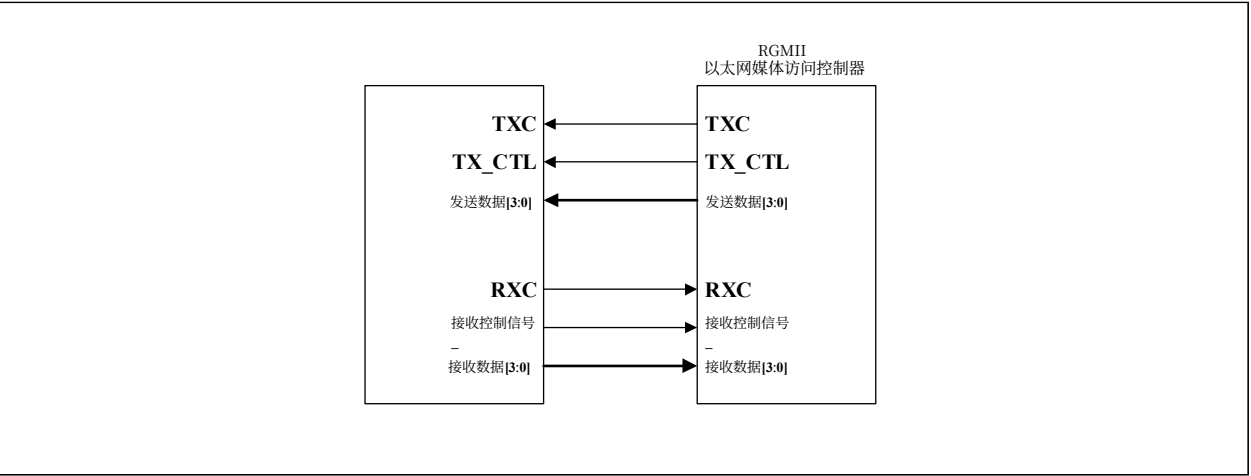
The above default RGMII timings imply:

- **RXC** clock skew is set by the default register settings.
- **TXC** clock skew is provided by the MAC.
- No PCB delay is required for **TXC** and **RXC** clocks.

5.15.3 RGMII 信号图

RGMII 与媒体访问控制器的引脚连接如图 5-12 所示:

图 5-12: RGMII 接口



5.15.4 RGMII 时序

5.15.4.1 RGMII 内部延迟时序 / TC6 源端延迟（DoS）模式

默认情况下，上电或复位后，RGMII 时序符合 RGMII 2.0 版规范中内部延迟模式的时序要求，以及 TC6 源端延迟（DoS）模式的时序要求。

对于发送路径（MAC 到 PHY），器件不会在其 **TXC**、**发送_控制** 和 **发送数据[3: 0]** 输入引脚处添加任何本地延迟，并期望由 MAC 在片上提供 **TXC** 延迟。

应用说明：如果 MAC 未为 **TXC** 提供延迟，或提供的延迟不足，器件有两种选择。器件可以在 **TXC** 输入端添加固定的 2 ns 基于延迟锁定环的延迟。除了此固定延迟外，焊盘偏斜寄存器还可提供高达 5.8 ns 的片上延迟（可为 **TXC** 额外增加 ~4.5 ns，可从 **发送_控制** 和 **发送数据[3: 0]**中减去 ~1.3 ns）。

对于接收路径（PHY 到媒体访问控制器），器件会在 **接收时钟**输出引脚上相对于 **接收_控制** 和 **接收数据[3:0]** 输出引脚增加固定的 2 ns 基于延迟锁定环的延迟。

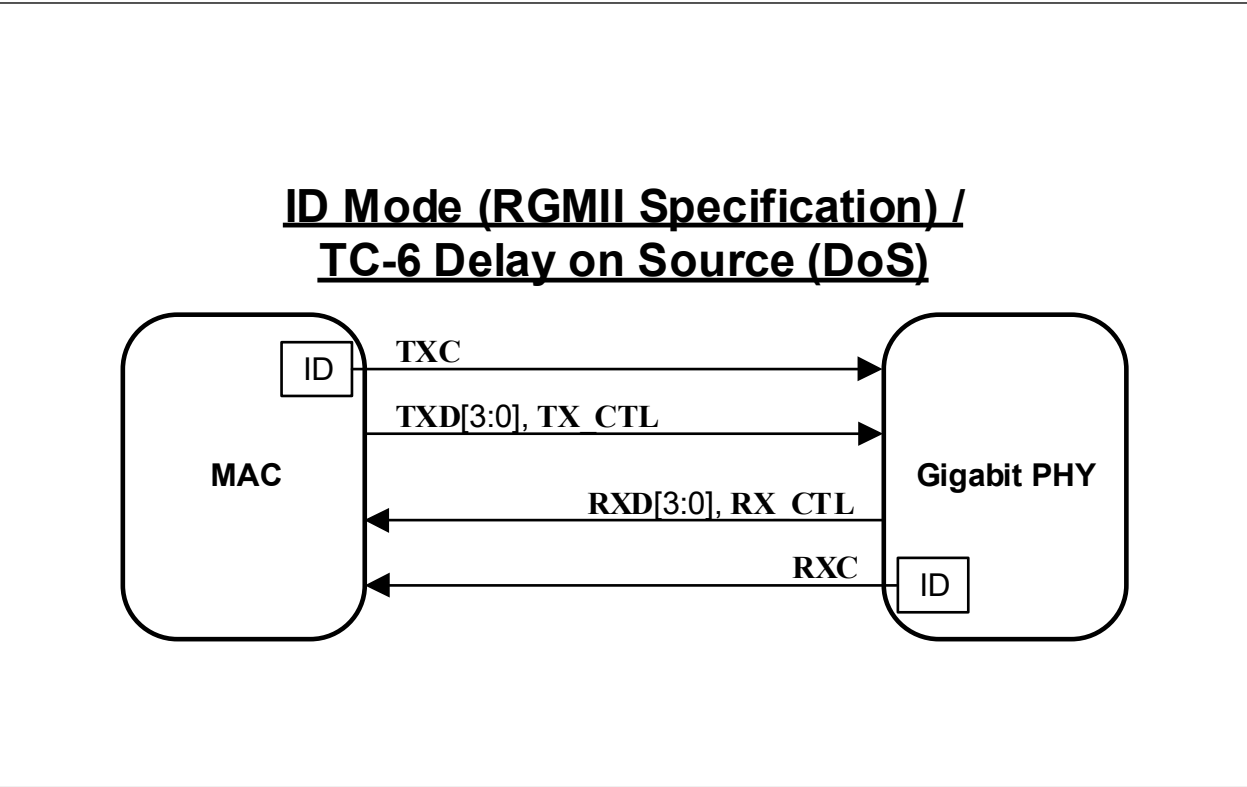
应用说明：如有需要，器件配有焊盘偏斜寄存器，可将 **接收时钟**片上延迟额外调整最多 5.8 ns（可为 **接收时钟**额外增加 ~4.5 ns，可从 **接收_控制** 和 **发送数据[3:0]**中减去 ~1.3 ns）。

上述默认 RGMII 时序意味着：

- **接收时钟**时钟偏斜由默认寄存器设置确定。
- **TXC**时钟偏斜由媒体访问控制器提供。
- **TXC**和 **接收时钟**时钟无需 PCB 延迟。

The following figure illustrates the RGMII Version 2.0 Specification Internal Delay / TC6 Delay on Source (DoS) operation.

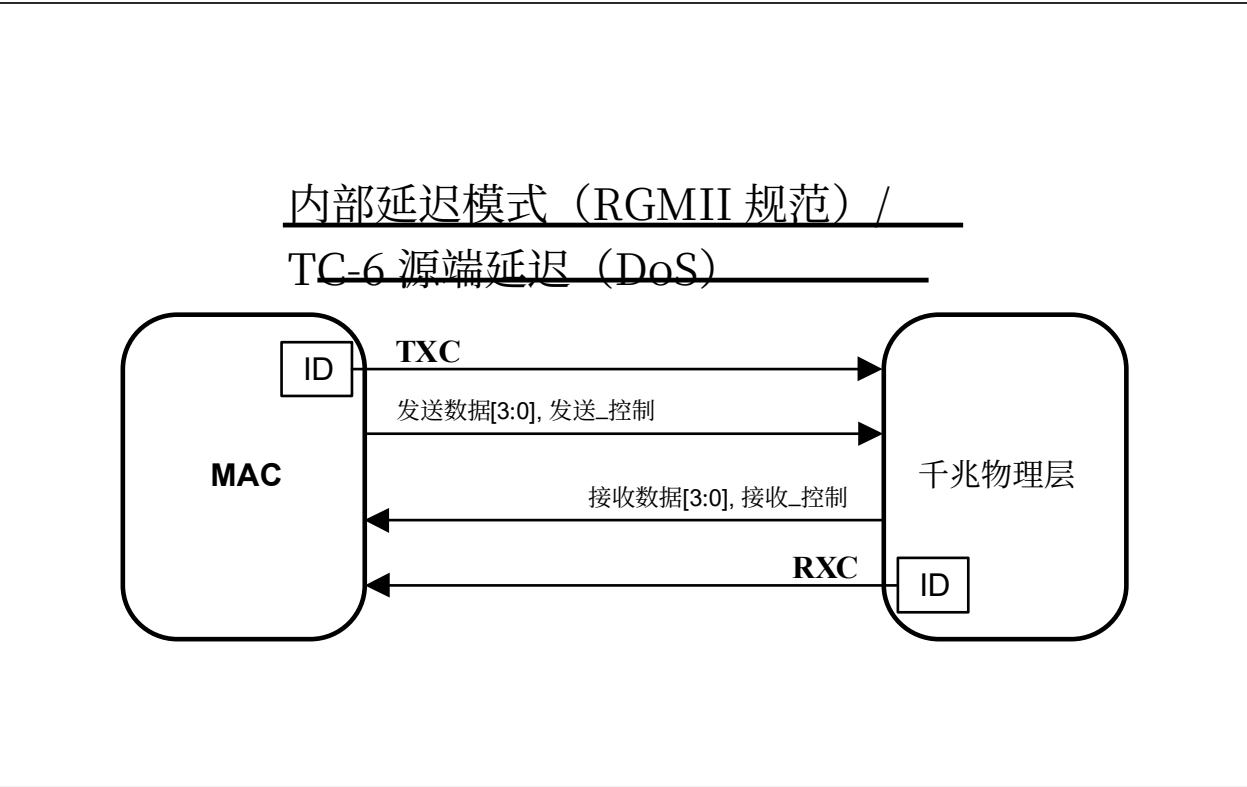
FIGURE 5-13: RGMII VERSION 2.0 INTERNAL DELAY / TC6 DELAY ON SOURCE (DoS) OPERATION



When computing the RGMII timing relationships, delays along the entire data path must be aggregated to determine the total delay to be used for comparison between RGMII pins within their respective timing group. For the transmit data path, total delay includes MAC output delay, MAC-to-PHY PCB routing delay, and PHY input delay and skew setting (if any). For the receive data path, the total delay includes PHY output delay, PHY-to-MAC PCB routing delay, and MAC input delay and skew setting (if any).

下图说明了 RGMII 2.0 版规范中的内部延迟 / TC6 源端延迟（DoS）操作。

图 5-13: RGMII 2.0 版内部延迟/TC6 源端延迟（DoS）操作



计算 RGMII 时序关系时，必须汇总整个数据路径上的延迟，以确定用于比较各自时序组内 RGMII 引脚的总延迟。对于发送数据路径，总延迟包括媒体访问控制器输出延迟、MAC 到 PHY 的 PCB 布线延迟、PHY 输入延迟以及偏斜设置（如果有）。对于接收数据路径，总延迟包括 PHY 输出延迟、PHY 到 MAC 的 PCB 布线延迟、媒体访问控制器输入延迟以及偏斜设置（如果有）。

Figure 5-14, Figure 5-15 and Table 5-6 from the RGMII v2.0 Specification are provided as references to understanding RGMII v1.3 external delay and RGMII v2.0 on-chip delay timing operation.

FIGURE 5-14: RGMII V2.0 SPEC (MULTIPLEXING AND TIMING DIAGRAM – ORIGINAL RGMII (V1.3) WITH EXTERNAL DELAY)

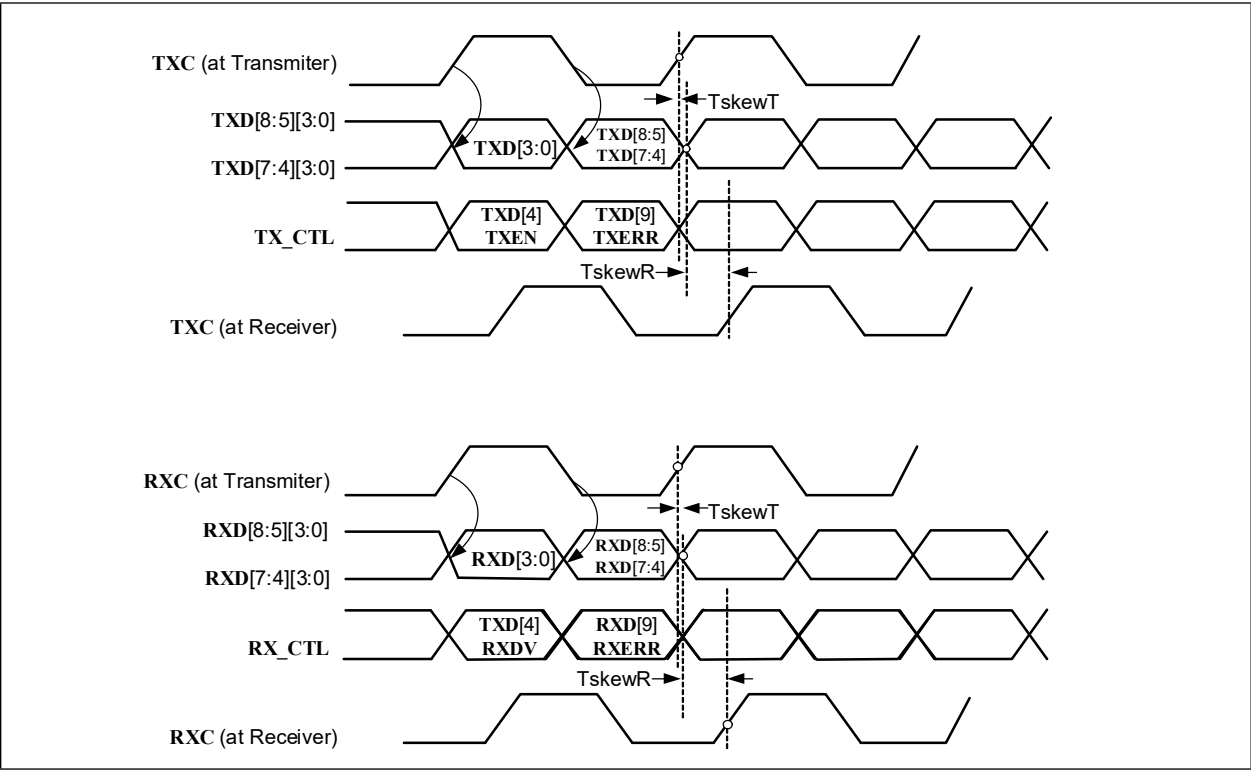


图 5-14、图 5-15 和 表 5-6 取自 RGMII v2.0 规范，可作为理解 RGMII v1.3 外部延迟和 RGMII v2.0 片上延迟时序操作的参考。

图 5-14: RGMII V2.0 规范（复用与时序图 – 原始 RGMII (v1.3)，带外部延迟）

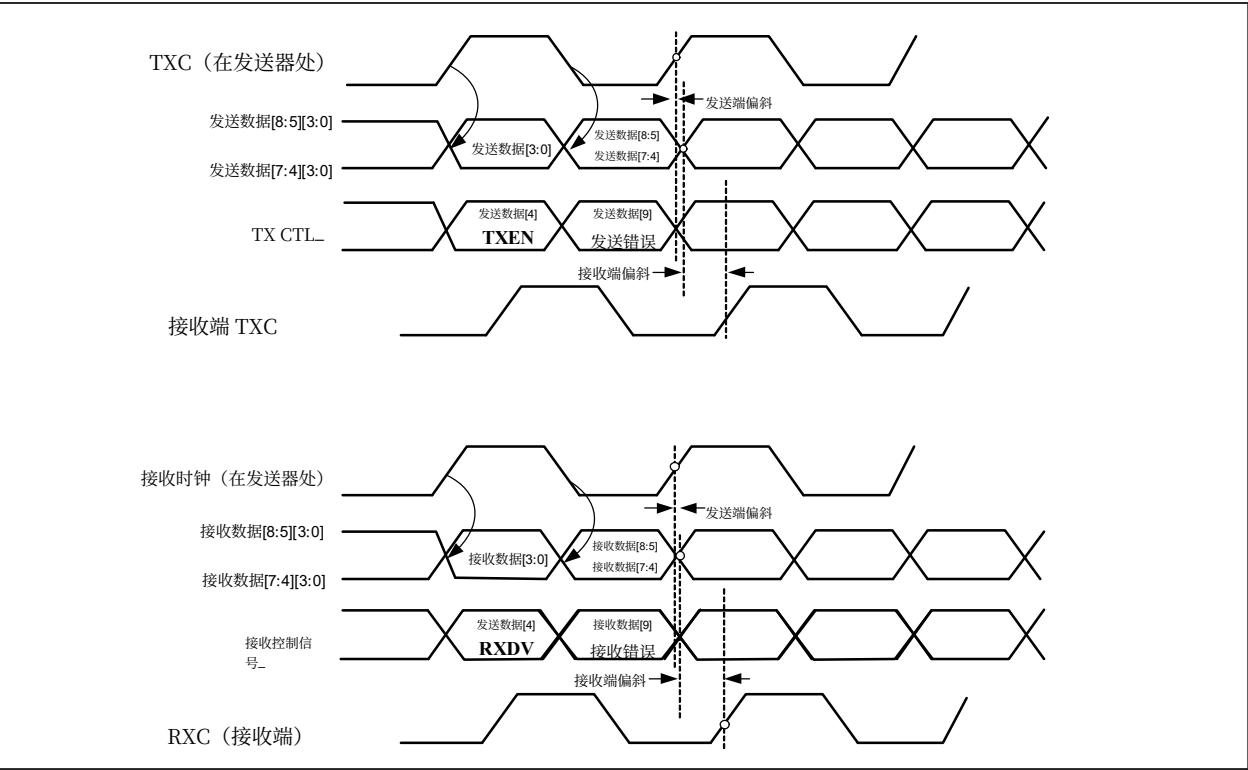
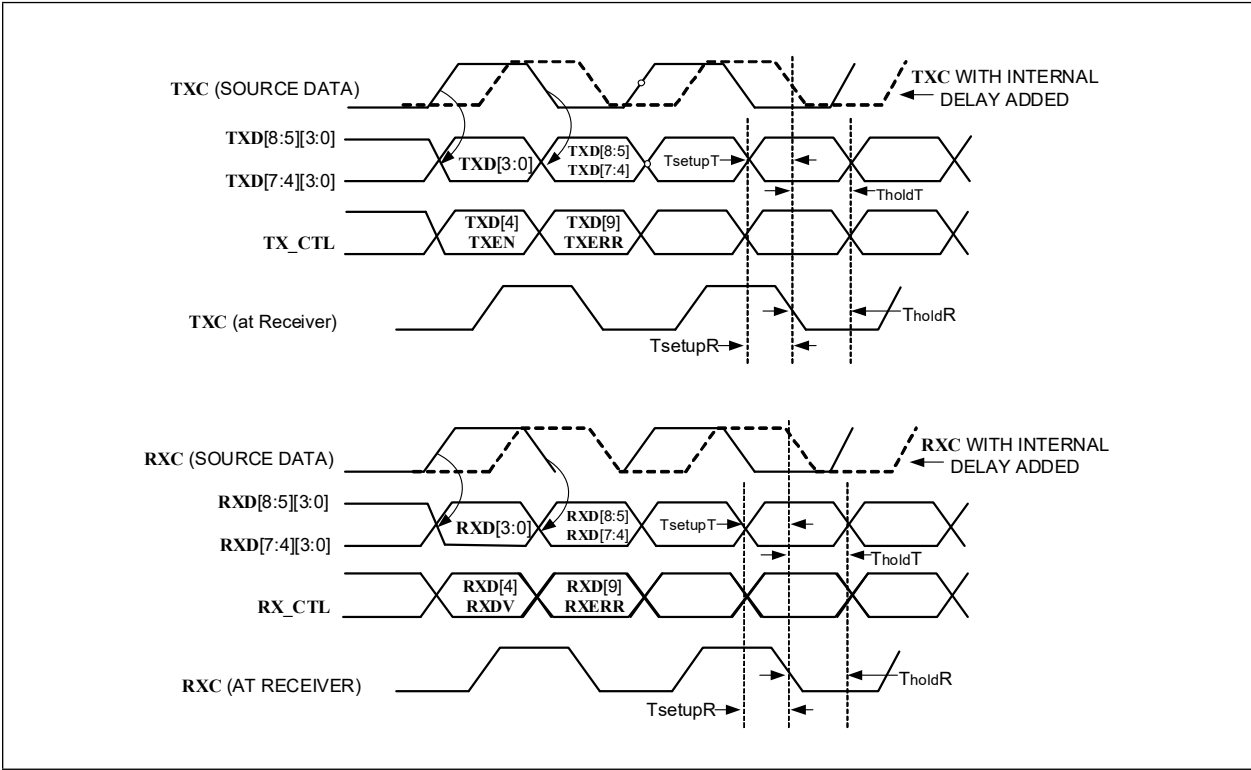


FIGURE 5-15: RGMII V2.0 SPEC (MULTIPLEXING AND TIMING DIAGRAM – RGMII-ID (V2.0) WITH INTERNAL CHIP DELAY)



The following notes provide clarification for [Figure 5-15](#):

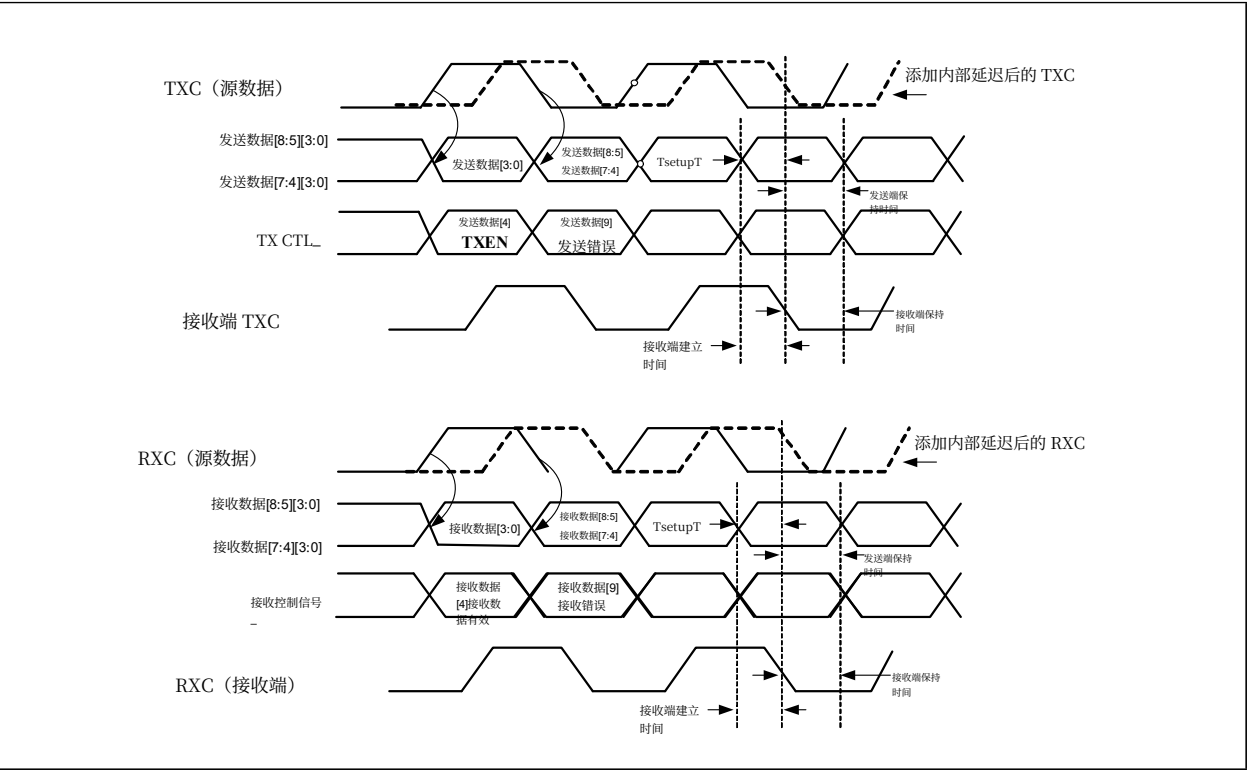
TXC (SOURCE DATA), solid line, is the MAC GTX_CLK clock output timing per RGMII v1.3 Specification (PCB delay line required or PHY internal delay required)

TXC (SOURCE DATA) WITH INTERNAL DELAY ADDED, dotted line, is the MAC GTX_CLK clock output timing per RGMII v2.0 Specification (no PCB delay required and no PHY internal delay required)

RXC (SOURCE DATA), solid line, is the PHY RX_CLK clock output timing per RGMII v1.3 Specification (PCB delay line required or MAC internal delay required)

RXC (SOURCE DATA) WITH INTERNAL DELAY ADDED, dotted line, is the PHY RX_CLK clock output timing per RGMII v2.0 Specification (no PCB delay required and no MAC internal delay required)

图 5-15: RGMII V2.0 规范 (复用与时序图 – RGMII-ID (V2.0) (带芯片内部延迟))



以下注释对 [图 5-15](#) 进行了说明：

TXC（源数据） 实线表示 MAC GTX_时钟时钟输出时序符合 RGMII v1.3 规范（需要 PCB 延迟线或 PHY 内部延迟）

添加内部延迟的 TXC（源数据） 虚线表示 MAC GTX_时钟时钟输出时序符合 RGMII v2.0 规范（不需要 PCB 延迟，也不需要 PHY 内部延迟）

RXC（源数据） 实线表示 PHY 接收_时钟时钟输出时序符合 RGMII v1.3 规范（需要 PCB 延迟线或 MAC 内部延迟）

添加内部延迟的 RXC（源数据），虚线表示符合 RGMII v2.0 规范的 PHY 接收_时钟时钟输出时序（无需 PCB 延迟，也无需 MAC 内部延迟）

TABLE 5-6: RGMII V2.0 SPECIFICATION

Parameter	Description	Min.	Typ.	Max.	Units
T _{skew} T	Data-to-clock output skew (at transmitter) per RGMII v1.3 (external delay)	-500	—	500	ps
T _{skew} R	Data-to-clock input skew (at receiver) per RGMII v1.3 (external delay)	1.0	—	2.6	ns
T _{setup} T	Data-to-clock output setup (at transmitter – integrated delay)	1.2	2.0	—	ns
T _{hold} T	Clock-to-data output hold (at transmitter – integrated delay)	1.2	2.0	—	ns
T _{setup} R	Data-to-clock input setup (at receiver – integrated delay)	1.0	2.0	—	ns
T _{hold} R	Clock-to-data input hold (at receiver – integrated delay)	1.0	2.0	—	ns
t _{cyc} (1000BASE-T)	Clock cycle duration for 1000BASE-T	7.2	8.0	8.8	ns
t _{cyc} (100BASE-TX)	Clock cycle duration for 100BASE-TX	36	40	44	ns
t _{cyc} (10BASE-T)	Clock cycle duration for 10BASE-T	360	400	440	ns

The RGMII Version 2.0 Specification defines the RGMII data-to-clock skews only for 1000 Mbps operation, which uses both clock edges for sampling the data and control signals at the 125 MHz clock frequency (8 ns period). For 10/100 Mbps operations, the data signals are sampled on the rising clock edge and the control signals are sampled on both clock edges. With slower clock frequencies, 2.5 MHz (400 ns period) for 10 Mbps and 25 MHz (40 ns period) for 100 Mbps, the RGMII data-to-clock skews for 10/100 Mbps operations will have greater timing margins than for 1000 Mbps operation, and therefore can be relaxed from 2.6 ns (maximum) for 1000 Mbps to 160 ns (maximum) for 10 Mbps and 16 ns (maximum) for 100 Mbps.

5.15.4.2 TC6 Delay on Destination (DoD) mode

In order to support the Delay on Destination (DoD) mode specified by the OPEN Alliance SIG TC6 - RGMII EPL (Electrical-Physical Layer) Recommendation for Automotive Application, the internal delay on the **RXC** is disabled and the internal delay on the **TXC** is enabled. This results in data and control, at the device pins, nominally centered about the clock edges. A bounded skew allows the data and control to occur before or after the clock edges.

For the transmit path (MAC to PHY), the device adds a fixed 2ns, DLL based delay locally at its **TXC** input with respect to **TX_CTL** and **TXD[3:0]** input pins. This effectively provides positive setup and hold timings to the internal flip flops.

For the receive path (PHY to MAC), the device does not any additional delay locally to the **RXC** output pin. The MAC will provide a delay on its input clock pin.

The above default RGMII timings imply:

- RXC** clock skew is provided by the MAC.
- TXC** clock skew is set by the default register settings.
- No PCB delay is required for **TXC** and **RXC** clocks since both devices add internal delay on their input clocks.

表 5-6：RGMII V2.0 规范

参数	说明	Min.	Typ.	Max.	单位
T _{skew} T	根据 RGMII v1.3（外部延迟）的数据到时钟输出偏斜（发射器处）	-500	—	500	ps
接收端偏斜	根据 RGMII v1.3（外部延迟）的数据到时钟输入偏斜（接收器处）	1.0	—	2.6	ns
T _{setup} T	数据至时钟输出建立时间（在发射器处——集成延迟）	1.2	2.0	—	ns
T _{hold} T	时钟至数据输出保持时间（在发射器处——集成延迟）	1.2	2.0	—	ns
T _{setup} R	数据至时钟输入建立时间（在接收器处——集成延迟）	1.0	2.0	—	ns
T _{hold} R	时钟到数据输入保持时间（接收器处 – 集成延迟）	1.0	2.0	—	ns
t _{cyc} (1000BASE-T)	1000BASE-T 的时钟周期时长	7.2	8.0	8.8	ns
t _{cyc} (100BASE-TX)	100BASE-TX 的时钟周期时长	36	40	44	ns
t _{cyc} (10BASE-T)	10BASE-T 的时钟周期时长	360	400	440	ns

RGMII 2.0 版规范仅针对 1000 Mbps 运行定义了 RGMII 的数据到时钟偏斜。在该运行模式下，使用两个时钟边沿，以 125 MHz 时钟频率（周期为 8 ns）对数据和控制信号进行采样。对于 10/100 Mbps 运行，数据信号在时钟上升沿采样，控制信号则在两个时钟边沿采样。由于时钟频率较低，10 Mbps 时为 2.5 MHz（周期为 400 ns），100 Mbps 时为 25 MHz（周期为 40 ns），因此，10/100 Mbps 运行的 RGMII 数据到时钟偏斜比 1000 Mbps 运行具有更大的时序裕量。因此，偏斜上限可从 1000 Mbps 运行时的 2.6 ns，放宽至 10 Mbps 时的 160 ns 和 100 Mbps 时的 16 ns。

5.15.4.2 TC6 目的端延迟（DoD）模式

为了支持《OPEN 联盟 SIG TC6 - RGMII 电气物理层（Electrical-Physical Layer）汽车应用建议》中指定的目的端延迟（DoD）模式，**RXC**上的内部延迟被禁用，而**TXC**上的内部延迟被启用。因此，器件引脚处的数据和控制信号通常以时钟边沿为中心。有界偏斜允许数据和控制信号出现在时钟边沿之前或之后。

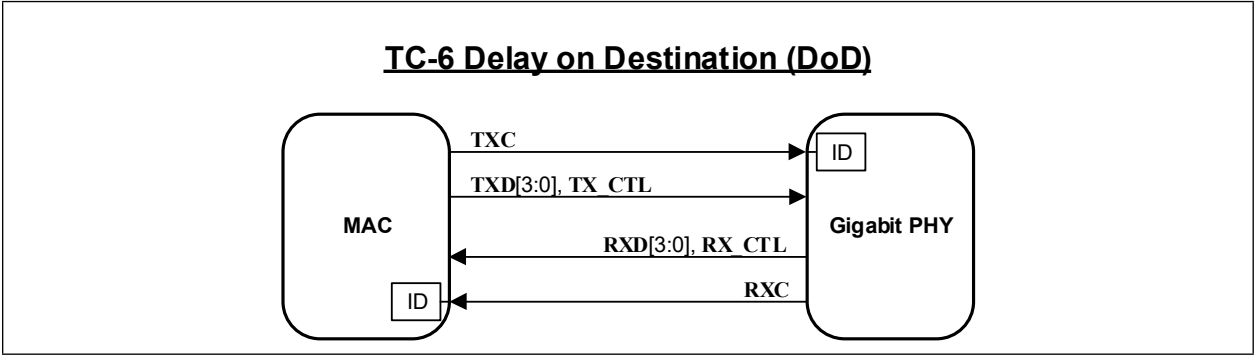
对于发送路径（MAC 到 PHY），器件会在**TXC**输入引脚处，相对于**TX_CTL**和**TXD[3:0]** 输入引脚，局部增加固定的 2 ns 基于延迟锁定环的延迟。这实际上为内部触发器提供了正的建立和保持时序。对于接收路径（PHY 到 MAC），器件不会在**RXC**输出引脚处局部增加任何额外延迟。MAC 将在其输入时钟引脚处提供延迟。

上述默认 RGMII 时序意味着：

- 接收时钟**的时钟偏斜由媒体访问控制器提供。
- TXC**的时钟偏斜由默认寄存器设置设定。
- TXC**和**接收时钟**均无需 PCB 延迟，因为两个器件都会在其输入时钟上增加内部延迟。

The following figure illustrates the TC6 Delay on Destination (DoD) operation.

FIGURE 5-16: TC6 DELAY ON DESTINATION (DoD) OPERATION



5.15.4.3 RGMII DLL BASED CLOCK SKEW

The delays on **RXC** (enabled by default) and optionally on **TXC** (bypassed by default) are provided by internal DLLs. When the RGMII delay is bypassed the delay chain is completely omitted and the DLL output clock is the same as the DLL input clock (except for one multiplexer delay). DLL bypass is controlled by the bypass rxdll and bypass txdll bits of the RX DLL Control Register and the TX DLL Control Register for **RXC** and **TXC** respectively.

When DLL tuning is enabled (normal operation) the DLL is dynamically tuned. The txdll_tap_adj and rxdll_tap_adj fields in the TX DLL Control Register and the RX DLL Control Register are used to statically account for the output multiplexer stage in the delay chain. The larger the value the less of a delay there is on the clock. The DLL tap select value is dynamically calculated and adjusts the delay chain. The txdll_tap_sel and rxdll_tap_sel fields provide the default DLL tap select values before the first tuning cycle occurs.

When DLL tuning is disabled, by setting the rxdll_tune_disable or txdll_tune_disable bits in the RX DLL Control Register or TX DLL Control Register respectively, the DLL is not dynamically tuned but is still used to provide a fixed delay. The txdll_tap_sel and rxdll_tap_sel settings are used as fixed delay values. The txdll_tap_adj and rxdll_tap_adj fields are not used.

The DLL must be reset whenever a new value is programmed in txdll_tap_sel or rxdll_tap_sel fields. This is accomplished by the respective rxdll_reset or txdll_reset bit in the RX / TX DLL Reset and BIST Control Register.

Note: These bits are not self-clearing and must be set then reset by software.

5.15.4.4 RGMII Pad Skew Registers

It is common to implement RGMII PHY-to-MAC designs that either PHY, MAC, or both PHY and MAC are not fully RGMII v2.0 compliant with on-chip clock delay. These combinations of mixed RGMII v1.3/v2.0 designs and plus sometimes non-matching RGMII PCB trace routings require a review of the entire RGMII system timings (PHY on-chip, PCB trace delay, MAC on-chip) to compute the aggregate clock delay and determine if the clock delay timing is met.

If timing adjustment is needed, pad skew registers are available for all RGMII pins (clocks, control signals, and data bits) to provide programming options to adjust or correct the timing relationship for each RGMII pin. Because RGMII is a source-synchronous bus interface, the timing relationship needs to be maintained only within the RGMII pin's respective timing group.

Clock Invert and Control Signal Pad Skew Register: **RX_CTL, TX_CTL**

RGMII RX Data Pad Skew Register: **RXD3, RXD2, RXD1, RXD0**

RGMII TX Data Pad Skew Register: **TXD3, TXD2, TXD1, TXD0**

Clock Pad Skew Register: **TXC, RXC**

The RGMII control signals and data bits have 4-bit skew settings, while the RGMII clocks have 5-bit skew settings.

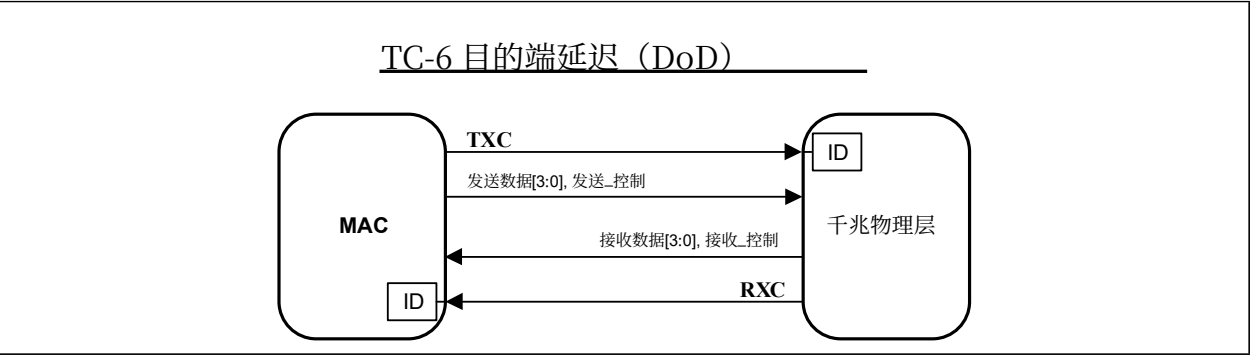
Each register value increment is approximately:

Data and control: 56ps (min) to 187ps (max)

RXC and **TXC**: 56ps (min) to 187ps (max)

下图展示了 TC6 目的端延迟（DoD）操作。

图 5-16: TC6 目的端延迟（DoD）操作



5.15.4.3 基于 RGMII DLL 的时钟偏斜

接收时钟（默认使能）以及可选的 **TXC**（默认旁路）的延迟由内部延迟锁定环提供。旁路 RGMII 延迟时，会完全省略延迟链，延迟锁定环输出时钟与延迟锁定环输入时钟相同（多路复用器延迟除外）。对于 **接收时钟** 和 **TXC**，延迟锁定环旁路分别由接收 DLL 控制寄存器和发送 DLL 控制寄存器中的 bypass rxdll 和 bypass txdll 位控制。

启用延迟锁定环调整时（正常运行），延迟锁定环会动态调整。发送 DLL 控制寄存器和接收 DLL 控制寄存器中的 txdll_tap_adj 和 rxdll_tap_adj 字段用于静态补偿延迟链中的输出多路复用器级。值越大，时钟延迟越小。DLL 抽头选择值会动态计算并调整延迟链。在第一个调整周期发生之前，txdll_tap_sel 和 rxdll_tap_sel 字段提供默认的 DLL 抽头选择值。

禁用 DLL 调整时，可分别通过设置接收 DLL 控制寄存器或发送 DLL 控制寄存器中的 rxdll_tune_disable 位或 txdll_tune_disable 位来实现。此时 DLL 不会动态调整，但仍用于提供固定延迟。txdll_tap_sel 和 rxdll_tap_sel 设置用作固定延迟值。txdll_tap_adj 和 rxdll_tap_adj 字段不使用。

每当在 txdll_tap_sel 或 rxdll_tap_sel 字段中写入新值时，都必须复位 DLL。该操作通过 RX/TX DLL 复位和 BIST 控制寄存器中相应的 rxdll_reset 或 txdll_reset 位完成。

注意： 这些位不会自动清零，必须由软件先置位再复位。

5.15.4.4 RGMII 焊盘偏斜寄存器

在 RGMII PHY 到 MAC 设计中，PHY、MAC 或两者都不完全符合带片上时钟延迟的 RGMII v2.0 规范，这种情况很常见。对于这些混合使用 RGMII v1.3/v2.0 设计、且有时 PCB 走线布线不匹配的组合，需要检查整个 RGMII 系统时序（PHY 片上延迟、PCB 走线延迟、MAC 片上延迟），以计算总时钟延迟并确定是否满足时钟延迟时序要求。

如需调整时序，可使用适用于所有 RGMII 引脚（时钟、控制信号和数据位）的焊盘偏斜寄存器，通过编程选项调整或校正每个 RGMII 引脚的时序关系。由于 RGMII 是源同步总线接口，因此只需在各 RGMII 引脚所属的时序组内保持时序关系。

时钟反相和控制信号焊盘偏斜寄存器：**接收_控制、发送_控制**

RGMII 接收数据焊盘偏斜寄存器：**RXD3、RXD2、RXD1、RXD0**

RGMII 发送数据焊盘偏斜寄存器：**TXD3、TXD2、TXD1、TXD0**

时钟焊盘偏斜寄存器：**TXC、接收时钟**

RGMII 控制信号和数据位采用 4 位偏斜设置，而 RGMII 时钟采用 5 位偏斜设置。

每个寄存器值增量大约为：

数据和控制：56ps（最小值）至 187ps（最大值）

接收时钟 和 **TXC**：56ps（最小值）至 187ps（最大值）

A lower value decreases the delay, while a higher value increases the delay.
Table 5-7 and Table 5-8 list the approximate delay for each pad skew (value) setting.

TABLE 5-7: ABSOLUTE DELAY FOR 5-BIT PAD SKEW SETTING

Pad Skew Value	Delay (ns)
0_0000	-0.39(min) / -1.31(max)
0_0001	-0.34(min) / -1.12(max)
0_0010	-0.28(min) / -0.94(max)
0_0011	-0.22(min) / -0.75(max)
0_0100	-0.17(min) / -0.56(max)
0_0101	-0.11(min) / -0.37(max)
0_0110	-0.06(min) / -0.19(max)
0_0111	No delay adjustment (default value)
0_1000	0.06(min) / 0.19(max)
0_1001	0.11(min) / 0.37(max)
0_1010	0.17(min) / 0.56(max)
0_1011	0.22(min) / 0.75(max)
0_1100	0.28(min) / 0.94(max)
0_1101	0.34(min) / 1.12(max)
0_1110	0.39(min) / 1.31(max)
0_1111	0.45(min) / 1.50(max)
1_0000	0.50(min) / 1.68(max)
1_0001	0.56(min) / 1.87(max)
1_0010	0.62(min) / 2.06(max)
1_0011	0.67(min) / 2.24(max)
1_0100	0.73(min) / 2.43(max)
1_0101	0.78(min) / 2.62(max)
1_0110	0.84(min) / 2.81(max)
1_0111	0.90(min) / 2.99(max)
1_1000	0.95(min) / 3.18(max)
1_1001	1.01(min) / 3.37(max)
1_1010	1.06(min) / 3.55(max)
1_1011	1.12(min) / 3.74(max)
1_1100	1.18(min) / 3.93(max)
1_1101	1.23(min) / 4.11(max)
1_1110	1.29(min) / 4.30(max)
1_1111	1.34(min) / 4.49(max)

较小的值会缩短延迟，较大的值则会增加延迟。表 5-7 和 表 5-8列出了每种焊盘偏斜（值）设置对应的大致延迟。

表 5-7：5 位焊盘偏斜设置的绝对延迟

焊盘偏斜值	延迟（ns）
0 0000_	-0.39（最小值） / -1.31（最大值）
0 0001_	-0.34（最小值） / -1.12（最大值）
0 0010_	-0.28（最小值） / -0.94（最大值）
0 0011_	-0.22（最小值） / -0.75（最大值）
0 0100_	-0.17（最小值） / -0.56（最大值）
0 0101_	-0.11（最小值） / -0.37（最大值）
0 0110_	-0.06（最小值） / -0.19（最大值）
0 0111_	无延迟调整（默认值）
0 1000_	0.06（最小值） / 0.19（最大值）
0 1001_	0.11（最小值） / 0.37（最大值）
0 1010_	0.17（最小值） / 0.56（最大值）
0 1011_	0.22（最小值） / 0.75（最大值）
0 1100_	0.28（最小值） / 0.94（最大值）
0 1101_	0.34（最小值） / 1.12（最大值）
0 1110_	0.39（最小值） / 1.31（最大值）
0 1111_	0.45（最小值） / 1.50（最大值）
1 0000_	0.50（最小值） / 1.68（最大值）
1 0001_	0.56（最小值） / 1.87（最大值）
1 0010_	0.62（最小值） / 2.06（最大值）
1 0011_	0.67（最小值） / 2.24（最大值）
1 0100_	0.73（最小值） / 2.43（最大值）
1 0101_	0.78（最小值） / 2.62（最大值）
1 0110_	0.84（最小值） / 2.81（最大值）
1 0111_	0.90（最小值） / 2.99（最大值）
1 1000_	0.95（最小值） / 3.18（最大值）
1 1001_	1.01（最小值） / 3.37（最大值）
1 1010_	1.06（最小值） / 3.55（最大值）
1 1011_	1.12（最小值） / 3.74（最大值）
1 1100_	1.18（最小值） / 3.93（最大值）
1 1101_	1.23（最小值） / 4.11（最大值）
1 1110_	1.29（最小值） / 4.30（最大值）
1 1111_	1.34（最小值） / 4.49（最大值）

TABLE 5-8: ABSOLUTE DELAY FOR 4-BIT PAD SKEW SETTING

Pad Skew Value	Delay (ns) (Si B0) (est.)
0000	-0.39(min) / -1.31(max)
0001	-0.34(min) / -1.12(max)
0010	-0.28(min) / -0.94(max)
0011	-0.22(min) / -0.75(max)
0100	-0.17(min) / -0.56(max)
0101	-0.11(min) / -0.37(max)
0110	-0.06(min) / -0.19(max)
0111	No delay adjustment (default value)
1000	0.06(min) / 0.19(max)
1001	0.11(min) / 0.37(max)
1010	0.17(min) / 0.56(max)
1011	0.22(min) / 0.75(max)
1100	0.28(min) / 0.94(max)
1101	0.34(min) / 1.12(max)
1110	0.39(min) / 1.31(max)
1111	0.45(min) / 1.50(max)

The following examples show how to read/write to the Clock Pad Skew Register (MMD Address 2h, Register 8h) for the RGMII **TXC** and **RXC** skew settings. MMD register access is through the MMD Access Control Register and MMD Access Address/Data Register. For more programming details, refer to the MMD Registers section.

- Read back value of MMD Address 2h, Register 8h.
 - Write MMD Access Control Register = 0x0002 // Select MMD Device Address 2h
 - Write MMD Access Address/Data Register = 0x0008 // Select Register 8h of MMD Device Address 2h
 - Write MMD Access Control Register = 0x4002 // Select register data for MMD Device Address 2h, Register 8h
 - Read MMD Access Address/Data Register // Read value of MMD Device Address 2h, Register 8h
- Write value 0x03FF (delay **TXC** and **RXC** pad skews to their maximum values) to the Clock Pad Skew Register (MMD Address 2h, Register 8h)
 - Write MMD Access Control Register = 0x0002 // Select MMD Device Address 2h
 - Write MMD Access Address/Data Register = 0x0008 // Select Register 8h of MMD Device Address 2h
 - Write MMD Access Control Register = 0x4002 // Select register data for MMD Device Address 2h, Register 8h
 - Write MMD Access Address/Data Register = 0x03FF // Write value 0x03FF to MMD Device Address 2h, Register 8h

5.15.5 RGMII IN-BAND STATUS

The device provides in-band status to the MAC during the inter-frame gap when **RX_CTL** is de-asserted. RGMII in-band status is always enabled after power-up.

The in-band status is sent to the MAC using the **RXD[3:0]** data pins, and is described in [Table 5-9](#).

表 5-8: 4 位焊盘偏斜设置的绝对延迟

焊盘偏斜值	延迟 (ns) (Si B0) (估算)
0000	-0.39 (最小值) / -1.31 (最大值)
0001	-0.34 (最小值) / -1.12 (最大值)
0010	-0.28 (最小值) / -0.94 (最大值)
0011	-0.22 (最小值) / -0.75 (最大值)
0100	-0.17 (最小值) / -0.56 (最大值)
0101	-0.11 (最小值) / -0.37 (最大值)
0110	-0.06 (最小值) / -0.19 (最大值)
0111	无延迟调整 (默认值)
1000	0.06 (最小值) / 0.19 (最大值)
1001	0.11 (最小值) / 0.37 (最大值)
1010	0.17 (最小值) / 0.56 (最大值)
1011	0.22 (最小值) / 0.75 (最大值)
1100	0.28 (最小值) / 0.94 (最大值)
1101	0.34 (最小值) / 1.12 (最大值)
1110	0.39 (最小值) / 1.31 (最大值)
1111	0.45 (最小值) / 1.50 (最大值)

以下示例说明如何读取/写入时钟焊盘偏斜寄存器 (MMD 地址 2h, 寄存器 8h), 以设置 RGMII **TXC** 和 **接收时钟** 的偏斜。MMD 寄存器访问通过 MMD 访问控制寄存器和 MMD 访问地址/数据寄存器进行。有关更多编程详情, 请参阅 MMD 寄存器章节。

- 回读 MMD 地址 2h、寄存器 8h 的值。
 - 写入 MMD 访问控制寄存器 = 0x0002 // 选择 MMD 设备地址 2h
 - 写入 MMD 访问地址/数据寄存器 = 0x0008 // 选择 MMD 设备地址 2h 的寄存器 8h
 - 写入 MMD 访问控制寄存器 = 0x4002 // 选择 MMD 设备地址 2h 的寄存器数据, 寄存器 8h
 - 读取 MMD 访问地址/数据寄存器 // 读取 MMD 设备地址 2h 的寄存器 8h 的值
- 将值 0x03FF (将**TXC**和 **RXC** 焊盘偏斜延迟设置为最大值) 写入时钟焊盘偏斜寄存器 (MMD 地址 2h, 寄存器 8h)
 - 将 = 0x0002 写入 MMD 访问控制寄存器 // 选择 MMD 设备地址 2h
 - 将 = 0x0008 写入 MMD 访问地址/数据寄存器 // 选择 MMD 设备地址 2h 的寄存器 8h
 - 将 = 0x4002 写入 MMD 访问控制寄存器 // 为 MMD 设备地址 2h、寄存器选择数据: 寄存器 8h
 - 写入 MMD 访问地址/数据寄存器 = 0x03FF // 将值 0x03FF 写入 MMD 设备地址 2h、寄存器 8h

5.15.5 RGMII 带内状态

当 **接收_控制** 被置为无效时, 器件会在帧间隙期间向媒体访问控制器提供带内状态。上电后, RGMII 带内状态始终处于使能状态。

带内状态通过以下方式发送到媒体访问控制器: **接收数据[3: 0]** 数据引脚, 相关说明见[表 5-9](#)。

TABLE 5-9: RGMII IN-BAND STATUS

RX_CTL	RXD3	RXD[2:1]	RXD0
0 / 0 (equivalent to RX_DV = 0 / RX_ER = 0)	Duplex Status 0 = Half-duplex 1 = Full-duplex	RX_CLK clock speed 00 = 2.5 MHz (10 Mbps) 01 = 25 MHz (100 Mbps) 10 = 125 MHz (1000 Mbps) 11 = Reserved	Link Status 0 = Link down 1 = Link up

5.15.6 RGMII EEE LOW POWER IDLE

The device supports EEE Low Power Idle on the RGMII interface. This is encoded as **RX_CTL** = 0/1 (equivalent to **RX_DV** = 0 / **RX_ER** = 1) and **RXD[3:0]** = 0x1 / 0x0 (equivalent to **RXD[7:0]** = 0x01) for receive and as **TX_CTL** = 0/1 (equivalent to **TX_EN** = 0 / **TX_ER** = 1) and **TXD[3:0]** = 0x1 / 0x0 (equivalent to **TXD[7:0]** = 0x01) for transmit.

5.16 Start of Frame (SOF) Indication

The device supports the generation of an SOF pulse for the receive and transmit paths. The SOF pulse is generated when the Start of Frame Delimiter (SFD octet 0xD5 immediately following the preamble) is detected by the PCS and can be output onto any enabled GPIO pin by setting the corresponding bits in the General Purpose IO Data Select 1 Register (GPIO_DATA_SEL1) or the General Purpose IO Data Select 2 Register (GPIO_DATA_SEL2).

The SOF pulse is always active high.

Note: The GPIO needs to be enabled and its direction set as an output via the General Purpose IO Enable Register (GPIO_EN) and the General Purpose IO Direction Register (GPIO_DIR). The GPIO Buffer Type (GPIO_BUF) field in the General Purpose IO Buffer Type Register (GPIO_BUF) also applies.

5.16.1 802.3BR FRAME PREEMPTION

The SOF pulse can also be configured to detect 802.3br-2016 Start mPacket Delimiters (SMDs). When the SOF_preemption_enable bits in MMD2 Register 75 are set to “10”, the SOF pulse is generated for any of the SMD values listed in 802.3br-2016 *except* for the continuation frame: SMD-V (0x07), SMD-R (0x19), SMD-E (0xD5) and SMD-S[0,1,2,3] (0xE6, 0x4C, 0x7F or 0xB3). When the SOF_preemption_enable bits are set to “11”, the SOF pulse is also generated for the continuation SMD values: SMD-C[0,1,2,3] (0x61, 0x52, 0x9E or 0x2A).

5.16.2 SOF LATENCY

There is a fixed timing relationship from the time the transmitted SFD arrives at the PCS to when it appears on the line and from when the received SFD arrives on the line to when it is detected by the PCS. The latencies between the SFD pulse and the wire are as follows:

表 5-9：RGMII 带内状态

RX_CTL	RXD3	接收数据[2:1]	RXD0
0 / 0 (等同于 接收_DV = 0 / 接收_错误 = 0)	双工状态 0 = 半双工 1 = 全双工	RX_时钟 时钟速率 00 = 2.5 兆赫（10 兆比特每秒） 01 = 25 兆赫（100 Mbps） 10 = 125 兆赫（1000 兆比特/秒） 11 = 保留	链路状态 0 = 链路断开 1 = 链路已连接

5.15.6 RGMII EEE 低功耗空闲

该器件支持 RGMII 接口上的节能以太网低功耗空闲模式。接收时，该模式编码为**接收_控制** = 0/1（等同于**接收_DV** = 0 / **接收_错误** = 1）和**接收数据 [3: 0]** = 0x1 / 0x0（等同于**接收数据 [7: 0]** = 0x01）；发送时，该模式编码为**发送_控制** = 0/1（等同于**发送_使能** = 0 / **TX_ER** = 1）和**发送数据[3: 0]** = 0x1 / 0x0（等同于**发送数据[7: 0]** = 0x01）。

5.16 帧起始 (SOF) 指示

该器件支持为接收和发送路径生成 SOF 脉冲。当物理编码子层检测到帧起始定界符（紧跟在前导码之后的 SFD 八位字节 0xD5）时，便会生成 SOF 脉冲。通过在通用输入/输出数据选择 1 寄存器 (GPIO_DATA_SEL1) 或通用输入/输出数据选择 2 寄存器 (GPIO_DATA_SEL2) 中设置相应位，可将该脉冲输出到任意已使能的 GPIO 引脚。

SOF 脉冲始终为高电平有效。

注：需要通过通用输入/输出使能寄存器 (GPIO_EN) 使能 GPIO，并通过通用输入/输出方向寄存器 (GPIO_DIR) 将其方向设置为输出。通用输入/输出缓冲器类型寄存器 (GPIO_BUF) 中的 GPIO 缓冲器类型 (GPIO_BUF) 字段同样适用。

5.16.1 802.3BR 帧抢占

SOF 脉冲还可配置为检测 802.3br-2016 mPacket 起始定界符（SMDs）。当 MMD2 寄存器 75 中的 SOF_抢占_使能位设为 “10” 时，对于 802.3br-2016 中列出的任意 SMD 值，都会生成 SOF 脉冲，但续传帧除外：SMD-V (0x07)、SMD-R (0x19)、SMD-E (0xD5) 和 SMD-S[0,1,2,3] (0xE6, 0x4C、0x7F 或 0xB3)。当 SOF_抢占_使能位设为 “11” 时，对于续传 SMD 值也会生成 SOF 脉冲：SMD-C[0,1,2,3] (0x61、0x52、0x9E 或 0x2A)。

5.16.2 SOF 延迟

发送的 SFD 从到达 PCS 到在线路上出现，以及接收的 SFD 从在线路上到达至被 PCS 检测到，这两者之间都存在固定的时序关系。SFD 脉冲与线路之间的延迟如下：

TABLE 5-10: SOF LATENCY

Speed	TX SFD Latency (start of SFD pulse to start of SFD on MDI output)	TX Variation per Link Up	TX Variation per Frame	RX SFD Latency (start of SFD on MDI input to start of SFD pulse)	RX Variation per Link Up	RX Variation per Frame
1000Mbps	138ns	0ns	0ns	430ns	0ns	+/-4ns
100Mbps (fixed mode Note 5-1)	140ns	0ns	0ns	615ns	0ns	+/-4ns
100Mbps (variable mode Note 5-1)	140ns	<2ns	0ns	488ns, 492ns, 498ns, 504ns, 508ns, 512ns, 520ns, 524 ns	8 @ 4ns steps	<2ns
10Mbps	654ns	0ns	0ns	2277ns, 2377ns, 2477ns, 2577ns	4 @ 100ns steps	+/-4ns

Note 5-1 For 100BASE-TX, at the expense of an increased variation between various link establishments, the receive latency may be reduced by clearing the Fixed 100BT latency bit in the Reserved Register.

5.17 MII Management (MIIM) Interface

The device supports the IEEE 802.3 MII management interface, also known as the Management Data Input / Output (MDIO) interface. This interface allows upper-layer devices to monitor and control the state of the device. An external device with MIIM capability is used to read the PHY status and/or configure the PHY settings. More details about the MIIM interface can be found in Clause 22.2.4 of the IEEE 802.3 Specification.

The MIIM interface consists of the following:

- A physical connection that incorporates the clock line (MDC) and the data line (MDIO).
- A specific protocol that operates across the physical connection mentioned earlier, which allows an external controller to communicate with one or more devices. Each device is assigned a unique PHY address between 0h through 7h via PHYAD[2:0] strapping pins.
- A 32-register address space for direct access to IEEE-defined registers and vendor-specific registers, and for indirect access to MMD addresses and registers. See the Register Map section.

[Table 5-11](#) shows the MII management frame format for the device.

TABLE 5-11: MII MANAGEMENT FRAME FORMAT

	Preamble	Start of Frame	Read/Write OP Code	PHY Address Bits [4:0]	REG Address Bits [4:0]	TA	Data Bits [15:0]	Idle
Read	32 1's	01	10	AAAAA	RRRRR	Z0	DDDDDDDD_DDDDDDDD	Z
Write	32 1's	01	01	AAAAA	RRRRR	10	DDDDDDDD_DDDDDDDD	Z

The LAN8840 is limited to PHY addresses 0h through 7h.

5.17.1 ALL PHYS ADDRESS

Normally, the Ethernet PHY is accessed at the PHY address set by the **PHYAD** strapping pins.

PHY Address 0h is optionally supported as the broadcast PHY address, which allows for a single write command to simultaneously program an identical PHY register for two or more PHY devices (for example, using PHY Address 0h to set the Basic Control Register to a value of 0x1940 to set Bit [11] to a value of one to enable software power-down).

PHY address 0 is enable (in addition to the PHY address set by the PHYAD strapping pins) when the All-PHYAD Enable bit in the Common Control Register is a 1.

表 5-10: SOF 延迟

速率	发送 SFD 延迟（从 SFD 脉冲开始到 MDI 输出端 SFD 开始） 的发送变化量	每帧接收 SFD 发送变化量	接收 SFD 脉冲开始到 SFD 开始到接收 SFD 脉冲开始） 每次链路建立的接收变化量	接收变化量 每帧		
1000 Mbps	138 纳秒	0ns	0ns	430 纳秒	0ns	+/-4 纳秒
100兆比特 每秒（固定 模式 注释 5-1 ）	140纳秒	0ns	0ns	615纳秒	0ns	+/-4纳秒
100兆比特 每秒（可变 模式 注释 5-1 ）	140纳秒	<2ns	0ns	488纳秒、492纳秒、 498纳秒、504纳秒、 508纳秒、512纳秒、 520纳秒、524纳秒	8 个， 每步 4 纳秒	<2ns
10兆比特每 秒	654纳秒	0ns	0ns	2277纳秒、2377纳秒、 2477纳秒、2577纳秒	每步 100 纳秒， 2377纳秒	+/-4纳秒

注释 5-1 对于 100BASE-TX，可通过清除保留寄存器中的固定 100BT 延迟位来缩短接收延迟，但代价是各次链路建立之间的差异增大。

5.17 MII 管理（MIIM）接口

该器件支持 IEEE 802.3 MII 管理接口，也称为管理数据输入/输出（MDIO）接口。此接口允许上层器件监视和控制该器件的状态。具有 MIIM 功能的外部器件可用于读取 PHY 状态和/或配置 PHY 设置。有关 MIIM 接口的更多详细信息，请参阅 IEEE 802.3 规范的第 22.2.4 条。

MIIM 接口由以下部分组成：

- 包含时钟线（MDC）和数据线（MDIO）的物理连接。
- 一种特定协议，通过前述物理连接运行，使外部控制器能够与一个或多个器件通信。每个器件都通过 PHYAD[2:0] 绑带引脚分配一个介于 0h 到 7h 之间的唯一 PHY 地址。

• 一个包含 32 个寄存器的地址空间，用于直接访问 IEEE 定义的寄存器和厂商专用寄存器，以及间接访问 MMD 地址和寄存器。请参阅寄存器映射部分。

[表 5-11](#) 显示了该器件的 MII 管理帧格式。

表 5-11: MII 管理帧格式

	前导码	读取/写入帧操作码的起始位		PHY寄存器地址 地址位 [4:0] 位 [4:0]		TA	数据位 [15:0]	Idle
Read	32 个 1	01	10	AAAAA	RRRRR	Z0	DDDDDDDD _DDDDDDDD	Z
书写	32 个 1	01	01	AAAAA	RRRRR	10	DDDDDDDD _DDDDDDDD	Z

LAN8840 的 PHY 地址范围仅限于 0h 到 7h。

5.17.1 所有 PHY 地址

通常，通过 **PHYAD** 跨接引脚设置的 PHY 地址访问以太网 PHY。

可选择支持 PHY 地址 0h 作为广播 PHY 地址，从而只需一条写命令即可同时为两个或更多 PHY 设备写入相同的 PHY 寄存器（例如，使用 PHY 地址 0h 将基本控制寄存器设为 0x1940，使位 [11] 的值为 1，以启用软件断电）。

当通用控制寄存器中的 All-PHYAD 使能位为 1 时，PHY 地址 0 也会使能（除 PHYAD 配置引脚设置的 PHY 地址外）。

5.17.2 MDIO OUTPUT DRIVE MODE

The MDIO output pin drive mode is controlled by the MDIO Buffer Type bit in Output Control Register. When set to a 0, the MDIO output is open-drain. When set to a 1, the MDIO output is push-pull.

5.17.3 HIGHER-SPEED MDIO OPERATION

The MDIO bus can operate at standard speeds up to 2.5 MHz, as well higher speeds up to 8.33 MHz.

Default register values enable standard MDIO operation, that is MDIO operation up to 2.5 MHz MDC frequency with multiple PHYs on the MDIO bus, using an open-drain MDIO driver. Due to the use of an open-drain driver an external 1kΩ to 4.7kΩ pull-up resistor is required.

Up to 8.33 MHz, open-drain MDIO operation is supported however it is limited to a single PHY on the MDIO bus and requires the used of a 1kΩ external pull-up resistor.

The device can be configured to use a push-pull MDIO driver. The reconfiguration process requires initially running the MDIO bus at 2.5 MHz since an open-drain MDIO driver is used by default. The push-pull MDIO driver allows operation up to 8.33 MHz with the 1kΩ external pull-up resistor.

5.17.4 CLAUSE 45 PROTECTION

To facilitate interoperability with other PHYs that are managed with Clause 45 of the IEEE 802.3 Specification, the MIIM Interface will ignore management frames that do not contain a Start of Frame field equal to '01'.

APPLICATION NOTE: If preamble suppression is allowed (MF Preamble Suppression in the Basic Status Register equals 1) and the Start of Frame field is not equal '01', the MIIM Interface will ignore the entire frame. It will not falsely detect the remainder of the frame as a new frame.

5.18 Interrupt (INT_N)

The INT_N pin is an optional interrupt signal that is used to inform the external controller that there has been a status update in the device. The Interrupt Enable Register contains the interrupt control bits that enable and disable the conditions for asserting the INT_N signal. The Interrupt Status Register contains the interrupt status bits that indicate which interrupt conditions have occurred. Most interrupt status bits are cleared upon reading. Some bits are read only and have a lower level register to indicate individual sources.

Note: Bits in the Interrupt Status Register are set by the interrupt events regardless of the value of the corresponding bits in the Interrupt Enable Register.

The INT_N buffer type is selectable between open-drain and push-pull and is configured by the INT Buffer Type field in the Output Control Register. The default is open-drain.

The Intr Polarity Invert bit in the Control Register sets the interrupt level to active high or active low. The default is active low. If the buffer type is set to open-drain, the polarity is forced to be active low.

5.19 LED Support

The device provides five programmable LED output pins, LED5 through LED1, which are configurable to support three LED modes. The LED mode is configured by the LED_MODE strap-in as well as the LED Mode bit in the UNH Test Register and is defined as follows:

- **Individual-LED Mode:** LED Mode = 1, LED_MODE strap input high (pulled up)
- **Tri-Color LED Mode:** LED Mode = 1, LED_MODE strap input low (pulled down) or left floating
- **Enhanced LED Mode:** LED Mode = 0, LED_MODE strap input is unused

Each LED output pin can directly drive an LED with a series resistor (typically 220Ω to 470Ω).

5.17.2 MDIO 输出驱动模式

MDIO 输出引脚的驱动模式由输出控制寄存器中的 MDIO 缓冲器类型位控制。该位设为 0 时，MDIO 输出为开漏；设为 1 时，MDIO 输出为推挽。

5.17.3 高速 MDIO 操作

MDIO 总线可按标准速率运行，最高可达 2.5 兆赫；也可按更高速率运行，最高可达 8.33 兆赫。

默认寄存器值可启用标准 MDIO 操作，即在 MDIO 总线上连接多个 PHY 时，以最高 2.5 MHz 的 MDC 频率进行 MDIO 操作，并使用开漏 MDIO 驱动器。由于使用开漏驱动器，因此需要外接 1kΩ 至 4.7kΩ 上拉电阻。

支持最高 8.33 MHz 的开漏 MDIO 操作，但 MDIO 总线上只能连接一个 PHY，并且需要使用 1kΩ 外接上拉电阻。

可将器件配置为使用推挽 MDIO 驱动器。由于默认使用开漏 MDIO 驱动器，因此重新配置过程需要先以 2.5 MHz 的频率运行 MDIO 总线。推挽 MDIO 驱动器配合 1kΩ 外接上拉电阻，可支持最高 8.33 MHz 的操作。

5.17.4 第45条款保护

为便于与其他使用 IEEE 802.3 规范第45条款进行管理的 PHY 互操作，MIIM 接口将忽略帧起始字段不等于“01”的管理帧。

应用说明：如果允许前导码抑制（基本状态寄存器中的 MF前导码抑制等于 1），且帧起始字段不等于“01”，则 MIIM 接口将忽略整个帧，不会错误地将帧的剩余部分检测为新帧。

5.18 中断 (INT_N)

INT_N引脚是可选的中断信号，用于通知外部控制器器件发生了状态更新。中断使能寄存器包含用于使能和禁用触发 INT_N信号的条件的中断控制位。中断状态寄存器包含用于指示发生了哪些中断条件的中断状态位。大多数中断状态位在读取后会清除。有些位为只读位，并通过下级寄存器指示各个来源。

注意：无论中断使能寄存器中相应位的值如何，中断状态寄存器中的位都会由中断事件置位。

INT_N缓冲器类型可选开漏或推挽，并通过输出控制寄存器中的 INT缓冲器类型字段进行配置。默认类型为开漏。

控制寄存器中的中断极性反转位用于设置中断电平为高电平有效或低电平有效。默认情况下为低电平有效。如果缓冲区类型设置为开漏，则极性强制为低电平有效。

5.19 LED支持

该器件提供五个可编程LED输出引脚，LED5 至 LED1，可配置为支持三种LED模式。LED模式由 LED_MODE绑带输入以及 UNH 测试寄存器中的 LED模式位配置，定义如下：——

- **独立 LED 模式：**LED模式 = 1，LED_MODE绑带输入为高电平（上拉）
- **三色 LED 模式：**LED模式 = 1，LED_MODE绑带输入为低电平（下拉）或保持悬空
- **增强型 LED 模式：**LED模式 = 0，LED_MODE绑带输入未使用

每个 LED输出引脚都可以通过串联电阻（通常为 220Ω 至 470Ω）直接驱动一个发光二极管。

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5.19.1 INDIVIDUAL-LED MODE

In Individual-LED mode, [Table 5-12](#) specifies the LED functionality. The LED function is controlled by the mr_led_sel field in the PCS Loop-back Swap/Polarity Control Register.

Note: LED_MODE strap is pull-down by default (Tri-Color Mode). LED_MODE must be pulled up to enable Individual-LED Mode.

Note: The LEDs are forced OFF when the Isolate (PHY_ISO) bits in the Basic Control Register is set.

Note: The LEDs are forced OFF when the Power Down bit in the Basic Control Register is set.

TABLE 5-12: INDIVIDUAL-LED MODE - PIN DEFINITION

LED Pin	Pin State Note 5-2	LED Definition	mr_led_sel=11b	mr_led_sel=10b	mr_led_sel=01b	mr_led_sel=00b
LED5	H	OFF	No Link 100	No Link 100	No Link 100	rsvd
	L	ON	Link 100	Link 100	Link 100	
	Toggle	Blinking	-	Activity 100	-	
LED4	H	OFF	No Link 1000	No Link 1000	No Link 1000	rsvd
	L	ON	Link 1000	Link 1000	Link 1000	
	Toggle	Blinking	-	Activity 1000	-	
LED3	H	OFF	Link Off or Half Duplex	Link Off or Half Duplex	Link Off or Half Duplex	rsvd
	L	ON	Full Duplex and Link On (any speed)	Full Duplex and Link On (any speed)	Full Duplex and Link On (any speed)	
	Toggle	Blinking	Collision	Collision	Collision	
LED2	H	OFF	Link Off			
	L	ON	Link On (any speed)			
LED1	H	OFF	No Activity			
	Toggle	Blinking	Activity (RX, TX)			

Note 5-2 Assuming active low LEDs.

5.19.2 TRI-COLOR LED MODE

In Tri-color LED mode, the link and activity status are indicated by the **LED2** pin for 1000BASE-T; by the **LED1** pin for 100BASE-TX; and by both **LED2** and **LED1** pins, working in conjunction, for 10BASE-T. This is summarized in [Table 5-13](#).

Note: This mode applies only to **LED1** and **LED2**, the other LEDs operate in Individual-LED mode.

Note: The LEDs are forced OFF when the Isolate (PHY_ISO) bits in the Basic Control Register is set.

Note: The LEDs are forced OFF when the Power Down bit in the Basic Control Register is set.

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5.19.1 独立 LED 模式

在独立 LED 模式下，[表 5-12](#)规定了 LED 功能。LED 功能由 PCS 回环交换/极性控制寄存器中的 mr_led_sel 字段控制。

注意：发光二极管_模式引脚默认下拉（三色模式）。必须将发光二极管_模式引脚上拉，才能启用单独发光二极管模式。

注意：设置基本控制寄存器中的隔离（PHY_ISO）位后，LED 将强制关闭。

注意：设置基本控制寄存器中的掉电位后，LED 将强制关闭。

表 5-12：独立 LED 模式 - 引脚定义

LED引脚	引脚状态 注5-2	LED 定义	mr 发光二极管 sel=11b - -	mr 发光二极管 sel=10 b_ -	mr 发光二极管 sel=01b - -	mr 发光二极管 sel=00 b_ -
LED5	H	OFF	无链路 100	无链路 100	无链路 100	rsvd
	L	ON	链路 100	链路 100	链路 100	
	切换	闪烁	-	活动 100	-	
LED4	H	OFF	未连接 1000	未连接 1000	未连接 1000	rsvd
	L	ON	已连接 1000	已连接 1000	已连接 1000	
	切换	闪烁	-	活动 1000	-	
LED3	H	OFF	链路关闭或半 双工	链路关闭或半 双工	链路关闭或半 双工	rsvd
	L	ON	全双工且 链路已连接 (任意速率)	全双工且 链路已连接 (任意速率)	全双工且 链路已连接 (任意速率)	
	切换	闪烁	冲突	冲突	冲突	
LED2	H	OFF	链路断开			
	L	ON	链路已连接（任意速率）			
LED1	H	OFF	无活动			
	切换	闪烁	活动（接收、发送）			

注5-2 假设使用低电平有效 LED。

5.19.2 三色 LED 模式

在三色 LED 模式下，1000BASE-T 的链路和活动状态由 **LED2** 引脚指示；100BASE-TX 的链路和活动状态由 **LED1** 引脚指示；对于 10BASE-T，则由 **LED2** 和 **LED1** 引脚协同指示。详见 [表 5-13](#)。

注：此模式仅适用于**LED1**和**LED2**，其他 LED 以独立 LED 模式运行。

注：当基本控制寄存器中的隔离（PHY_ISO）位被置位时，LED 将被强制关闭。

注意：设置基本控制寄存器中的掉电位后，LED 会被强制关闭。

TABLE 5-13: TRI-COLOR LED MODE - PIN DEFINITION

LED Pin (State) Note 5-3		LED Pin (Definition)		Link/Activity
LED2	LED1	LED2	LED1	
H	H	OFF	OFF	Link Off
L	H	ON	OFF	1000 Link/No Activity
Toggle	H	Blinking	OFF	1000 Link/Activity (RX, TX)
H	L	OFF	ON	100 Link/No Activity
H	Toggle	OFF	Blinking	100 Link/Activity (RX, TX)
L	L	ON	ON	10 Link/No Activity
Toggle	Toggle	Blinking	Blinking	10 Link/Activity (RX, TX)

Note 5-3 Assuming active low LEDs.

5.19.3 ENHANCED LED MODE

Enhanced LED mode is enabled when the device LED Mode bit in the UNH Test Register is cleared. In Enhanced LED mode, each LED can be configured to display different status information that can be selected by setting the corresponding LED Configuration field of the LED Mode Select Register. The modes are shown in Table 5-14. The blink/pulse-stretch and other LED settings can be configured via the LED Behavior Register.

Note: This mode applies only to LED1, LED2, LED3 and LED4. LED5 is disabled in this mode.

Note: The LEDs are forced OFF when the Power Down bit in the Basic Control Register is set.

TABLE 5-14: LED MODE AND FUNCTION SUMMARY

Mode	Name	Description
0	Link/Activity	1 (led off) = No link in any speed on any media interface. 0 (led on) = Valid link at any speed on any media interface. Blink or pulse stretch (led turns off) = Valid link at any speed on any media interface with activity present.
1	Link1000/Activity	1 (led off) = No link at 1000BASE-T. 0 (led on) = Valid link at 1000BASE-T. Blink or pulse stretch (led turns off) = Valid link at 1000BASE-T with activity present.
2	Link100/Activity	1 (led off) = No link at 100BASE-TX. 0 (led on) = Valid link at 100BASE-TX. Blink or pulse stretch (led turns off) = Valid link at 100BASE-TX with activity present.
3	Link10/Activity	1 (led off) = No link at 10BASE-T. 0 (led on) = Valid link at 10BASE-T. Blink or pulse stretch (led turns off) = Valid link at 10BASE-T with activity present.
4	Link100/1000/Activity	1 (led off) = No link at 100BASE-TX or 1000BASE-T. 0 (led on) = Valid link at 100BASE-TX or 1000BASE-T. Blink or pulse stretch (led turns off) = Valid link at 100BASE-TX or 1000BASE-T, with activity present.
5	Link10/1000/Activity	1 (led off) = No link at 10BASE-T or 1000BASE-T. 0 (led on) = Valid link at 10BASE-T or 1000BASE-T. Blink or pulse stretch (led turns off) = Valid link at 10BASE-T or 1000BASE-T, with activity present.

表 5-13：三色 LED 模式 - 引脚定义

LED引脚（状态） 注 5-3		LED引脚（定义）		链路/活动
LED2	LED1	LED2	LED1	
H	H	OFF	OFF	链路断开
L	H	ON	OFF	1000 链路/无活动
切换	H	闪烁	OFF	1000 链路/活动（接收、发送）
H	L	OFF	ON	100 链路/无活动
H	切换	OFF	闪烁	100 链路/活动（接收、发送）
L	L	ON	ON	10 链路/无活动
切换	切换	闪烁	闪烁	10 链路/活动（接收、发送）

注 5-3 假设 LED 为低电平有效。

5.19.3 增强型 LED 模式

清除 UNH 测试寄存器中的器件 LED 模式位后，即可启用增强型 LED 模式。在增强型 LED 模式下，可通过设置 LED 模式选择寄存器中相应的 LED 配置字段，将每个 LED 配置为显示不同的状态信息。模式如 表 5-14所示。可通过 LED 行为寄存器配置闪烁/脉冲拉伸及其他 LED 设置。

注：此模式仅适用于 LED1、LED2、LED3 和 LED4。LED5 在此模式下禁用。

注意：当基本控制寄存器中的掉电位设置为开启时，LED 会被强制关闭。

表 5-14：LED 模式和功能摘要

Mode	Name	说明
0	链路/活动	1（发光二极管关闭）= 任何媒体接口上均无任何速率的链路。0（发光二极管开启）= 任何媒体接口上均存在有效链路，速率不限。 闪烁或脉冲延展（发光二极管熄灭）= 任何媒体接口上均存在有效链路，且有活动。
1	千兆链路/活动	1（发光二极管关闭）= 1000BASE-T 无链路。0（发光二极管开启）= 1000BASE-T 存在有效链路。 闪烁或脉冲延展（发光二极管熄灭）= 1000BASE-T 存在有效链路，且有活动。
2	百兆链路/活动	1（发光二极管关闭）= 100BASE-TX 无连接。 0（发光二极管开启）= 100BASE-TX 链路有效。 闪烁或脉冲延展（发光二极管关闭）= 100BASE-TX 链路有效，且有活动。
3	Link10/活动	1（发光二极管关闭）= 10BASE-T 无连接。 0（发光二极管开启）= 10BASE-T 链路有效。 闪烁或脉冲延展（发光二极管熄灭）= 10BASE-T 链路有效且存在活动。
4	Link100/1000/活动	1（发光二极管关闭）= 100BASE-TX 或 1000BASE-T 无连接。 0（发光二极管开启）= 100BASE-TX 或 1000BASE-T 链路有效。 闪烁或脉冲延展（发光二极管熄灭）= 100BASE-TX 或 1000BASE-T 链路有效且存在活动。
5	Link10/1000/活动	1（发光二极管关闭）= 10BASE-T 或 1000BASE-T 无连接。0（发光二极管开启）= 10BASE-T 或 1000BASE-T 链路有效。闪烁或脉冲延展（发光二极管熄灭）= 10BASE-T 或 1000BASE-T 链路有效且存在活动。

TABLE 5-14: LED MODE AND FUNCTION SUMMARY (CONTINUED)

Mode	Name	Description
6	Link10/100/Activity	1 (led off) = No link at 10BASE-T or 100BASE-TX. 0 (led on) = Valid link at 10BASE-T or 100BASE-TX. Blink or pulse stretch (led turns off) = Valid link at 10BASE-T or 100BASE-TX, with activity present.
7	RESERVED	RESERVED
8	Duplex/Collision	1 (led off) = Link established in half-duplex mode, or no link established. 0 (led on) = Link established in full-duplex mode. Blink or pulse stretch (led turns on) = Link established in half-duplex mode but collisions are present.
9	Collision	1 (led off) = No collisions detected. Blink or pulse stretch (led turns on) = Collision detected.
10	Activity	1 (led off) = No activity present. Blink or pulse stretch (led turns on) = Activity present. (becomes TX activity present if the LED Activity Output Select bit in the LED Behavior Register is set to 1.)
11	RESERVED	RESERVED
12	Auto-Negotiation Fault Parallel Detect Fault	1 (led off) = No Parallel Detect fault present. 0 (led on) = Parallel Detect fault occurred.
13	RESERVED	RESERVED
14	Force LED Off	1 (led off) = De-asserts the LED.
15	Force LED On	0 (led on) = Asserts the LED.

5.19.3.1 LED Behavior

Using the LED Behavior Register, the following LED behaviors can be configured.

- [LED Combine](#)
- [LED Blink or Pulse-Stretch](#)
- [Rate of LED Blink or Pulse-Stretch](#)
- [LED Pulsing Enable](#)

5.19.3.1.1 LED Combine

Enables an LED to display the status for a combination of primary and secondary modes. This can be enabled or disabled for each LED pin via the LED Combination Disables field of the LED Behavior Register. For example, a copper link running in 1000BASE-T mode with activity present can be displayed with one LED by configuring an LED pin to Link1000/Activity mode. The LED asserts when linked to a 1000BASE-T partner and also blinks or performs pulse-stretch when activity is either transmitted by the PHY or received by the Link Partner. When disabled, the combine feature only provides status of the selected primary function. In this example, only Link1000 asserts the LED, and the secondary mode, activity, does not display if the combine feature is disabled.

5.19.3.1.2 LED Blink or Pulse-Stretch

This behavior is used for activity and collision indication. This can be uniquely configured for each LED pin via the LED Pulse Stretch Enables field of the LED Behavior Register. Activity and collision events can occur randomly and intermittently throughout the link-up period.

Blink is a 50% duty cycle oscillation of asserting and de-asserting an LED pin.

表 5-14：LED 模式和功能摘要（续）

Mode	Name	说明
6	链路 10/100/活动	1（发光二极管关闭）= 10BASE-T 或 100BASE-TX 无连接。0（发光二极管开启）= 10BASE-T 或 100BASE-TX 链路有效。闪烁或脉冲延展（发光二极管熄灭）= 10BASE-T 或 100BASE-TX 链路有效，且有活动。
7	保留	保留
8	双工/冲突	1（发光二极管关闭）= 已建立半双工模式链路，或未建立链路。0（发光二极管开启）= 已建立全双工模式链路。闪烁或脉冲延展（发光二极管点亮）= 已建立半双工模式链路，但存在冲突。
9	冲突	1（发光二极管关闭）= 未检测到冲突。 闪烁或脉冲延展（发光二极管点亮）= 检测到冲突。
10	活动	1（发光二极管关闭）= 无活动。闪烁或脉冲延展（发光二极管亮起）= 有活动。（如果 LED 行为寄存器中的 LED 活动输出选择位设为 1，则表示有发送活动。）
11	保留	保留
12	自动协商故障 并行检测故障	1（发光二极管关闭）= 未出现并行检测故障。 0（发光二极管开启）= 出现并行检测故障。
13	保留	保留
14	强制关闭 LED	1（发光二极管关闭）= 关闭发光二极管。
15	强制点亮 LED	0（发光二极管开启）= 使发光二极管点亮。

5.19.3.1 LED 行为

使用 LED 行为寄存器，可以配置以下 LED 行为。

- [LED 组合](#)
- [LED 闪烁或脉冲延展](#)
- [LED 闪烁或脉冲延展速率](#)
- [LED 脉冲使能](#)

5.19.3.1.1 LED 组合

使 LED 能够显示主模式和次模式的组合状态。可通过 LED 行为寄存器的 LED 组合禁用字段，针对每个 LED 引脚启用或禁用此功能。例如，运行在 1000BASE-T 模式且有活动的铜缆链路，可通过将 LED 引脚配置为 Link1000/活动模式，使用一个 LED 显示其状态。当链路连接到 1000BASE-T 对端时，LED 亮起；当 PHY 发送或链路伙伴接收活动信号时，LED 还会闪烁或执行脉冲延展。禁用此功能时，组合功能仅提供所选主功能的状态。在此示例中，只有 Link1000 会使 LED 亮起；如果禁用了组合功能，则不会显示次模式（活动）的状态。

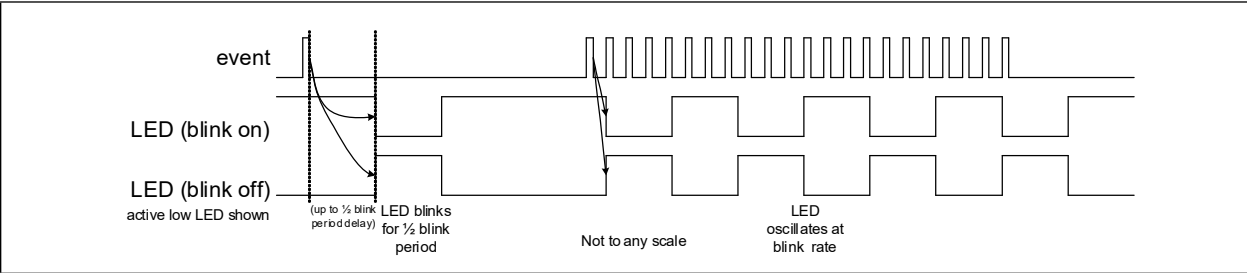
5.19.3.1.2 LED 闪烁或脉冲延展

此行为用于指示活动和冲突。可通过 LED 行为寄存器的 LED 脉冲延展使能字段，为每个 LED 引脚单独配置此行为。在链路建立期间，活动和冲突事件可能随机、间歇地发生。

闪烁是通过置位和取消置位 LED 引脚实现的占空比为 50% 的振荡。

As shown in [Figure 5-17](#), for a single event, the LED will blink (either on or off depending on the LED function) for half of the blink period. For continual events, the LED will oscillate at the blink rate.

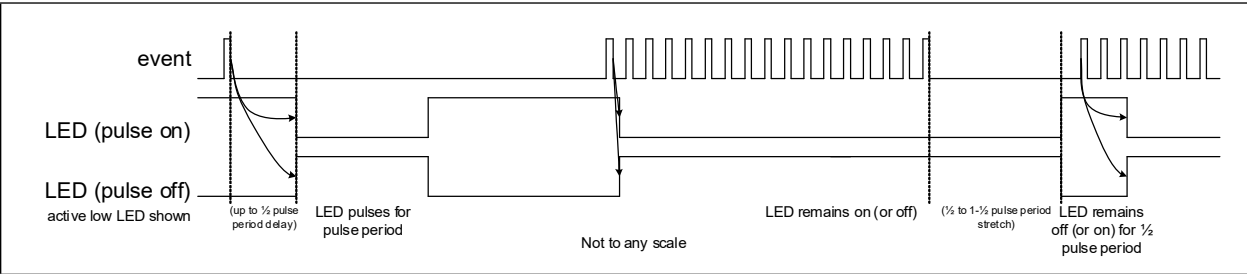
FIGURE 5-17: LED BLINK PATTERN



Pulse-stretch guarantees that an LED is asserted and de-asserted for a specific period of time when activity is either present or not present.

As shown in [Figure 5-18](#), for a single event, the LED will pulse (either on or off depending on the LED function) for the full pulse period. For continual events, the LED will remain on (or off) and will extend from a half to one and a half pulse periods once the events terminate. Once off (or on), the LED will remain in that state for at least a half pulse period.

FIGURE 5-18: LED PULSE PATTERN



The blink / pulse stretch rate can be configured, as detailed in [Section 5.19.3.1.3, "Rate of LED Blink or Pulse-Stretch"](#).

5.19.3.1.3 Rate of LED Blink or Pulse-Stretch

This behavior controls the LED blink rate or pulse-stretch length when the blink/pulse-stretch is enabled (LED Pulse Stretch Enables) on an LED pin. This can be uniquely configured for each LED pin via the LED Blink / Pulse-Stretch Rate field of the LED Behavior Register. The blink rate, which alternates between a high and low voltage level at a 50% duty cycle, can be set to 2.5 Hz, 5 Hz, 10 Hz, or 20 Hz. For pulse-stretch, the rate can be set to 50 ms, 100 ms, 200 ms, or 400 ms.

5.19.3.1.4 LED Pulsing Enable

To provide additional power savings, the LEDs (when asserted) can be modulated at 5 kHz, 20% duty cycle, by setting the LED Pulsing Enable bit of the LED Behavior Register.

5.19.4 LED POLARITY AND BUFFER TYPE

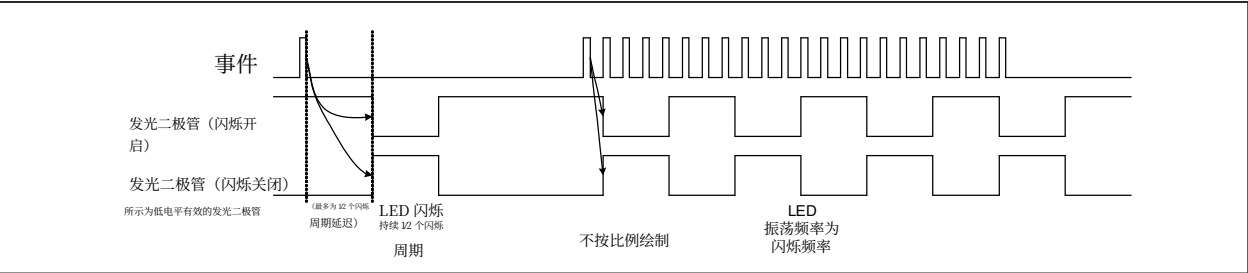
The LED polarity is configured by the LED Polarity field in the Output Control Register. The default of this register is set by the strap values on the LED pins themselves, effectively making the polarity self setting.

The design of the LED polarity control takes into account that the LEDs are forced OFF with the Power Down and Isolate (PHY_ISO) bits in the Basic Control Register, no matter the polarity.

The LED buffer type is selectable between open-drain/open-source and push-pull and is configured by the LED Buffer Type field in the Output Control Register. If open-drain/open-source is selected, the polarity then determines open-drain (active low) and open-source (active high).

如图 5-17所示，对于单个事件，发光二极管将在半个闪烁周期内闪烁（开启或关闭，取决于 LED 功能）。对于连续事件，发光二极管将以闪烁频率振荡。

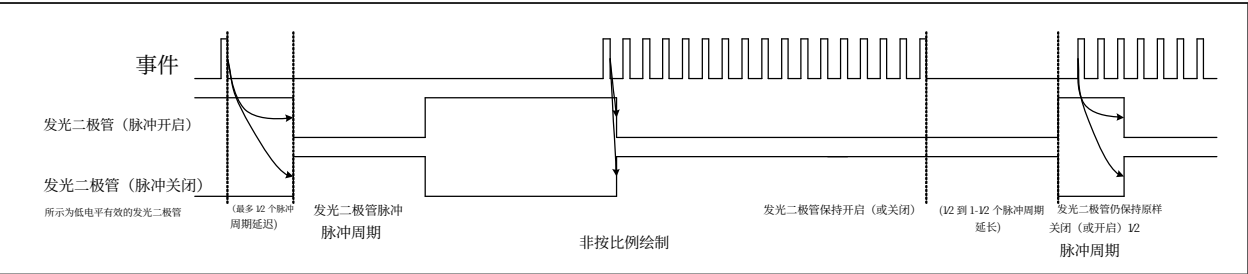
图 5-17：LED 闪烁模式



脉冲延展可确保在存在或不存在活动时，发光二极管在特定时间段内保持置位或复位。

如图 5-18所示，对于单个事件，发光二极管将在整个脉冲周期内闪烁（根据发光二极管功能开启或关闭）。对于持续发生的事件，发光二极管将保持开启（或关闭），并在事件终止后延续半个至一个半脉冲周期。关闭（或开启）后，发光二极管将至少保持该状态半个脉冲周期。

图 5-18：LED 脉冲模式



闪烁/脉冲延展速率可进行配置，详情请参见第 5.19.3.1.3 节“LED 闪烁或脉冲延展速率”。

5.19.3.1.3 LED 闪烁或脉冲延展速率

当 LED 引脚上的闪烁/脉冲延展已使能（LED 脉冲延展使能）时，此行为用于控制 LED 闪烁速率或脉冲延展时长。每个 LED 引脚都可以通过 LED 行为寄存器中的 LED 闪烁/脉冲延展速率字段单独配置。闪烁频率可设置为 2.5 Hz、5 Hz、10 Hz 或 20 Hz，在 50% 占空比下交替输出高、低电压电平。脉冲延展速率可设置为 50 ms、100 ms、200 ms 或 400 ms。

5.19.3.1.4 LED 脉冲使能

为进一步节省功耗，可通过设置 LED 行为寄存器中的 LED 脉冲使能位，将已使能的 LED 调制为 5 kHz、20% 占空比。

5.19.4 LED 极性和缓冲器类型

LED 极性由输出控制寄存器中的 LED 极性字段配置。该寄存器的默认值由 LED 引脚本身的绑带值设置，因此极性实际上可以自动设置。

LED 极性控制的设计考虑到了：无论极性如何，基本控制寄存器中的断电并隔离（PHY_ISO）位都会强制关闭 LED。

LED 缓冲器类型可在漏极开路/源极开路和推挽之间选择，并通过输出控制寄存器中的 LED 缓冲器类型字段进行配置。如果选择漏极开路/源极开路，则由极性决定采用漏极开路（低电平有效）还是源极开路（高电平有效）。

5.20 GPIOs

The GPIO controller is comprised of 7 programmable input / output pins that are shared with other pins (see [Section 3.4, "Pin Alternate Functions"](#)). These pins are individually configurable via the GPIO registers.

For a pin to function as its GPIO, the GPIO must be enabled via the corresponding bit in the General Purpose IO Enable Register (**GPIO_EN**).

When configured as an input, via the General Purpose IO Direction Register (GPIO_DIR), the pin’s pull-up is enabled. The corresponding General Purpose IO Data Register (GPIO_DATA) bit reflects the current state of the GPIO input.

Extreme care must be taken on strap input pins that may be used for General Purpose Inputs. The General Purpose Inputs must be conditioned or otherwise disabled such that they do not drive incorrect strap input values during the strap loading time.

When configured as an output, the output buffer type is selected by the corresponding bit in the General Purpose IO Buffer Type Register (GPIO_BUF). Push/pull and open-drain output buffers are supported for each GPIO.

When functioning as an open-drain driver, the GPIO output pin is driven low when the corresponding bit in the General Purpose IO Data Register (GPIO_DATA) is cleared to 0 and is not driven when set to 1.

When a GPIO is enabled as a push/pull output, the value output to the GPIO pin is controlled via the corresponding bit in the General Purpose IO Data Register (GPIO_DATA).

When a GPIO is set to an output, the pin’s pull-up is disabled, however the pin’s input buffer remains enabled. A read of the General Purpose IO Data Register (GPIO_DATA) still returns the state of the GPIO input (not the previous value written to the register).

Note: Unless otherwise determined by the normal pin function, the pull-up is only enabled if the pin is set as a GPIO input.

APPLICATION NOTE: Internal pull-up resistors prevent unconnected inputs from floating. Do not rely on internal resistors to drive signals external to the device. When connected to a load that must be pulled high, an external resistor must be added.

When functioning as an output, the pin’s pull-up is disabled. An open-drain output may require an external pull-up depending on the application.

5.20.1 GPIO INTERRUPTS

Each GPIO provides the ability to trigger a unique GPIO interrupt in the General Purpose IO Interrupt Status Register (GPIO_INT_STS). Reading the GPIO Interrupt (**GPIO_INT**) bits of this register provides the current status of the corresponding interrupt. Each interrupt is enabled by setting the corresponding General Purpose IO Interrupt Enable Register (GPIO_INT_EN) bit. The GPIO Controller aggregates the enabled interrupt values into an internal signal that is sent to the main interrupt logic and is reflected via the GPIO Interrupt bit in the Interrupt Status Register.

Note: Bits in the General Purpose IO Interrupt Status Register (GPIO_INT_STS) are set by the interrupt events regardless of the value of the corresponding bits in the General Purpose IO Interrupt Enable Register (GPIO_INT_EN).

As interrupts, GPIO inputs are level sensitive and must be active for greater than ~85 nS to be recognized (sampled by 2 consecutive 25MHz edges).

APPLICATION NOTE: Upon reset, GPIOs that were outputs may generate an active interrupt status as the system settles - typically when a low GPIO pin slowly rises due to the internal pull-up. The interrupt status bits within the General Purpose IO Interrupt Status Register (GPIO_INT_STS) should be cleared as part of the device initialization software routine.

5.20.1.1 GPIO Interrupt Polarity

The interrupt polarity can be set for each individually via the General Purpose IO Interrupt Polarity Register (GPIO_INT_POL). When set, a high logic level on the GPIO pin will set the corresponding interrupt bit in the General Purpose IO Interrupt Status Register (GPIO_INT_STS). When cleared, a low logic level on the GPIO pin will set the corresponding interrupt bit.

5.20 通用输入输出端口

GPIO 控制器由 7 个与其他引脚共用的可编程输入/输出引脚组成（请参阅 [第 3.4 节“引脚复用功能”](#)）。这些引脚可通过 GPIO 寄存器分别进行配置。

要使引脚用作通用输入/输出端口，必须通过通用输入/输出使能寄存器（**GPIO_使能**）中的相应位使能该通用输入/输出端口。

通过通用输入/输出方向寄存器（GPIO_DIR）将引脚配置为输入时，其上拉功能会使能。相应的通用输入/输出数据寄存器（GPIO_DATA）位会反映 GPIO 输入的当前状态。

对于可能用作通用输入的绑带输入引脚，必须格外谨慎。必须对通用输入进行调节或以其他方式禁用，以免它们在绑带加载时间内驱动错误的绑带输入值。

配置为输出时，输出缓冲器类型由通用输入/输出缓冲器类型寄存器（GPIO_BUF）中的相应位选择。每个 GPIO 均支持推挽输出缓冲器和开漏输出缓冲器。

作为开漏驱动器工作时，如果通用输入/输出数据寄存器（GPIO_DATA）中的相应位清零为 0，则 GPIO 输出引脚被驱动为低电平；如果该位设为 1，则不驱动该引脚。

将 GPIO 使能为推挽输出时，输出到 GPIO 引脚的值由通用输入/输出数据寄存器（GPIO_DATA）中的相应位控制。

将 GPIO 设为输出时，该引脚的上拉会被禁用，但引脚的输入缓冲器仍保持使能。读取通用输入/输出数据寄存器（GPIO_DATA）时，返回的仍是 GPIO 输入的状态（而非之前写入该寄存器的值）。

注意：除非正常引脚功能另有规定，否则仅当引脚设置为 GPIO 输入。

应用说明：内部上拉电阻可防止未连接的输入端处于悬空状态。请勿依赖内部电阻驱动器外部的信号。连接到必须被上拉至高电平的负载时，必须加装外部电阻。

作为输出时，引脚的上拉功能会被禁用。开漏输出可能需要根据应用情况加装外部上拉电阻。

5.20.1 通用输入/输出中断

每个通用输入/输出都可以在通用输入输出中断状态寄存器（GPIO_INT_STS）中触发一个独立的通用输入/输出中断。读取此寄存器中的通用输入/输出中断（**通用输入/输出_INT**）位，可获取相应中断的当前状态。设置相应的通用输入输出中断使能寄存器（GPIO_INT_EN）位即可使能每个中断。GPIO 控制器将已使能的中断值汇总为一个内部信号，并将其发送至主中断逻辑；该信号也会通过中断状态寄存器中的通用输入/输出中断位反映出来。

注意：通用输入输出中断状态寄存器（GPIO_INT_STS）中的位由中断事件置位，与通用输入输出中断使能寄存器（GPIO_INT_EN）中对应位的值无关。

作为中断时，通用输入/输出输入端为电平敏感，必须处于活动状态超过 ~85 nS 才能被识别（由连续两个 25MHz 边沿采样）。

应用说明：复位后，随着系统逐渐稳定，曾配置为输出的通用输入输出端口可能会产生活动状态的中断状态——通常是低电平 GPIO 引脚因内部上拉而缓慢升高时。应在设备初始化软件例程中清除通用输入输出中断状态寄存器（GPIO_INT_STS）中的中断状态位。

5.20.1.1 GPIO 中断极性

可通过通用输入输出中断极性寄存器 (GPIO_INT_POL) 单独设置每个中断的极性。设置后，GPIO 引脚上的高逻辑电平会在通用输入输出中断状态寄存器 (GPIO_INT_STS) 中置位相应的中断位。清除后，GPIO 引脚上的低逻辑电平会置位相应的中断位。

5.20.2 GPIO ALTERNATE USAGE

Each GPIO has the ability to be used as for an alternate function. Once enabled as a GPIO, the output source is selected by the bits in the General Purpose IO Data Select 1 Register (GPIO_DATA_SEL1) and in the General Purpose IO Data Select 2 Register (GPIO_DATA_SEL2). The output buffer type (General Purpose IO Buffer Type Register (GPIO_BUF) applies to all of output sources.

5.21 Self-Test Frame Generation and Checking

The device is capable of generating and checking frames.

Normally, when connected to a link partner, a valid link-up indication must be established before test frames are generated. For certain cases, for example 10BASE-T compliance testing, a link partner may not be present. The Force_-self_test_pgen_en bit in the Self-Test PGEN Enable Register may be used to override the link-up requirement.

If desired, the device can be connected to a loopback connector or a loopback cable. Pair A should be connected to pair B, and pair C to pair D, as shown in [Figure 5-11](#).

If a loopback connector is to be used,

- Auto-negotiation should be disabled, full duplex set and the desired link speed set via the Basic Control Register.
- For 10BASE-T and 100BASE-TX:
 - Auto-MDIX should be disabled and the desired configuration (MDI vs. MDIX) selected via the swapoff and mdi_set bits in the Digital Debug Control 1 Register. Selecting MDI vs. MDIX will test different sections of the PHY.
- For 1000BASE-T:
 - Set the Leader-Follower configuration to leader by setting the Master/Slave Manual Configuration Enable and Master/Slave Manual Configuration Value bits in the Auto-Negotiation Master Slave Control Register.
 - Enable the 1000BASE-T connector loopback by setting the Ext_lpbk bit in the Reserved Register.

Two frame generation modes can be enabled. Single-Stream and Multiple-Stream.

5.21.1 SINGLE-STREAM FRAME GENERATION

In Single-Stream mode frames are sent at line-rate with a minimum Inter Frame Gap (IFG).

The Single-Stream frame has a length between 64 and 1518 bytes (including the MAC source and destination addresses, the Type/Length Field and the FCS). The frame length can be randomly selected or can be set to a fixed value by using the Self_test_packet_fixed_size_enable and Self_test_packet_fixed_size_length fields in the Self-Test PGEN Configuration Register. Note the Self_test_packet_fixed_size_length field does not include the MAC source and destination addresses, the Type/Length Field or the FCS.

The frame's MAC source and destination, Type/Length Field and payload can be set to all ones or all zeros as selected by the Self_test_packet_type[1:0] field in the Self-Test Enable Register.

If not set to all ones or all zeros, the MAC source and destination can be fully random using the Self_test_packet_random_MAC field in the Self-Test PGEN Configuration Register. If not fully random, one of two MAC source and destination address pairs is randomly selected from the DES1/SRC1 Register or the DES2/SRC2 Register.

If not set to all ones or all zeros, the Type/Length Field is set to the frame length selected above (not including the 18 bytes of MAC source and destination addresses, the Type/Length Field and the FCS).

If not set to all ones or all zeros, the frame payload is random.

The preamble, SFD and FCS are normal values.

Single-Stream frame generation is enabled by setting a frame count into the Self-Test Packet Count LO Register and Self-Test Packet Count HI Register and then setting the following in order:

- Force_self_test_pgen_en bit in the Self-Test PGEN Enable Register if needed to bypass link-up
- Self_test_en bit in the Self-Test Enable Register
- Self_test_frame_cnt_en bit in the Self-Test Frame Count Enable Register
- Self_test_pgen_en bit in the Self-Test PGEN Enable Register

5.21.2 MULTIPLE-STREAM FRAME GENERATION

In Multiple-Stream mode bursts of frames are sent at line-rate with a minimum IFG and with a random idle time between each burst. EEE LPI can be enabled during the idle time.

5.20.2 GPIO 复用功能

每个 GPIO 均可用作替代功能。一旦启用为 GPIO，输出源由通用输入/输出数据选择 1 寄存器 (GPIO_DATA_SEL1) 和通用输入/输出数据选择 2 寄存器 (GPIO_DATA_SEL2) 中的位选择。输出缓冲器类型（通用输入/输出缓冲器类型寄存器 (GPIO_BUF)）适用于所有输出源。

5.21 自测试帧生成与检查

该器件能够生成和检查帧。

通常，连接到链路伙伴时，必须先建立有效的链路接通指示，才能生成测试帧。在某些情况下，例如进行 10BASE-T 合规性测试时，链路伙伴可能不存在。可使用自测试 PGEN 使能寄存器中的 Force_-self_test_pgen_en 位来覆盖链路建立要求。

如果需要，可以将器件连接到回环连接器或回环电缆。应将 A 对线连接到 B 对线，并将 C 对线连接到 D 对线，如[图 5-11所示](#)。

如果要使用回环连接器，

- 应禁用自动协商，设置为全双工，并通过基本控制寄存器设置所需的链路速度。
- 对于 10BASE-T 和 100BASE-TX:
 - 应禁用自动 MDIX，并通过数字调试控制 1 寄存器中的 swapoff 和 mdi_set 位选择所需配置（介质相关接口或介质无关交叉接口）。选择介质相关接口或介质无关交叉接口将测试 PHY 的不同部分。
- 对于 1000BASE-T:
 - 在自动协商主从控制寄存器中设置主从手动配置使能位和主/从手动配置值位，将主从配置设为主端。

- 在保留寄存器中设置 Ext_lpbk 位，以使能 1000BASE-T 连接器环回。

可以启用两种帧生成模式：单流和多流。

5.21.1 单流帧生成

在单流模式下，帧以线速发送，帧间隙（IFG）为最小值。

单流帧的长度介于 64 到 1518 字节之间（包括 MAC 源地址和目的地址、类型/长度字段以及帧校验序列）。帧长度可以随机选择，也可以通过自检 PGEN 配置寄存器中的 Self_test_packet_fixed_size_enable 和 Self_test_packet_fixed_size_length 字段设置为固定值。请注意，Self_test_packet_fixed_size_length 字段不包括 MAC 源地址和目的地址、类型/长度字段或帧校验序列。

帧的 MAC 源地址和目的地址、类型/长度字段以及有效载荷可根据自测试使能寄存器中的 Self_test_packet_type[1:0] field 字段的设置，全部置为 1 或全部置为 0。

如果未设置为全部 1 或全部 0，则可使用自检 PGEN 配置寄存器中的 Self_test_packet_random_MAC 字段，使 MAC 源地址和目的地址完全随机。如果不是完全随机，则会从 DES1/SRC1 寄存器或 DES2/SRC2 寄存器中随机选择一对 MAC 源地址和目的地址。

如果未设置为全 1 或全 0，则类型/长度字段将设为上方所选的帧长度（不包括 MAC 源地址和目的地址、类型/长度字段及 FCS 共 18 字节）。

如果未设置为全 1 或全 0，则帧载荷为随机数据。

前导码、帧起始定界符和 FCS 均为正常值。

通过在自检数据包计数低位寄存器和自检数据包计数高位寄存器中设置帧计数，然后按以下顺序进行设置，即可启用单流帧生成：

- 强制_自_检_PGEN_使能位（位于自测试 PGEN 使能寄存器中），如需绕过链路建立，则设置此位
- 自_检_使能位（位于自测试使能寄存器中）
- 自_测试_帧_计数_使能位于自测试帧计数使能寄存器中
- 自_测试_PGEN_使能位于自测试 PGEN 使能寄存器中

5.21.2 多流帧生成

在多流模式下，以线速发送突发帧，帧间隙最小，并且每个突发之间的空闲时间随机。可在空闲时间内使能节能以太网低功耗空闲。

The frame count for each burst is randomly selected as mask by the Frame Counter Mask Register.

The frame's MAC source and destination, Type/Length Field and payload can be set to all ones or all zeros as selected by the Self_test_packet_type[1:0] field in the Self-Test Enable Register.

If not set to all ones or all zeros, the MAC source and destination can be fully random using the Self_test_packet_random_MAC field in the Self-Test PGEN Configuration Register. If not fully random, one of two MAC source and destination address pairs is randomly selected at the beginning of each burst from the DES1/SRC1 Register or the DES2/SRC2 Register.

If not set to all ones or all zeros, the Type/Length Field is set to the frame length selected above (not including the 18 bytes of MAC source and destination addresses, the Type/Length Field and the FCS).

If not set to all ones or all zeros, the frame payload is random.

At the end of each burst a random idle time occurs. This is followed by a random period of `EEE LPI` (sleep) and then followed by a random wake-up recover time. The random time of sleep can be controlled with the Sleep Timer Counter Mask Register and the Sleep Counter Minimum Register. The sending of `EEE LPI` is enabled with the `St_no_lpi` bit in the Multi stream Control Register. Using the PHYs carrier sense `EEE` timer in lieu of the random wake-up timer can be enabled with the `Crs_use_timer` bit in the Multi stream Control Register.

Multiple-Stream mode is enabled by setting the following:

- `Force_self_test_pgen_en` bit in the Self-Test PGEN Enable Register if needed to bypass link-up
- `Self_test_en` bit in the Self-Test Enable Register
- `St_multi_stream_en` bit in the Multi stream Control Register
- `St_frame_ctrl_en` bit in the Multistream Start Register

Multiple-Stream mode runs continuously until disabled by clearing the `St_frame_ctrl_en` bit.

Optionally the Multiple-Stream mode can run for approximately 1 second by setting the `Ipef_en` bit in the Multi stream Control Register.

The cumulative count of frames that was sent for each destination / source pair can be read via the Self-Test Source 1/2 Count HI Extended, HI and LO Registers. These counters can be reset by setting and clearing the `clr_src_frame` bit in the Clear Source Counters Register.

5.21.3 FRAME CHECKING

There are three frame CRC checkers available, one on TX data from the MAC and two on the receive data from the line.

A count of good and errored frames from the MAC TX is available in the TX Correct Count HI Extended, HI and LO Registers and the TX Error Count HI and LO Registers. These counters are enabled by setting the `GMII_TX_CRC_check_en` bit in the Self-Test Enable Register. Clearing the `GMII_TX_CRC_check_en` bit clears the counters.

Two sets of good and errored frames counters received are available in the Self-Test Stream 1/2 Correct Count HI Extended, HI and LO Registers and the Self-Test Stream 1/2 Error Count HI and LO Registers.

Normally both sets of counters will count all frames. When the `St2_trig_en` or `St1_trig_en` bit in the Multi stream Control Register is set, the corresponding counter set will be qualified by matching the MAC destination address in the frame to the source address set in the DES1/SRC1 Register or in the DES2/SRC2 Register.

APPLICATION NOTE: In a PHY to PHY test, the DES1 and SRC1 (and DES2 and SRC2) settings in each PHY would be reversed such that PHY A sends frames to the MAC destination address that PHY B has set as it source. Hence the counters qualify the received frame's destination address with the local SRC addresses.

These counters are enabled by setting the `Self_test_CRC_checker_enable` bit in the Self-Test Enable Register. Clearing the `Self_test_CRC_checker_enable` bit clears the counters.

The counters have an option to automatically clear on a link down event. This is enabled by setting the `Self_test_clear_counters_on_link_down` bit in the Self-Test Enable Register.

每个突发的帧计数会随机选取，并由帧计数器屏蔽寄存器进行屏蔽。

帧的 MAC 源地址和目的地址、类型/长度字段及有效载荷可设置为全 1 或全 0，具体由自测试使能寄存器中的 `Self_test_packet_type[1:0]` 字段选择。

如果未设置为全 1 或全 0，则可使用自检 PGEN 配置寄存器中的 `Self_test_packet_random_MAC` 字段将 MAC 源地址和目的地址设置为完全随机。如果未设置为完全随机，则在每个突发开始时，从 DES1/SRC1 寄存器或 DES2/SRC2 寄存器中随机选择两组 MAC 源地址和目的地址。

如果未设置为全 1 或全 0，则类型/长度字段将设为上述选定的帧长度（不包括 18 字节的 MAC 源地址和目的地址、类型/长度字段以及帧校验序列）。

如果未设置为全 1 或全 0，则帧载荷为随机值。

每次突发结束时都会出现一段随机的空闲时间。随后是随机时长的节能以太网低功耗空闲（休眠），接着是随机的唤醒恢复时间。休眠的随机时长可通过休眠计时器计数器屏蔽寄存器和休眠计数器最小值寄存器进行控制。通过多流控制寄存器中的 `St_no_lpi` 位使能节能以太网低功耗空闲的发送。通过多流控制寄存器中的 `Crs_use_timer` 位，可启用 PHY 载波侦听 `EEE` 计时器，以替代随机唤醒计时器。

通过设置以下内容可启用多流模式：

- Self-Test PGEN 使能寄存器中的 `Force_self_test_pgen_en` 位（如需绕过链路建立）
- 自测试使能寄存器中的 `Self_test_en` 位
- 多流控制寄存器中的 `St_multi_stream_en` 位
- 多流启动寄存器中的 `St_帧控制_使能位`

多流模式会持续运行，直到清除 `St_帧控制_使能位` 将其禁用。

也可以通过设置多流控制寄存器中的 `Ipef_使能位`，使多流模式运行约 1 秒。

每个目的地址/源地址对已发送的帧累计计数，可通过自测试源 1/2 计数高位扩展、高位和低位寄存器读取。设置并清除清除源计数器寄存器中的 `clr_源_帧位`，即可复位这些计数器。

5.21.3 帧检查

共有三个帧 CRC 校验器：一个用于来自媒体访问控制器的 TX 数据，另外两个用于来自线路的接收数据。

MAC 发送端正确帧和错误帧的计数值可在发送正确计数高位扩展、高位和低位寄存器以及发送错误计数高位和低位寄存器中查看。通过在自测试使能寄存器中设置 `GMII_TX_CRC_check_en` 位，可使能这些计数器。清除 `GMII_TX_CRC_check_en` 位会清零这些计数器。

接收的正确帧和错误帧有两组计数器，分别位于自测试流 1/2 正确计数高位扩展、高位和低位寄存器以及自测试流 1/2 错误计数高位和低位寄存器中。

通常，两组计数器都会对所有帧进行计数。当设置多流控制寄存器中的 `St2_trig_en` 位或 `St1_trig_en` 位时，相应的计数器组将通过匹配帧中的 MAC 目的地址与 DES1/SRC1 寄存器或 DES2/SRC2 寄存器中设置的源地址来筛选帧。

应用说明：在 PHY 到 PHY 测试中，每个 PHY 上的 DES1 和 SRC1（以及 DES2 和 SRC2）设置会互换，因此 PHY A 会向 PHY B 设置为源地址的 MAC 目的地址发送帧。因此，计数器会根据本地 SRC 地址筛选接收帧的目的地址。

通过在自测试使能寄存器中设置自测试_测试_CRC_校验器_使能位来启用这些计数器。清除自测试 CRC 校验器使能位会清零这些计数器。 _ _ _ _

计数器可设置为在链路断开事件发生时自动清零。设置自测试使能寄存器中的 `Self_test_clear_counters_on_link_down` 位即可启用此功能。

5.22 Power Management

The device incorporates a number of power-management modes and features that provide methods to consume less energy. These are discussed in the following sections.

5.22.1 SMART POWER SAVING

For shorter cable lengths (< ~70 meters) the signal to noise ratio is sufficiently high to allow the reduction of ADC resolution as well as DPS taps, Based on the detected cable length, the device automatically reduces power consumption by approximately 20mW.

5.22.2 ENERGY-DETECT POWER-DOWN MODE (EDPD)

The device supports an Energy-Detect-Power-Down (EDPD) mode to save power when there is no link partner sending signals.

In EDPD Mode, the device shuts down all transceiver blocks, except for the transmitter and energy detect circuits. Power can be reduced further by extending the time interval between the transmissions of link pulses to check for the presence of a link partner. The periodic transmission of link pulses is needed to ensure the device and its link partner, when operating in the same low-power state and with Auto MDI/MDI-X disabled, can wake up when the cable is connected between them.

By default, EDPD mode is disabled after power-up. EDPD is enabled by setting the p_edpd_en in the EDPD Control Register within the MMD address space.

EDPD operation may be adjusted via the p_edpd_mask_timer[1:0], p_edpd_timer[1:0] and p_EDPD_random_dis fields in the EDPD Control Register within the MMD address space.

The energy detection change status can be read from the Interrupt Status Register. The current energy detection status can be read from the EDPD low power bit in the EDPD Control Register.

While the p_edpd_en bit is set, the cable link status will be down, and the energy detection normally monitors pairs A and B for energy. If the link speed is forced to 1000Mbps (by disabling auto-negotiation and setting the speed manually), the energy detection monitors all four pairs. For cable diagnostic purposes, individual wire pairs may be monitored by forcing the link speed to 1000Mbps and selecting the wire pair using the EDPD Wire Pair Selection bits in Analog Control Register 8.

The device’s PLL is normally enabled during EDPD. It can be set to be disabled during EDPD by setting the DGT_edpd_pll_dis bit in MMD2 Spare 115 Register.

Normally, previous register setting are maintained when EDPD mode is cleared. With the PLL disabled, a device reset occurs following the removal of EDPD (or if the DGT_edpd_pll_dis bit is cleared during EDPD), in which case register settings will return to their defaults.

APPLICATION NOTE: If the reset is caused by the clearing of the DGT_edpd_pll_dis bit, the p_edpd_en bit will be cleared due to the device reset.

5.22.3 SOFTWARE POWER-DOWN MODE (SPD)

The device supports a software power down (SPD) mode. This mode is used to power down the device when it is not in use after power-up. Software power-down mode is enabled by writing a one to the Power Down bit in the Basic Control Register. The device may also be placed into software power-down by default by setting the **MODE[4:0]** configuration straps to ‘0100x’. The device exits the SPD state after a zero is written to the Power Down bit.

In the SPD state, the device disables most internal functions.

During SPD, the crystal oscillator and PLL are enabled and the internal (125MHz and 250MHz) clocks are gated, The standard registers (0 through 31) and the MII Management Interface operate using the crystal clock.

Previous register settings are maintained during and following the removal of SPD.

The following remain operational during SPD:

- MII Management Interface
 - Only access to the standard registers (0 through 31) is supported.
 - Access to MMD address spaces other than MMD address space 1 is possible if the spd_clock_gate_override bit is set.
 - Access to MMD address space 1 is not possible.
- Voltage Regulator Controller (LDO)

5.22 电源管理

该器件集成了多种电源管理模式和功能， 可通过不同方法降低能耗。以下各节将对此进行讨论。

5.22.1 智能节能

对于较短的电缆长度（< ~70 米）， 信噪比足够高， 因此可以降低模数转换器分辨率以及 DPS 抽头。器件会根据检测到的电缆长度自动将功耗降低约 20 mW。

5.22.2 能量检测掉电模式（EDPD）

当没有链路伙伴发送信号时， 器件支持能量检测掉电（EDPD）模式以节省功耗。

在 EDPD 模式下， 器件会关闭所有收发器模块， 但发射器和能量检测电路除外。通过延长发送链路脉冲以检测链路伙伴是否存在的时间间隔， 可以进一步降低功耗。定期发送链路脉冲是必要的， 以确保器件及其链路伙伴在处于相同低功耗状态且禁用自动 MDI/MDI-X 时， 能在连接两者的电缆接入后唤醒。

默认情况下，EDPD 模式在上电后处于禁用状态。通过在 MMD 地址空间内的 EDPD 控制寄存器中设置 p_edpd_en， 可启用 EDPD。

可通过 MMD 地址空间内 EDPD 控制寄存器中的 p_edpd_mask_timer[1:0], p_edpd_timer[1:0] 和 p_EDPD_random_dis 字段调整 EDPD 操作。

能量检测状态的变化可通过中断状态寄存器读取。当前的能量检测状态可通过 EDPD 控制寄存器中的 EDPD 低功耗位读取。

当 p_edpd_en 位设置为 1 时， 电缆链路状态将为断开， 能量检测通常会监测 A 和 B 线对的能量。如果将链路速度强制设为 1000 Mbps（通过禁用自动协商并手动设置速率）， 能量检测将监测全部四对线。出于电缆诊断目的， 可通过将链路速度强制设为 1000 Mbps， 并使用模拟控制寄存器 8 中的 EDPD 线对选择位选择线对， 来监测单个线对。

在 EDPD 期间， 器件的锁相环通常处于使能状态。通过设置 MMD2 Spare 115 寄存器中的 DGT_edpd_pll_dis 位， 可将其设为在 EDPD 期间禁用。

通常， 清除 EDPD 模式后会保留先前的寄存器设置。如果锁相环已禁用， 则在退出 EDPD 后（或在 EDPD 期间清除 DGT_edpd_pll_dis 位时）会发生设备复位， 此时寄存器设置将恢复为默认值。

应用说明： 如果复位是由清除 DGT_edpd_pll_dis 位引起的， 则 p_edpd_en 位将因设备复位而被清除。

5.22.3 软件断电模式 (SPD)

该器件支持软件断电 (SPD) 模式。此模式用于在上电后器件不使用时将其断电。向基本控制寄存器中的掉电位写入 1， 即可启用软件断电模式。也可以通过将 **模式[4:0]** 配置绑带设置为 “0100x”， 使器件默认进入软件断电模式。向掉电位写入 0 后， 器件将退出 SPD 状态。

在 SPD 状态下， 器件会禁用大多数内部功能。

在 SPD 期间， 晶体振荡器和锁相环保持使能状态， 内部（125 兆赫和 250 兆赫）时钟则被门控。标准寄存器（0 至 31）和 MII 管理接口使用晶体时钟运行。

在 SPD 移除期间及移除后， 之前的寄存器设置会保持不变。

SPD 期间以下功能仍可正常运行：

- MII 管理接口
 - 仅支持访问标准寄存器（0 至 31）。
 - 如果设置了 spd_clock_gate_override 位， 则可以访问 MMD 地址空间 1 以外的 MMD 地址空间。
 - 无法访问 MMD 地址空间 1。
- 电压调节器控制器（低压差稳压器）

- The LDO controller can be disabled by setting the active low LDO enable bit in Analog Control Register 11. An external source of 1.1V is necessary for operation in this case.
- PLL
 - Normally the PLL is enabled during SPD. It may be disabled by setting the spd_pll_disable bit described below.
- Crystal Oscillator
 - Normally the Crystal Oscillator is enabled during SPD. It may be disabled by setting the enXTALb bit described below.
- Bandgap
 - This is always enabled.
- Internal Slow Oscillator
 - This is always enabled.

The following are normally disabled during SPD:

- ADC / PGA / TX / common bias circuits
- DLL
- TX and RX clocks
 - If the above mentioned spd_clock_gate_override bit is set, TX and RX clocks would be enabled. They may alternately be stopped by setting the Isolate (PHY_ISO) bit in the Basic Control Register.
- **CLK125_NDO** pin
 - If the above mentioned spd_clock_gate_override bit is set, **CLK125_NDO** would be enabled (if previously enabled for clock output). It may alternately be disabled by clearing the clk125 Enable bit in Common Control Register.

5.22.3.1 SPD Extra Power Savings

To achieve a lower power usage, the PLL maybe disable during SPD by setting the spd_pll_disable bit in the Digital Debug Control 2 Register prior to entering SPD. The device may also be placed into software power-down with the PLL disabled by default by setting the **MODE[4:0]** configuration straps to ‘01001’.

With the PLL disabled, a device reset occurs following the removal of SPD (or if the spd_pll_disable bit is cleared during SPD). Register settings will return to their defaults, determined by the Operation Mode Strap Override Low Register and the Operation Mode Strap Override High Register.

APPLICATION NOTE: If either the spd_pll_dis_mode or spd_pll_en_mode bits in the Operation Mode Strap Override Low Register are set, the device will return to SPD mode, potentially with the PLL disabled. In order to avoid this logical loop, software should clear the spd_pll_dis_mode and spd_pll_en_mode bits before exiting SPD.

To further reduce power usage, the crystal oscillator maybe disable by setting the enXTALb bit in Analog Control Register 1 after setting the spd_pll_disable bit and entering SPD.

Since the MII Management Interface operates using the crystal clock, once this bit is set, the device will become inaccessible. A pin reset or power cycle is required to resume operation.

5.23 Wake-On-LAN

Wake-On-LAN (WOL) is normally a MAC-based function to wake up a host system (for example, an Ethernet end device, such as a PC) that is in standby power mode. Wake-up is triggered by receiving and detecting a special packet (commonly referred to as the “magic packet”) that is sent by the remote link partner. The device can perform the same WOL function if the MAC address of its associated MAC device is entered into the device registers for magic-packet detection. When the device detects the magic packet, it wakes up the host by driving its power management event (PME_N) output pin low.

By default, the WOL function is disabled. It is enabled by setting the enabling bit and configuring the associated registers for the selected wake-up detection method.

The device provides four methods to trigger a wake-up:

- Magic-packet detection
- Customized-packet detection
- Link status change detection

- 可通过设置模拟控制寄存器 11 中低电平有效的 LDO 使能位来禁用 LDO 控制器。在此情况下，运行需要外部 1.1V 电源。
- 锁相环
 - 通常，锁相环在 SPD 期间处于使能状态。可通过设置下文所述的 spd_pll_disable 位将其禁用。
- 晶体振荡器
 - 通常，晶体振荡器在 SPD 期间处于使能状态。可通过设置下文所述的 enXTALb 位将其禁用。

- 带隙
 - 此项始终处于使能状态。
- 内部慢速振荡器
 - 此项始终使能。

以下功能在 SPD 期间通常处于禁用状态：

- 模数转换器 / 可编程增益放大器 / 发送 / 公共偏置电路
- 延迟锁定环
- 发送和接收时钟
 - 如果设置了上述 spd_clock_gate_override 位，则发送和接收时钟将启用。也可以通过设置基本控制寄存器中的隔离（PHY_ISO）位来停止这些时钟。
- **CLK125_NDO** 引脚
 - 如果设置了上述 spd_clock_gate_override 位，则会使能 **CLK125_NDO**（如果之前已将其使能为时钟输出）。也可以通过清除通用控制寄存器中的 clk125 使能位来禁用它。

5.22.3.1 SPD 额外节能

要降低功耗，可在进入 SPD 模式前，通过设置数字调试控制 2 寄存器中的 spd_pll_disable 位，在 SPD 期间禁用锁相环。也可以将**模式[4: 0]**配置绑带设为“01001”，使器件进入软件断电状态，并默认禁用锁相环。

禁用锁相环后，退出 SPD（或在 SPD 期间清除 spd_pll_disable 位）时会发生设备复位。寄存器设置将恢复为默认值，这些默认值由操作模式绑带覆盖低位寄存器和操作模式绑带覆盖高位寄存器决定。

应用说明：如果操作模式绑带覆盖低位寄存器中的 spd_pll_dis_mode 或 spd_pll_en_mode 位被设置，器件将返回 SPD 模式，此时锁相环可能处于禁用状态。为避免这一逻辑循环，软件应在退出 SPD 前清除 spd_pll_dis_mode 和 spd_pll_en_mode 位。

为进一步降低功耗，可在设置 spd_pll_disable 位并进入 SPD 后，通过设置模拟控制寄存器 1 中的 enXTALb 位来禁用晶体振荡器。由于 MII 管理接口使用晶体时钟运行，因此设置此位后，器件将无法访问。要恢复运行，必须执行引脚复位或电源循环。

5.23 网络唤醒

网络唤醒（WOL）通常是一种基于 MAC 的功能，用于唤醒处于待机电源模式的主机系统（例如以太网终端设备，如个人电脑）。远端链路伙伴发送的特殊数据包（通常称为“魔术包”）被接收并检测到后，即会触发唤醒。如果将关联的 MAC 设备的 MAC 地址写入器件寄存器以用于魔术包检测，该器件也可执行相同的 WOL 功能。检测到魔术包后，器件会将其电源管理事件（PME_N）输出引脚拉低，从而唤醒主机。

默认情况下，网络唤醒功能处于禁用状态。通过设置使能位并配置所选唤醒检测方法对应的寄存器，即可启用该功能。

该器件提供四种触发唤醒的方法：

- 魔术包检测
- 自定义数据包检测
- 链路状态变化检测

- Energy Detection

5.23.1 MAGIC-PACKET DETECTION

The magic packet’s frame format starts with 6 bytes of 0xFFh and is followed by 16 repetitions of the MAC address of its associated MAC device (local MAC device).

When the magic packet is detected from its link partner, the device asserts its PME_N output pin low.

The following MMD Address 2h registers are provided for magic-packet detection:

- Magic-packet detection is enabled by writing a ‘1’ to the En_mpkt_pmen bit in the Wake-On-LAN Control Register (MMD Address 2h, Register 10h, Bit [6]).
- The MAC address (for the local MAC device) is written to and stored in the Wake-On-LAN-MAC-LO Register, Wake-On-LAN-MAC-MI Register and Wake-On-LAN-MAC-HI Register (MMD Address 2h, Registers 11h – 13h).

The device does not generate the magic packet. The magic packet must be provided by the external system.

Magic packet detection status can be read from the Wake-on-LAN Magic Packet Receive Status Register.

Frame preemption reassembly (Section 5.11) is not supported for magic packet detection. The magic packet detection logic will ignore any packet that does not contain the standard SFD.

5.23.2 CUSTOMIZED-PACKET DETECTION

The customized packet has associated register/bit masks to select which byte, or bytes, of the first 64 bytes of the packet to use in the CRC calculation. After the device receives the packet from its link partner, the selected bytes for the received packet are used to calculate the CRC. The calculated CRC is compared to the expected CRC value that was previously written to and stored in the device registers. If there is a match, the device asserts its PME_N output pin low.

Four customized packets are provided to support four types of wake-up scenarios. A dedicated set of registers is used to configure and enable each customized packet.

The following MMD registers are provided for customized-packet detection:

- Each of the four customized packets is enabled via the Wake-On-LAN Control Register (MMD Address 2h, Register 10h):
 - Bit [2] // For customized packets, type 0
 - Bit [3] // For customized packets, type 1
 - Bit [4] // For customized packets, type 2
 - Bit [5] // For customized packets, type 3
- 32-bit expected CRCs are written to and stored in:
 - MMD Address 2h, Registers 14h – 15h // For customized packets, type 0
 - MMD Address 2h, Registers 16h – 17h // For customized packets, type 1
 - MMD Address 2h, Registers 18h – 19h // For customized packets, type 2
 - MMD Address 2h, Registers 1Ah – 1Bh // For customized packets, type 3
- Masks to indicate which of the first 64-bytes to use in the CRC calculation are set in:
 - MMD Address 2h, Registers 1Ch – 1Fh // For customized packets, type 0
 - MMD Address 2h, Registers 20h – 23h // For customized packets, type 1
 - MMD Address 2h, Registers 24h – 27h // For customized packets, type 2
 - MMD Address 2h, Registers 28h – 2Bh // For customized packets, type 3

Customized packet detection status can be read from the Wake-on-LAN Custom Packet Receive Status Register.

Frame preemption reassembly (Section 5.11) is not supported for customized packet detection. The customized packet detection logic will ignore any packet that does not contain the standard SFD.

5.23.3 LINK STATUS CHANGE DETECTION

If link status change detection is enabled, the device asserts its PME_N output pin low whenever there is a link status change using the following bits in the Wake-On-LAN Control Register (MMD Address 2h, Register 10h):

- Bit [0] - En_link_up_pmen // For link-up detection
- Bit [1] - En_link_dn_pmen // For link-down detection

The link change status can be read from the Interrupt Status Register. The current link status can be read from the Basic Status Register.

- 能量检测

5.23.1 魔术包检测

魔术包的帧格式以 6 字节的 0xFFh 开头，后跟其关联的 MAC 设备（本地 MAC 设备）的 MAC 地址重复 16 次。

检测到来自链路伙伴的魔术包时，器件会将其 PME_N 输出引脚拉低。

提供以下 MMD 地址 2h 寄存器用于魔术包检测：

- 通过在网络唤醒控制寄存器（MMD 地址 2h，寄存器 10h，位 [6]）的 En_mpkt_pmen 位写入 “1”，即可使能魔术包检测。
- MAC 地址（本地 MAC 设备的地址）写入并存储在网络唤醒 MAC 低位寄存器、网络唤醒 MAC 中位寄存器和网络唤醒 MAC 高位寄存器中（MMD 地址 2h，寄存器 11h – 13h）。

该器件不会生成魔术包。魔术包必须由外部系统提供。

可从网络唤醒魔术包接收状态寄存器读取魔术包检测状态。

魔术包检测不支持帧抢占重组（第 5.11 节）。魔术包检测逻辑将忽略任何不包含标准帧起始定界符的数据包。

5.23.2 自定义数据包检测

自定义数据包具有相应的寄存器/位掩码，用于选择数据包前 64 个字节中的一个或多个字节，以用于 CRC 计算。器件从链路伙伴接收到数据包后，将使用接收数据包中选定的字节计算 CRC。计算出的 CRC 将与先前写入并存储在设备寄存器中的预期 CRC 值进行比较。如果匹配，器件会将其 PME_N 输出引脚拉低。

提供四种自定义数据包，以支持四种唤醒场景。每种自定义数据包都使用一组专用寄存器进行配置和使能。

提供以下 MMD 寄存器用于自定义数据包检测：

- 四种自定义数据包均通过网络唤醒控制寄存器（MMD 地址 2h，寄存器 10h）使能： -

- | | |
|---------------------------------------|-------------------|
| 位 [2] | // 自定义数据包请填写 0 |
| - 位 [3] | // 自定义数据包请填写 1 |
| - 位 [4] | // 自定义数据包请填写 2 |
| - 位 [5] | // 对于自定义数据包，输入 3 |
| • 32 位预期 CRC 写入并存储在： | |
| - MMD 地址 2h，寄存器 14h – 15h | // 对于自定义数据包，输入 0 |
| - MMD 地址 2h，寄存器 16h – 17h | // 对于自定义数据包，输入 1 |
| - MMD 地址 2h，寄存器 18h – 19h | // 对于自定义数据包，类型 2 |
| - MMD 地址 2h，寄存器 1Ah – 1Bh | // 对于自定义数据包，类型 3 |
| • 用于指示 CRC 计算中采用前 64 个字节中的哪些字节的掩码设置在： | |
| - MMD 地址 2h，寄存器 1Ch – 1Fh | // 对于自定义数据包，类型为 0 |
| - MMD 地址 2h，寄存器 20h – 23h | // 对于自定义数据包，类型为 1 |
| - MMD 地址 2h，寄存器 24h – 27h | // 对于自定义数据包，类型为 2 |
| - MMD 地址 2h，寄存器 28h – 2Bh | // 对于自定义数据包，类型为 3 |

可从网络唤醒自定义数据包接收状态寄存器读取自定义数据包检测状态。

不支持对自定义数据包检测进行帧抢占重组（第 5.11 节）。自定义数据包检测逻辑会忽略任何不包含标准帧起始定界符的数据包。

5.23.3 链路状态变化检测

如果启用了链路状态变化检测，则每当链路状态发生变化时，器件都会使用网络唤醒控制寄存器（MMD 地址 2h，寄存器 10h）中的以下位，将其 PME_N 输出引脚拉低：

- 位 [0] - 使能_链路_up_pmen // 用于链路建立检测
- 位 [1] - 使能_链路_断开_使能 // 用于链路断开检测

可从中断状态寄存器读取链路变化状态。可从基本状态寄存器读取当前链路状态。

Note: Clearing the link change status by reading the Interrupt Status Register DOES NOT clear the link status change detection that asserts **PME_N**. In order to clear the **PME_N** indication, the En_link_up_pmen and/or En_link_dn_pmen control bits must be cleared.

5.23.4 ENERGY STATUS CHANGE DETECTION

During Energy Detect Power Down ([Section 5.22.2](#)) if the energy status change detection is enabled, the device asserts its **PME_N** output pin low whenever there is an energy detection status change using the following bits in the Wake-On-LAN Control Register (MMD Address 2h, Register 10h):

- Bit [11] - En_energy_detect_pmen // For energy detected
- Bit [12] - En_energy_off_pmen // For energy not detected

The energy detection change status can be read from the Interrupt Status Register. The current energy detection status can be read from the EDPD low power bit in the Analog Control Register 8.

5.23.5 PME_N OUTPUT SIGNAL

The **PME_N** output signal may be mapped to any enabled GPIO (including but not limited to **INT_N** and **LED1**) by setting the corresponding bits in the General Purpose IO Data Select 1 Register (GPIO_DATA_SEL1) or the General Purpose IO Data Select 2 Register (GPIO_DATA_SEL2).The GPIO needs to be enabled and its direction set as an output via the General Purpose IO Enable Register (**GPIO_EN**) and the General Purpose IO Direction Register (GPIO_DIR).

The PME buffer type is selectable between open-drain and push-pull and is configured by the GPIO Buffer Type (GPIO_O_BUF) field in the General Purpose IO Buffer Type Register (GPIO_BUF). The default is open-drain.

The PME Polarity bit in the Output Control Register sets the **PME_N** level to active high or active low. The default is active low.

The Pmen_out field in the Wake-On-LAN Control Register (MMD Address 2h, Register 10h, Bits [15:14]) defines the output functions for PME.

When asserted, the **PME_N** output is de-asserted by disabling the register bit that enabled the PME trigger source (magic packet, customized packet, link status change, energy detect status change).

5.24 NAND Tree Support

The device provides parametric NAND tree support for fault detection between chip I/Os and board. NAND tree mode is enabled at power-up/reset with the **MODE**[4:0] strap-in pins set to ‘00100’.

The NAND tree consists of a series chain of NAND gates with each input pin NANDed with the output of the previous NAND. [Table 5-15](#) lists the NAND tree pin order. Input (1) is the start of the tree and is NANDed with a constant “1”. Note the LAN8840 has an additional final inversion.

TABLE 5-15: NAND TREE TEST PIN ORDER

Pin	Description
LED5	Input
LED4	Input
LED3	Input
LED2	Input
LED1	Input
TXD0	Input
TXD1	Input
TXD2	Input
TXD3	Input
TXC	Input
TX_CTL	Input
RXD3	Input
RXD2	Input

注意：读取中断状态寄存器以清除链路变化状态，并不会清除触发 **PME_N** 的链路状态变化检测。若要清除 **PME_N** 指示，必须清除 En_link_up_pmen 和/或 En_link_dn_pmen 控制位。

5.23.4 能量状态变化检测

在能量检测掉电（[第 5.22.2 节](#)）期间，如果启用了能量状态变化检测，则每当能量检测状态发生变化时，器件都会使用唤醒局域网控制寄存器（MMD 地址 2h，寄存器 10h）中的以下位，将其 **PME_N** 输出引脚拉低：

- 位 [11] - 使能_能量_检测_PME // 检测到能量时
- 位 [12] - 使能_能量_关闭_PME // 未检测到能量时

能量检测状态的变化可从中断状态寄存器中读取。当前能量检测状态可通过模拟控制寄存器 8 中的 EDPD 低功耗位读取。

5.23.5 PME _N 输出信号

PME_N 输出信号可映射到任意已使能的通用输入/输出（包括但不限于 **INT_N** 和 **LED1**），方法是在通用输入/输出数据选择 1 寄存器 (GPIO_DATA_SEL1) 或通用输入 / 输出数据选择 2 寄存器 (GPIO_DATA_SEL2) 中设置相应的位。通用输入 / 输出需要通过通用输入/输出使能寄存器 (**GPIO_EN**) 启用，并通过通用输入/输出方向寄存器 (GPIO_DIR) 将方向设为输出。

PME 缓冲器类型可在开漏和推挽之间选择，并通过通用输入/输出缓冲器类型寄存器 (GPIO_BUF) 中的 GPIO 缓冲器类型 (GPI- O_BUF) 字段进行配置。默认类型为开漏。

输出控制寄存器中的 PME 极性位用于将 **PME_N** 电平设为高电平有效或低电平有效。默认值为低电平有效。

网络唤醒控制寄存器（MMD 地址 2h，寄存器 10h，位 [15:14]）中的 Pmen_out 字段定义了 PME 的输出功能。

触发后，通过禁用用于使能 PME 触发源（魔术包、自定义数据包、链路状态变化、能量检测状态变化）的寄存器位，**PME_N** 输出将被取消置位。

5.24 NAND 树支持

该器件支持参数化 NAND 树，用于检测芯片 I/O 与电路板之间的故障。上电/复位时，将 **MODE**[4:0] 配置引脚设为 “00100” 即可启用 NAND 树模式。

NAND 树由一系列串联的 NAND 门组成，每个输入引脚都与前一个 NAND 门的输出进行 NAND 运算。[表 5-15](#)列出了 NAND 树引脚顺序。输入 (1) 是树的起点，与常数 “1” 进行 NAND 运算。请注意，LAN8840 还有一次额外的最终反相。

表 5-15: NAND 树测试引脚顺序

Pin	说明
LED5	输入
LED4	输入
LED3	输入
LED2	输入
LED1	输入
TXD0	输入
TXD1	输入
TXD2	输入
TXD3	输入
TXC	输入
TX_CTL	输入
RXD3	输入
RXD2	输入

TABLE 5-15: NAND TREE TEST PIN ORDER (CONTINUED)

Pin	Description
RXD1	Input
RXD0	Input
RX_CTL	Input
RXC	Input
INT_N	Input
MDC	Input
MDIO	Input
CLK125_NDO	Output

To test any given pin, the pin is toggled while holding the lower ranked pins low and the higher ranked pins high. Depending on the position in the chain (odd or even), the output of the tree will reflect the tested input pin or its inverse (odd positions are inverted with respect to the output pin).

APPLICATION NOTE: Since the NAND tree output is on the **CLK125_NDO** pin, the pin must be enabled for output by using the **CLK125_EN** strap input.

5.25 PHY Resets

5.25.1 SOFTWARE RESET

The Gigabit Ethernet PHY may be reset by software by using the IEEE 802.3 standard PHY Soft Reset (**RESET**) bit in the Basic Control Register. This resets all the PHY and all its registers to their default state, with the following exceptions.

- The Variable IO are not re-tuned
(but IO Pad Tune Control Register (**IO_TUNE_CTRL**) is reset)
- The RGMII DLLs are not re-tuned
(but RX DLL Control Register and TX DLL Control Register are reset)
- The crystal oscillator amplitude and termination resistor are not re-trimmed
- Input straps are not re-loaded from the pins
The Operation Mode Strap Override Low Register and the Operation Mode Strap Override High Register are not reset
(Registers that default based on the input straps or on the override registers are reset)
- Analog Control Register 1, Analog Control Register 2, Analog Control Register 9, Analog Control Register 10, Analog Control Register 11, Analog Control Register 12 and Analog Control Register 13 are not reset
(these may be reset with the AFE Soft Reset bit in the Digital Debug Control 2 Register)

An additional software reset is available by using the bit Software Reset bit in the Control Register. This resets all of the PHY except for its registers.

5.25.2 PHY POWER ON READY

Four voltages are monitored.

- +1.1V Analog Power Supply (on **VDDAL**) set for ~0.8V (typical)
 - +1.1V Digital Core Power Supply (on **VDD**) set for ~0.9V (typical)
 - +2.5 / 3.3V Analog Power Supply (on **VDDAH**) set for ~2.1V (typical)
 - Variable I/O Power Supply (on **VDDIO**) set for ~1.5V (typical)
- See [Table 6-11, "POR Thresholds"](#) for the D.C. characteristics of POR.

表 5-15: NAND 树测试引脚顺序（续）

Pin	说明
RXD1	输入
RXD0	输入
接收控制信号_	输入
RXC	输入
INT_N	输入
MDC	输入
MDIO	输入
125 MHz 时钟 _NDO	输出

要测试任意给定的引脚，在将较低优先级的引脚保持为低电平、较高优先级的引脚保持为高电平的同时，切换该引脚。根据它在链中的位置（奇数或偶数），树的输出将反映被测输入引脚或其反相信号（奇数位置相对于输出引脚反相）。

应用说明： 由于 NAND 树输出位于**时钟125_NDO**引脚上，因此必须使用**时钟125_使能**绑带输入来使能该引脚作为输出。

5.25 PHY 复位

5.25.1 软件复位

可通过软件使用 IEEE 802.3 标准的基本控制寄存器中的 PHY 软复位 (**复位**) 位来复位千兆以太网 PHY。此操作会将所有 PHY 及其寄存器复位为默认状态，但以下情况除外。

- 可变 IO 不会重新调谐
(但 IO 焊盘调谐控制寄存器 (**IO_调谐_控制**) 会复位)
- RGMII 延迟锁定环不会重新调谐
(但接收 DLL 控制寄存器和发送 DLL 控制寄存器会复位)
- 晶体振荡器的振幅和端接电阻不会重新微调
- 输入绑带不会从引脚重新加载
操作模式绑带覆盖低位寄存器和操作模式绑带覆盖高位寄存器不会复位

(根据输入绑带或覆盖寄存器确定默认值的寄存器会复位)
- 模拟控制寄存器 1、模拟控制寄存器 2、模拟控制寄存器 9、模拟控制寄存器 10、模拟控制寄存器 11、模拟控制寄存器 12 和模拟控制寄存器 13 不会复位（可通过数字调试控制 2 寄存器中的 AFE 软复位位复位这些寄存器）

还可以通过控制寄存器中的软件复位位执行额外的软件复位。此操作会复位 PHY 的所有部分，但不包括其寄存器。

5.25.2 PHY 上电就绪

监测四种电压。

- +1.1V 模拟电源（在 **VDDAL** 上）设为 ~0.8V（典型值）
 - +1.1V 数字内核电源（在 **VDD** 上）设为 ~0.9V（典型值）
 - +2.5 / 3.3V 模拟电源（在 **VDDAH** 上）设为 ~2.1V（典型值）
 - 可变 I/O 电源（在 **VDDIO** 上）设为 ~1.5V（典型值）
- 请参见 [表 6-11](#)，“[上电复位阈值](#)”，了解上电复位的直流特性。